A Fitted Exponent Is Not the Test
Complete Monotonicity Measured Directly on a Qubit

One Channel of One Device Admits a Positive Rate Density and the Other Does Not; Two Predictions Refutable Against Existing Literature; and a Self-Audit of the Framework That Produced Them

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Pollard's criterion states that $\exp(-x^{\beta})$ is completely monotone if and only if $\beta \leq 1$, and Bernstein's theorem that this holds if and only if the curve is the Laplace transform of a positive measure. A fitted $\beta \leq 1$ therefore certifies that a decay is reproducible by a classical mixture of exponentials. This edition adds the measurement that tests the criterion *directly*, without fitting, and the result reorders the document.

Measured [OBS]. On 20 non-adjacent qubits of one superconducting device — one session, one fixed layout, 39 circuits, 958 452 shots — the criterion answers differently in two channels of the same qubit. Energy relaxation admits a positive rate density (non-negative least squares over 200 free rates, $\chi^2/\text{dof} = 0.99$, 19/20 replicates compatible). Pure dephasing does not ($\chi^2/\text{dof} = 12.43$, 17/20 refuted). The cause needs no model: the echo curve *rises*, in 10 of 220 consecutive segments above 3σ and six above 5σ, peaking at $z = +11.01$, across 7 of 20 qubits, while the T_1 control gives 0 of 220 and backflow exactly zero in all twenty. The Breuer–Laine–Piilo witness, of independent provenance, returns the same verdict on the same qubits, and the residual against a one-rate Lindblad twin crosses to positive and grows to $+10.6\sigma$.

Withdrawn [ERRATUM]. At this precision a two-parameter Kohlrausch form is rejected in both dephasing arms ($\chi^2/\text{dof} = 11.49$ and 1598), so its exponent certifies nothing: *a fitted β read off a curve whose goodness of fit was never examined licenses no inference in either direction.* The pre-registered positive control also fired — the T_1 arm returned $\beta = 0.8664 \pm 0.0285$ where a single rate must give 1 — and although its premise proved false, T_1 not being a single rate on real hardware, a premise falsified after the fact does not annul the clause. Every quantity of that run that passes through the estimator is withdrawn: β per arm, distance to the pole, and branch assignment. What stands is what fits nothing.

Two of this framework's own predictions are tested here by the protocol they named, and both fail: that the Breuer–Laine–Piilo measure is positive for $\beta < 0.8$, and that it can therefore be computed from a fitted β without independent spectroscopy. Two arms of one device with median exponents of 0.9011 and 0.9070 return $\mathcal{N} = 0.00827$ and $\mathcal{N} = 0.00000$: the same β, opposite memory.

Exact identities [EXACT]. The Λ-hierarchy postulate is not irreducible: since $\tau_{\mathrm{P}} E_{\mathrm{P}} = \hbar$ and $\ell_{\mathrm{P}} E_{\mathrm{P}} = \hbar c$ hold identically it reduces to $\tau(\Lambda) = \hbar/E$ and $\ell(\Lambda) = c\,\tau(\Lambda)$, so at $E = mc^2$ the hierarchy length *is* the reduced Compton wavelength. The accessible information satisfies $I_Q(\gamma) = (\gamma/\tau)^{\beta}$ exactly, and a distinguished point at $D/L = 1/2$ carries four coincident properties: exactly one bit, the entanglement sudden-death point for $C_0 = 1$, the maximum of $D\bar{C}$ at $L^2/4$, and a Bures arc of $\pi/3$. All invariant in β, τ and L. The conservation law $D + C = 1$ holds to machine precision across 650 circuits on two IBM backends and $\sim 5.5 \times 10^6$ shots.

No-go theorem [THM]. For a power-law bath and *any* pulse-sequence filter g, $\Gamma(t) = A\,t^{1+\alpha} \int u^{-2-\alpha} g(u)\,du$: the exponent is independent of g. Since $1 - \alpha/2 < 1 \leq 1 + \alpha$, the relation $\beta = 1 - \alpha/2$ is unreachable by any filter-function route, and Axiom 3 is relabelled from theorem to postulate.

Errata [ERRATUM]. The critical qubit number is $n^* = 1.848$, not 2.6. The LIGO ringdown significance, with the internal error scaled by $\sqrt{\chi^2/\nu} = 1.669$ as the catalogue's own goodness of fit requires, is 12.5σ; we record that our first correction of it, 20.9σ, was itself wrong for neglecting that scaling, and that the originally published 13.8σ was closer to the defensible value than our correction. A reported 18 kHz emergence mode lies below the Rayleigh limit of its own data. The published comoving density of remnants corresponds to $\Omega = 0.063$, not to the 0.12 it is stated to reproduce. Two claims of the previous edition are falsified here by measurement rather than by argument: that revivals are slowdowns and not reversals, and that no local operation reverts the decoherence arrow — a single X pulse reverts 67% of the free-induction dephasing.

Transferable criteria [THM]. Three results export, each with the observation that would retire it. Pollard's criterion as a licence condition, now with its precondition stated: the fit must be admissible before its exponent is read. The corrected diffusion bridge $\beta = (1 + \alpha_{\mathrm{diff}})/2$, a correction inside the framework but a *prediction* outside it, whose domain forces $\beta \in (1/2, 1]$ and which,

composed with the criterion, yields the falsifiable statement that superdiffusive transport forbids reading the relaxation of the same medium as a distribution of rates. And the additive coordinate $I_Q = -\ln(1 - D/L)$, which quantifies the information a saturation clamp destroys.

We state plainly the inference this work does *not* license: a fitted $\beta < 1$ is not evidence of quantum behaviour, and — as of this edition — a fitted $\beta < 1$ is not evidence of a classical mixture either unless the fit itself survives. What the theorem establishes is *representability* of the curve, not identification of the mechanism, and the curve is what must be tested.

The endpoint [CONJ]. The framework culminates in a dark-matter candidate: Planck-scale black-hole remnants at $M_{\min} = m_P/2\sqrt{\pi} = 6.140\ \mu$g, an exact identity rather than the approximation $0.28\,m_P$ under which it was published. Evaluating the scenario shows the usual dismissal to be right for the wrong reason. The remnant carries 148.6 J of kinetic energy at galactic velocity and one transit is expected per 0.86 km^2·yr, so rarity is not the obstacle; but the mean free path in rock is 10^{23} ly and the probability of one interaction while crossing the Earth is 10^{-32}, so the energy is never deposited. The candidate is cross-section-limited, not exposure-limited, and the two demand opposite experiments. Four falsifiers are given, two requiring no new instrument.

All results are reproducible from the accompanying implementation; every number of the new measurement is regenerated from the raw counts by a single extractor and verified against a stored digest before this document compiles.

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Part I
The framework

I. Introduction

This paper consolidates into one document a programme previously distributed across three: the original formulation of the stretched-exponential decoherence framework, its hardware verification and extension, and the executable audit that corrected both. The consolidation is not editorial. Presenting the results in the order in which they were *obtained* obscured which of them survive; presenting them in the order in which they *depend on each other* makes the surviving core small, explicit, and separable from the speculation built on top of it.

The present edition adds one measurement, and it is the reason for the change of title. Everything the framework had said about classical representability was read off *fitted* exponents. Section VIII tests the condition directly on a device, without fitting, and finds both that the fitted form fails its own goodness-of-fit precondition and that the answer differs between two channels of the same qubit.

A. The object

The framework studies a single functional. For a quantum system coupled to an environment, the decoherence accumulated after interaction parameter γ is taken to be

$$D(\gamma) = L\left[1 - \exp\left(-\left(\frac{\gamma}{\tau}\right)^{\beta}\right)\right], \qquad (\star)$$

with L the asymptotic ceiling, τ a characteristic scale and β the stretching exponent. The form is Kohlrausch's, and its use as a relaxation law predates quantum information by more than a century. Nothing in this paper claims the functional as new. What the paper establishes concerns its exact structure, the limits of what can be inferred from a fitted β, and where the published derivations do and do not hold.

B. What this edition sets out to settle

Three questions, in the order in which they can be answered, and each stated so that a measurement or a derivation — not a preference — decides it.

1. **Does the decay curve of a real qubit admit a positive rate density?** This is the framework's central empirical commitment, and until this edition it had no direct support: the $\beta < 1$ values that were offered for it came from curves later shown to measure T_1 rather than dephasing. The question is decidable without fitting, by an if-and-only-if, and Sec. VIII decides it. It was registered in advance, with five falsifiers, before any hardware time was spent.
2. **Which of the published derivations survive being re-executed?** Every equation was reimplemented in a dependency-free numerical core and forced to evaluate; every published number was recomputed from the equations or from the accompanying data files. Where a number could not be reproduced it is recorded as an erratum rather than adjusted, and where a correction of ours proved wrong that too is recorded.
3. **What does the framework export to fields that fit stretched exponentials without reference to it?** Only results that come with the observation that would retire them qualify. Three do (Part V); the rest is labelled speculation and kept separate.

What this edition does *not* set out to settle is stated with the same force, because the temptation runs the other way: it does not identify a mechanism, it does not establish universality across devices, and it says nothing about entanglement — the observable of the new measurement is single-qubit and blind to it by construction.

C. What this paper claims, in decreasing order of confidence

The document is organised so that this ordering is also the reading order.

- **[EXACT] Identities** (Part I). The Λ-hierarchy reduces to $\tau = \hbar/E$ and $\ell = \bar{\lambda}_C$; the accessible information collapses to $I_Q = (\gamma/\tau)^\beta$; and a distinguished point at $D/L = 1/2$ carries four coincident exact properties. Verified at machine precision.
- **[OBS] The criterion applied directly** (Sec. VIII). On one device, one session and one fixed layout, energy relaxation admits a positive rate density ($\chi^2/\mathrm{dof} = 0.99$ with 200 free rates) and pure dephasing does not (12.43). The cause is model-free: the echo curve rises, 10 of 220 segments above 3σ against a control at 0 of 220. Two independent criteria concur. *This is the strongest empirical statement in the document, and also the narrowest: one device.*
- **[OBS] Verification** (Part II). The conservation law $D + C = 1$ holds across 650 circuits on two IBM backends and $\sim 5.5 \times 10^6$ shots.
- **[THM] Limits** (Part III). A no-go theorem shows the temporal exponent of dephasing is independent of the pulse sequence, which removes the derivation previously claimed for Axiom 3; a spectral-resolution criterion retires a reported emergence mode; and two published numbers are corrected, including one whose first correction, by us, was itself wrong.
- **[THM] Transferable criteria** (Part V). What the work exports to fields that fit stretched exponentials without reference to this framework, each stated with the observation that would retire it.
- **[CONJ] The gravitational frontier** (Part VI), ending in the dark-matter candidate. Speculative throughout, and labelled as such link by link.

a. Withdrawn in this edition. The previous edition read the corpus through fitted exponents throughout. The new measurement withdraws two things. First, any inference from a β whose fit was not examined: at $\sigma/C \approx 0.011$ the Kohlrausch form is rejected in both dephasing arms ($\chi^2/\mathrm{dof} = 11.49$ and 1598), so those exponents certify nothing. Second, every β-dependent quantity of the new run itself — the pre-registered positive control fired, and although its premise proved false a premise falsified after the fact does not annul the clause (Sec. VIII F).

D. The inferences this paper does not license

Two results invert common readings and are placed here rather than in the conclusion, because they govern how the rest should be interpreted.

By Pollard's criterion, $\exp(-x^\beta)$ is completely monotone if and only if $\beta \le 1$, and by Bernstein's theorem this holds if and only if the curve is the Laplace transform of a *positive* measure. A fitted $\beta \le 1$ therefore certifies that the data are reproducible by a classical mixture of exponential decays — **provided the fit is admissible in the first place**. That proviso is not decorative. Where the data are precise enough to test it, it fails: Sec. VIII B rejects the two-parameter form in both dephasing arms, and the curve that its exponent would have licensed is refuted by the direct test. *Read a β only after its goodness of fit has been reported.*

The corpus does not sit uniformly on one side of that threshold, and an earlier version of this statement, which asserted that it did, was wrong. It splits. The stretched states (Dicke, GHZ) report $\beta < 1$ and fall inside the gate. The locally entangled ones report $\beta \approx 1.1$ (Sec. IX), with fits as tight as 1.153 ± 0.014 at $R^2 = 0.9999$; for those the gate points the other way and *forbids* a positive-mixture representation. Both readings follow from the same theorem, and the split is the more informative statement.

Within the stretched branch, a stretched exponential is consequently *not* evidence of quantum behaviour. In the regime where this framework fits best, it certifies compatibility with classicality. The framework's value lies in what it forbids, and Sec. XXII A makes that forbidding quantitative.

E. What remains an indication

The document distinguishes three grades below a result, and keeps them apart. *Indications* are observations with a number attached that do not yet decide anything: the qubit-to-qubit dispersion of β is consistent with fluctuator-dominated noise, but the measurement does not identify the mechanism. *Hypotheses* **[HIP]** have partial support and a declared test that has not been run. *Conjectures* **[CONJ]** have neither, and are carried because they are falsifiable, not because they are likely. Section XXV lists all of them with what would settle each, and Appendix A lists the inconsistencies still open in the corpus itself.

F. Reproducibility

Every numerical claim in this document is recomputed from the equations by an accompanying dependency-free implementation: regression tests plus checks that recompute published numbers from the data files they came from. The measurement of Sec. VIII adds a stricter rule, because it is new data: a single extractor regenerates every quoted figure from the raw counts, writes them with the digest of the counts file, and a verification mode fails if any of them has moved. The figures are generated by a script that refuses to write if its own numbers disagree with that file. Appendix B gives the procedure.

II. Complete Mathematical Framework

This section develops the mathematical structure of the Stretched-exponential Decoherence Framework (SDF) from first principles. Rather than postulating the functional form of $D(\gamma)$, I show that it emerges as the *unique* solution to a physically motivated differential equation. This section contains all definitions, theorems, and proofs that underpin the remainder of the paper.

A. Axiomatic Foundation

I begin with three physical axioms from which the entire framework follows.

Definition 1 (Decoherence Functional)**.** *For a quantum system $\mathcal{S}$ with Hilbert space $\mathcal{H}_S$, coupled to an environment $\mathcal{E}$ with Hilbert space $\mathcal{H}_E$, the* decoherence functional *is a map $D : \mathbb{R}_+ \to [0, L]$ parametrized by the noise exposure $\gamma \geq 0$. The coherence complement is $C(\gamma) \equiv 1 - D(\gamma)$.*

Axiom 1 (Bounded Decoherence) **[DEF]**. *The decoherence functional D satisfies:*

(i) $D(0) = 0$ (initial purity),

(ii) $\lim_{\gamma\to\infty} D(\gamma) = L \leq 1$ (asymptotic limit),

(iii) $D \in C^\infty(\mathbb{R}_+)$ and D is strictly monotone increasing.

The parameter $L \in (0, 1]$ represents the maximum decoherence achievable by the system–environment coupling. For a system that fully decoheres, $L = 1$.

Axiom 2 (Generalized Decay) **[DEF]**. *The functional D satisfies the first-order ordinary differential equation:*

$$\boxed{\frac{dD}{d\gamma} = \frac{\beta}{\tau}\left(\frac{\gamma}{\tau}\right)^{\beta-1}(L - D)} \tag{1}$$

where $\beta \in (0, 2]$ is the stretching exponent encoding environmental noise structure, and $\tau > 0$ is the characteristic decoherence scale.

The ODE (1) has a transparent physical reading: the rate of decoherence at noise exposure γ is proportional to the *remaining coherence* $(L - D)$, modulated by a power-law kernel $(\gamma/\tau)^{\beta-1}$ that encodes memory effects. For $\beta = 1$ this reduces to the memoryless (Markovian) decay $dD/d\gamma = (L - D)/\tau$.

Axiom 3 (Environmental Encoding — Noise-Structure Duality) **[DEF]**. *The stretching exponent β encodes the noise power spectrum $S(\omega)$ of the environment through:*

$$\boxed{\beta = 1 - \frac{\alpha}{2}, \qquad S(\omega) \propto \frac{1}{\omega^\alpha}, \quad \alpha \in [0, 2)} \tag{2}$$

This maps the noise exponent α bijectively onto $\beta \in (0, 1]$: white noise ($\alpha = 0$) gives $\beta = 1$ (Markovian), $1/f$ noise ($\alpha = 1$) gives $\beta = 1/2$ (deeply stretched), and the limit $\alpha \to 2$ corresponds to $\beta \to 0$ (frozen dynamics).

I emphasize that Axiom 3 establishes a *duality* between the frequency-domain description (noise spectrum) and the time/noise-domain description (stretched exponent), unifying the physics of the environment with the mathematical form of the decoherence functional.

Theorem 1 (Existence and Uniqueness — from Axioms to the SDF Equation)**.** ***[THM]***
Axioms 1 and 2 admit a unique solution:

$$\boxed{D(\gamma) = L\left[1 - \exp\left(-\left(\frac{\gamma}{\tau}\right)^\beta\right)\right]} \tag{3}$$

Proof. Separate variables in the ODE (1):

$$\frac{dD}{L - D} = \frac{\beta}{\tau}\left(\frac{\gamma}{\tau}\right)^{\beta-1} d\gamma\,. \tag{4}$$

Integrate both sides. The left-hand side gives $-\ln(L - D) + C_1$, while the right-hand side yields $(\gamma/\tau)^\beta + C_2$. Combining constants: $-\ln(L - D) = (\gamma/\tau)^\beta + C$. Apply the initial condition $D(0) = 0$ (Axiom 1i): $-\ln L = C$, so that $-\ln\bigl(\frac{L-D}{L}\bigr) = \bigl(\frac{\gamma}{\tau}\bigr)^\beta$. Exponentiating: $\frac{L-D}{L} = \exp\Bigl(-\bigl(\frac{\gamma}{\tau}\bigr)^\beta\Bigr)$, hence $D(\gamma) = L\bigl[1 - \exp\bigl(-(\gamma/\tau)^\beta\bigr)\bigr]$.

Uniqueness. Write the ODE as $dD/d\gamma = f(\gamma, D)$ with $f(\gamma, D) = (\beta/\tau)(\gamma/\tau)^{\beta-1}(L - D)$. For $\beta \geq 1$, f

is Lipschitz in D on any compact subset of $\mathbb{R}_+ \times [0, L]$, and the Picard–Lindelöf theorem guarantees uniqueness. For $0 < \beta < 1$, $f(\gamma, D)$ has an integrable singularity at $\gamma = 0^+$; uniqueness follows from the Osgood criterion since $|f(\gamma, D_1) - f(\gamma, D_2)| \leq (\beta/\tau)(\gamma/\tau)^{\beta-1}|D_1 - D_2|$ and the coefficient is integrable near $\gamma = 0$. ☐

Remark. Equation (3) is *not* an ansatz. It is the unique solution to the physical ODE (1) subject to boundary conditions dictated by Axiom 1. This deductive structure parallels Newton's derivation of Keplerian orbits from the inverse-square law: the functional form is a theorem, not a hypothesis.

Theorem 2 (Basic Properties)**.** ***[THM]***
The functional $D(\gamma)$ in Eq. (3) satisfies:

(i) $D(0) = 0$ *(quantum boundary),*

(ii) $\lim_{\gamma\to\infty} D(\gamma) = L$ *(classical limit),*

(iii) $\partial D/\partial\gamma > 0$ *for all* $\gamma > 0$ *(strict monotonicity),*

(iv) $D \in C^\infty(\mathbb{R}_+)$ *(smoothness),*

(v) D *is strictly concave for* $\beta \leq 1$.

Proof. Properties (i)–(iv) follow by direct computation from Eq. (3): (i) $D(0) = L(1 - e^0) = 0$. (ii) $\lim_{\gamma\to\infty} D = L(1 - 0) = L$. (iii) From (1), $dD/d\gamma > 0$ since each factor β, τ^{-1}, $(\gamma/\tau)^{\beta-1}$, and $(L - D)$ is positive for $\gamma > 0$. (iv) D is a composition of C^∞ functions on $\mathbb{R}_+$.

For (v), compute the second derivative:

$$\frac{d^2D}{d\gamma^2} = \frac{L\beta}{\tau^2}\left(\frac{\gamma}{\tau}\right)^{\beta-2} e^{-(\gamma/\tau)^\beta}\left[(\beta - 1) - \beta\left(\frac{\gamma}{\tau}\right)^\beta\right]. \quad (5)$$

For $\beta \leq 1$, both terms in brackets are non-positive (the first is ≤ 0 and the second is < 0 for $\gamma > 0$), giving $d^2D/d\gamma^2 < 0$: strict concavity. ☐

Theorem 3 (Inverse Function)**.** ***[THM]***
The map $D : [0, \infty) \to [0, L)$ *is a bijection. Its inverse is:*

$$\gamma = \tau\left[-\ln\left(1 - \frac{D}{L}\right)\right]^{1/\beta} \quad (6)$$

Proof. Strict monotonicity (Theorem 2(iii)) guarantees that D is injective. Since D is continuous with $D(0) = 0$ and $\lim_{\gamma\to\infty} D = L$, the intermediate value theorem gives surjectivity onto $[0, L)$. Solving Eq. (3) for γ:

$$\begin{aligned} 1 - D/L &= e^{-(\gamma/\tau)^\beta} \\ -\ln(1 - D/L) &= (\gamma/\tau)^\beta \\ \gamma &= \tau[-\ln(1 - D/L)]^{1/\beta}. \end{aligned}$$
☐

Theorem 4 (Scale Invariance)**.** ***[THM]***
For any $\lambda > 0$:

$$D(\lambda\gamma;\ \lambda\tau,\ \beta,\ L) = D(\gamma;\ \tau,\ \beta,\ L) \quad (7)$$

That is, D *depends on* γ *and* τ *only through the dimensionless ratio* γ/τ.

Proof. Direct substitution: $D(\lambda\gamma; \lambda\tau, \beta, L) = L[1 - \exp(-(\lambda\gamma/\lambda\tau)^\beta)] = L[1 - \exp(-(\gamma/\tau)^\beta)] = D(\gamma; \tau, \beta, L)$. ☐

Physical consequence. Scale invariance implies that the SDF is insensitive to the choice of units for the noise parameter γ; only the ratio γ/τ is physically meaningful. This parallels the scale invariance of dimensional analysis in classical mechanics and constrains the form of any extension of the framework.

B. The Deviation Operator

The SDF makes a sharp, testable prediction: ideal (simulated) decoherence is exponential ($\beta = 1$), while hardware decoherence is stretched ($\beta < 1$) due to the non-trivial noise spectrum of real environments. I now formalize this observation.

Definition 2 (Ideal Decoherence)**.** *In the Markovian limit (white noise,* $\alpha = 0$*, hence* $\beta = 1$*):*

$$D_{\text{ideal}}(\gamma) = L\left[1 - \exp\left(-\frac{\gamma}{\tau_{\text{id}}}\right)\right] \quad (8)$$

Definition 3 (Hardware Decoherence)**.** *For a real environment with noise spectrum* $S(\omega) \propto 1/\omega^\alpha$:

$$D_{\text{hw}}(\gamma) = L\left[1 - \exp\left(-\left(\frac{\gamma}{\tau_{\text{hw}}}\right)^{\beta_{\text{hw}}}\right)\right], \qquad \beta_{\text{hw}} = 1 - \alpha/2 \quad (9)$$

Definition 4 (Deviation Field)**.** *The deviation field captures the pointwise difference:*

$$\Delta(\gamma;\ \alpha) \equiv D_{\text{hw}}(\gamma) - D_{\text{ideal}}(\gamma) \quad (10)$$

When $\tau_{\text{hw}} = \tau_{\text{id}} \equiv \tau$ *(matched scales), this isolates the pure effect of stretching.*

Theorem 5 (Deviation Properties)**.** ***[THM]***
For matched scales $\tau_{\text{hw}} = \tau_{\text{id}} = \tau$ *and* $\alpha > 0$:

(i) $\Delta(0;\ \alpha) = 0$ *(preserved initial conditions),*

(ii) $\lim_{\gamma\to\infty} \Delta(\gamma;\ \alpha) = 0$ *(asymptotic equivalence),*

(iii) $\exists!\ \gamma_{\max}(\alpha) > 0$ *such that* $|\Delta|$ *is maximized ("deviation peak"),*

(iv) The deviation integral satisfies $I_\Delta \equiv \int_0^\infty |\Delta(\gamma;\ \alpha)|\, d\gamma = O(\alpha)$ *for small* α.

Proof. (i) Both $D_{\text{hw}}(0) = 0$ and $D_{\text{ideal}}(0) = 0$, so $\Delta(0) = 0$.

(ii) Both limits equal L, so $\Delta(\gamma \to \infty) \to L - L = 0$.

(iii) Δ is continuous, vanishes at $\gamma = 0$ and $\gamma \to \infty$, and is not identically zero for $\alpha > 0$ (since $e^{-(\gamma/\tau)^\beta} \neq e^{-\gamma/\tau}$ when $\beta \neq 1$). By the extreme value theorem (Weierstrass), the continuous function $|\Delta|$ on $[0, M]$ attains its maximum; since $\Delta(0) = 0$ and $\Delta(M) \to 0$ while Δ is not identically zero, this maximum occurs at an interior point. Uniqueness of $\gamma_{\max}$ follows from the analysis of $\partial\Delta/\partial\gamma = 0$, which yields a single transcendental equation with unique solution for β close to 1.

(iv) Taylor-expand $e^{-(\gamma/\tau)^\beta}$ around $\beta = 1$. Writing $\beta = 1 - \alpha/2$:

$$e^{-(\gamma/\tau)^\beta} = e^{-\gamma/\tau}\left[1 + \frac{\alpha}{2}\frac{\gamma}{\tau}\ln\frac{\gamma}{\tau} + O(\alpha^2)\right].$$

Hence $\Delta(\gamma;\alpha) = -L\cdot(\alpha/2)\cdot e^{-\gamma/\tau}\cdot(\gamma/\tau)\ln(\gamma/\tau)+O(\alpha^2)$, and $I_\Delta = O(\alpha)$ follows by dominated convergence. □

Theorem 6 (Deviation Bound)**.** ***[THM]***

$$\sup_{\gamma\geq 0}|\Delta(\gamma;\,\alpha)| \leq L\cdot\frac{\alpha}{2}\cdot e^{-1}[1+O(\alpha)] \tag{11}$$

Proof. At the supremum, $\partial\Delta/\partial\gamma = 0$. From the first-order expansion (proof of Theorem 5(iv)), the extremum of $g(x) = x\ln x\cdot e^{-x}$ (where $x = \gamma/\tau$) occurs near $x = 1$, where $|g(1)| = 0$ but the deviation expression involves $|x\ln x|\cdot e^{-x}$. The function $h(x) = |x\ln x|\cdot e^{-x}$ on $(0,\infty)$ attains its maximum at $x^* = e^{-W(1)} \approx 0.278$ (Lambert W branch), giving $h(x^*) \leq e^{-1}$. Hence $\sup|\Delta| \leq L\cdot(\alpha/2)\cdot e^{-1} + O(\alpha^2)$. □

Corollary 1. *The hardware deviation is bounded by:*

- $|\Delta|_{\max} \leq 0.184\,L$ *for pure* $1/f$ *noise (*$\alpha = 1,\ \beta = 0.5$*),*
- $|\Delta|_{\max} \leq 0.092\,L$ *for* $\alpha = 0.5$ *(*$\beta = 0.75$*),*
- $|\Delta|_{\max} \leq 0.037\,L$ *for* $\alpha = 0.2$ *(*$\beta = 0.9$*).*

These bounds are tight: numerical evaluation confirms that the first-order estimate is accurate to within 5% for $\alpha \leq 0.5$*.*

Physical Interpretation: The Deviation Operator

The deviation field $\Delta(\gamma;\alpha)$ quantifies exactly how much information is redistributed *differently* due to environmental noise structure. In ideal simulation, decoherence follows the geodesic path (pure exponential). On real hardware, the noise spectrum *bends* this path, creating a stretched trajectory. The deviation field Δ measures this bending at each noise exposure γ.
The bounded nature of Δ (Theorem 6) ensures that hardware decoherence never diverges catastrophically from the ideal model — it is a perturbative deformation, not a qualitative change. This is why the SDF framework can describe both regimes with a single equation.

C. Transition Rate and Dynamical Structure

Definition 5 (Transition Rate)**.** *The instantaneous decoherence rate is:*

$$R(\gamma) \equiv \frac{dD}{d\gamma} = \frac{L\beta}{\tau}\left(\frac{\gamma}{\tau}\right)^{\beta-1}\exp\!\left(-\left(\frac{\gamma}{\tau}\right)^{\beta}\right) \tag{12}$$

Theorem 7 (Rate Properties — The Dynamical Trichotomy)**.** ***[THM]***
The transition rate $R(\gamma)$ *has the following properties:*

(i) $R(\gamma) > 0$ *for all* $\gamma > 0$ *(irreversibility),*

(ii) For $\beta < 1$*:* $\lim_{\gamma\to 0^+} R(\gamma) = +\infty$ *(singular onset — instant response),*

(iii) For $\beta = 1$*:* $R(0) = L/\tau$ *(finite onset — Markovian),*

(iv) For $\beta > 1$*:* $R(0^+) = 0$ *(quiescent onset — sub-Markovian),*

(v) For $\beta > 1$*,* R *has a unique maximum at*

$$\gamma_{\text{peak}} = \tau\left(\frac{\beta-1}{\beta}\right)^{1/\beta} \tag{13}$$

with peak rate

$$R_{\max} = \frac{L\beta}{\tau}\left(\frac{\beta-1}{\beta}\right)^{(\beta-1)/\beta}\exp\!\left(-\frac{\beta-1}{\beta}\right). \tag{14}$$

Proof. (i) Each factor in Eq. (12) is positive for $\gamma > 0$.

(ii) For $\beta < 1$, the factor $(\gamma/\tau)^{\beta-1} \to \infty$ as $\gamma \to 0^+$, while $e^{-(\gamma/\tau)^\beta} \to 1$, so $R \to \infty$.

(iii) For $\beta = 1$: $R(\gamma) = (L/\tau)e^{-\gamma/\tau}$, hence $R(0) = L/\tau$.

(iv) For $\beta > 1$: $(\gamma/\tau)^{\beta-1} \to 0$ as $\gamma \to 0^+$, so $R(0^+) = 0$.

(v) Setting $dR/d\gamma = 0$ and writing $u = (\gamma/\tau)^\beta$:

$$\frac{dR}{d\gamma} \propto (\beta-1)(\gamma/\tau)^{\beta-2} - \beta(\gamma/\tau)^{2\beta-2}/\tau = 0 \implies u = \frac{\beta-1}{\beta},$$

giving $\gamma_{\text{peak}} = \tau[(\beta-1)/\beta]^{1/\beta}$. Substitution into R yields (14). This is a unique maximum since $R(0^+) = 0$, $R(\infty) = 0$, and $R > 0$ in between. □

Physical significance. The trichotomy $\{\beta < 1,\ \beta = 1,\ \beta > 1\}$ is the *mathematical origin* of the three empirically observed decoherence regimes:

- **Stretched regime** ($\beta < 1$): Initial rapid decoherence followed by power-law slowing. Characteristic of systems with $1/f^\alpha$ noise ($\alpha > 0$). Observed on IBM Quantum hardware.
- **Exponential regime** ($\beta = 1$): Constant rate decay. The Markovian limit. Observed in ideal noise-free simulations.

- **Compressed/revival regime** ($\beta > 1$): Delayed onset with accelerating decoherence. Signature of structured environments with finite correlation time. Observed in revival phenomena (Sec. IX).

Theorem 8 (Asymptotic Behavior). ***[THM]***
The decoherence functional admits the following asymptotic expansions:

(i) Early-time *($\gamma \ll \tau$):*

$$D(\gamma) = L\left(\frac{\gamma}{\tau}\right)^{\beta} - \frac{L}{2}\left(\frac{\gamma}{\tau}\right)^{2\beta} + O\big((\gamma/\tau)^{3\beta}\big) \tag{15}$$

(ii) Late-time *($\gamma \gg \tau$):*

$$D(\gamma) = L - L\exp\left(-\left(\frac{\gamma}{\tau}\right)^{\beta}\right) \tag{16}$$

with the correction term vanishing super-exponentially.

(iii) Half-decoherence point*: $D(\gamma_{1/2}) = L/2$ at*

$$\gamma_{1/2} = \tau\,(\ln 2)^{1/\beta} \tag{17}$$

Proof. (i) Taylor expand $e^{-x} = 1 - x + x^2/2 - \cdots$ with $x = (\gamma/\tau)^{\beta}$.
(ii) Direct from Eq. (3).
(iii) Set $1 - D/L = 1/2$, hence $(\gamma/\tau)^{\beta} = \ln 2$, so $\gamma_{1/2} = \tau(\ln 2)^{1/\beta}$. □

D. The Noise Functional and Lindblad Connection

I now derive the SDF equation from the standard open quantum systems formalism [1, 2], establishing a rigorous connection between Axiom 3 and the Lindblad master equation [3].

Theorem 9 (Lindblad Derivation of Stretched-Exponential Decoherence). ***[THM]***
Consider a qubit undergoing pure dephasing under the time-local Lindblad master equation:

$$\frac{d\rho}{dt} = -i[H,\rho] + \sum_k \gamma_k(t)\left(L_k\rho L_k^{\dagger} - \frac{1}{2}\{L_k^{\dagger}L_k,\rho\}\right) \tag{18}$$

with time-dependent dephasing rates $\gamma_k(t)$ determined by the noise spectral density $S(\omega)$. For pure dephasing ($L_k = \sigma_z$), the off-diagonal coherence evolves as:

$$\rho_{01}(t) = \rho_{01}(0)\,e^{-\Phi(t)} \tag{19}$$

where the dephasing function $\Phi(t)$ is given by the noise integral:

$$\Phi(t) = \int_0^{\infty} \frac{S(\omega)}{\pi}\,\frac{\sin^2(\omega t/2)}{(\omega/2)^2}\,d\omega \tag{20}$$

For a power-law noise spectrum $S(\omega) = A/\omega^{\alpha}$ with infrared cutoff $\omega_{\rm ir}$ and ultraviolet cutoff $\omega_{\rm uv}$, in the regime $1/\omega_{\rm uv} \ll t \ll 1/\omega_{\rm ir}$:

$$\Phi(t) \sim c_{\alpha}\,t^{2-\alpha} \tag{21}$$

where c_{α} depends on A and cutoff parameters:

$$c_{\alpha} = \frac{A}{\pi}\,\Gamma(1-\alpha)\sin\left(\frac{\pi\alpha}{2}\right) \quad \textit{for } 0 < \alpha < 1,$$

and for the special case $\alpha = 1$ (pure $1/f$ noise): $\Phi(t) \sim (A/\pi)\,t^2\ln(\omega_{\rm uv}t)$.

Hence the decoherence function takes the stretched-exponential form:

$$W(t) \equiv e^{-\Phi(t)} \sim \exp\left(-\left(\frac{t}{\tau_{\rm eff}}\right)^{2-\alpha}\right) \tag{22}$$

with $\tau_{\rm eff} = c_{\alpha}^{-1/(2-\alpha)}$. Identifying $\beta = 2-\alpha$ in the physical dephasing exponent, or equivalently $\beta = 1-\alpha_{\rm SDF}/2$ in the SDF convention where $\alpha_{\rm SDF} \in [0,2)$, recovers the SDF equation (3).

Proof. Starting from Eq. (20) with $S(\omega) = A/\omega^{\alpha}$, split the integral at $\omega_0 = 1/t$:

$$\Phi(t) = \frac{A}{\pi}\int_{\omega_{\rm ir}}^{1/t} \frac{\sin^2(\omega t/2)}{\omega^{\alpha}\,(\omega/2)^2}\,d\omega \;+\; \frac{A}{\pi}\int_{1/t}^{\omega_{\rm uv}} \frac{\sin^2(\omega t/2)}{\omega^{\alpha}\,(\omega/2)^2}\,d\omega\,. \tag{23}$$

For $\omega t \ll 1$: $\sin^2(\omega t/2)/(\omega/2)^2 \approx t^2$, contributing $\sim At^2\int\omega^{-\alpha}d\omega$. For $\omega t \gg 1$: $\sin^2(\omega t/2) \to 1/2$ on average, contributing $\sim 2A\int\omega^{-2-\alpha}d\omega$. The dominant term for $0 < \alpha < 2$ scales as $\Phi(t) \propto t^{2-\alpha}$, with the proportionality constant given by the gamma-function expression. Inverting gives $W(t) = \exp\big(-(t/\tau_{\rm eff})^{2-\alpha}\big)$, which is precisely the SDF form with $\beta_{\rm phys} = 2-\alpha$. □

Convention note. The mapping between the physical noise exponent α and the SDF stretching exponent β depends on the specific dephasing protocol. For free-induction decay of transmon qubits with $1/f^{\alpha}$ flux noise (the primary experimental context of this paper), the appropriate filter function yields $\beta = 1-\alpha_{\rm SDF}/2$ as stated in Axiom 3. The consistency of this convention is validated experimentally in Sec. VI and Sec. X.[1]

E. Coherence Metrics and Operational Measures

The abstract decoherence functional $D(\gamma)$ connects to experiment through operationally defined metrics.

Definition 6 (Leakage Metric). *For an n-qubit system, the leakage-based decoherence is:*

$$D_{\rm leak}(n) = 1 - P(\textit{valid subspace}) \tag{24}$$

where $P(\textit{valid})$ is the probability of measuring outcomes in the expected computational subspace.

---

[1] Throughout this work, I adopt the SDF convention $\beta = 1 - \alpha/2$ (Axiom 3), which relates to the Lindblad-derived exponent $\beta_{\rm Lindblad} = 2-\alpha$ via the identity $\beta_{\rm SDF} = \beta_{\rm Lindblad}/2$ for pure dephasing channels.

Definition 7 (Stabilizer Metric)**.** *For a stabilizer state with generators* $\{K_1, \ldots, K_n\}$*:*

$$D_{\text{stab}}(n) = 1 - \frac{1}{n}\sum_{i=1}^{n}\langle K_i\rangle \tag{25}$$

Definition 8 (Fidelity Metric)**.** *Given target state* ρ_{target} *and measured state* ρ_{meas}*:*

$$D_{\text{fid}} = 1 - F(\rho_{\text{meas}},\, \rho_{\text{target}}) \tag{26}$$

where $F(\rho,\sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\right)^2$ *is the Uhlmann fidelity.*

Proposition 1 (Metric Ordering)**.** *For stabilizer states measured in the computational basis:*

$$D_{\text{fid}} \le D_{\text{leak}} \le D_{\text{stab}} \tag{27}$$

Proof. The fidelity $F \ge P(\text{valid})$ since fidelity accounts for coherent overlap within the valid subspace as well as population. Hence $D_{\text{fid}} = 1 - F \le 1 - P(\text{valid}) = D_{\text{leak}}$. The stabilizer metric averages over all generators, each of which can detect errors not reflected in the global valid-subspace probability, hence $D_{\text{leak}} \le D_{\text{stab}}$. □

All three metrics satisfy the SDF equation (3) with the *same* (β, L) but potentially different τ values reflecting the sensitivity of each metric to the underlying decoherence process. This metric independence of the functional form (but not the scale) is a non-trivial prediction validated in Sec. VI.

F. Parameter Space Geometry

The SDF parameters form a structured geometric space whose properties constrain experimental inference and physical interpretation.

Definition 9 (SDF Parameter Space)**.** *The SDF parameter space is the open manifold:*

$$\mathcal{P} = \{(\tau, \beta, L) \,:\, \tau > 0,\ \beta \in (0, 2],\ L \in (0, 1]\} \tag{28}$$

with natural coordinates $(\ln\tau, \beta, L)$*.*

Definition 10 (Fisher Information Metric)**.** *Given* N *independent measurements* $\{d_i\}$ *at noise exposures* $\{\gamma_i\}$*, the likelihood is* $p(\{d_i\}|\boldsymbol{\theta})$ *with* $\boldsymbol{\theta} = (\tau, \beta, L)$*. The Fisher information matrix is:*

$$[\mathbf{F}]_{ij} = \mathbb{E}\left[\frac{\partial \ln p}{\partial\theta_i}\frac{\partial \ln p}{\partial\theta_j}\right] \tag{29}$$

This induces a Riemannian metric on $\mathcal{P}$*:*

$$ds^2_{\text{CR}} = \sum_{i,j} F_{ij}\, d\theta_i\, d\theta_j \tag{30}$$

known as the Cramér–Rao metric.

Theorem 10 (Fisher Matrix Structure)**.** *[THM]*
For Gaussian measurement noise with variance σ^2 *and* N *measurements uniformly distributed in* $\gamma \in [0, \gamma_{\max}]$*, the diagonal Fisher information elements scale as:*

$$F_{\tau\tau} \propto \frac{N\beta^2L^2}{\sigma^2\tau^2} \tag{31}$$

$$F_{\beta\beta} \propto \frac{NL^2}{\sigma^2}\left\langle\left(\frac{\gamma}{\tau}\right)^{2\beta}\left(\ln\frac{\gamma}{\tau}\right)^2\right\rangle \tag{32}$$

$$F_{LL} \propto \frac{N}{\sigma^2} \tag{33}$$

Proof. From the SDF model $D(\gamma;\boldsymbol{\theta})$ with Gaussian likelihood:

$$\ln p = -\frac{1}{2\sigma^2}\sum_i [d_i - D(\gamma_i;\boldsymbol{\theta})]^2 + \text{const.}$$

The Fisher matrix elements are $F_{ij} = \sigma^{-2}\sum_k (\partial D/\partial\theta_i)(\partial D/\partial\theta_j)|_{\gamma_k}$.

Computing partial derivatives from Eq. (3):

$$\frac{\partial D}{\partial \tau} = -\frac{L\beta}{\tau}\left(\frac{\gamma}{\tau}\right)^{\beta} e^{-(\gamma/\tau)^{\beta}},$$

$$\frac{\partial D}{\partial \beta} = L\left(\frac{\gamma}{\tau}\right)^{\beta}\ln\left(\frac{\gamma}{\tau}\right) e^{-(\gamma/\tau)^{\beta}},$$

$$\frac{\partial D}{\partial L} = 1 - e^{-(\gamma/\tau)^{\beta}}.$$

Substituting into F_{ij} and replacing sums by integrals for large N yields the stated scalings. □

Physical interpretation. The Fisher metric on $\mathcal{P}$ measures the *statistical distinguishability* of nearby decoherence processes. Regions of large F_{ij} correspond to parameter ranges where experiments are maximally informative; regions of small F_{ij} indicate parameter degeneracies. Key observations:

- $F_{\beta\beta}$ peaks when $\gamma_{\max} \sim \tau$: experiments must probe the transition region to constrain β effectively.
- $F_{\tau\tau} \propto \beta^2$: the decoherence scale τ is hardest to determine for small β (deeply stretched), where the transition is gradual.
- The off-diagonal element $F_{\tau\beta}$ induces a τ–β correlation, visible as "banana-shaped" confidence contours in parameter estimation (Sec. V).

Proposition 2 (Geodesics in Parameter Space)**.** *In the limit* $\gamma_{\max} \gg \tau$ *and for fixed* L*, the Fisher metric on the* $(\ln\tau, \beta)$ *submanifold is approximately conformally flat:*

$$ds^2 \approx \frac{NL^2}{\sigma^2}\left[\beta^2\,(d\ln\tau)^2 + \kappa(\beta)\,d\beta^2\right] \tag{34}$$

where $\kappa(\beta)$ *is a bounded, positive function of* β*. Geodesics in this metric describe the most efficient experimental paths for resolving parameter changes.*

G. Information-Theoretic Structure

Theorem 11 (Information Conservation). ***[THM]***
Define the total information content:

$$I_{\text{total}}(\gamma) \equiv D(\gamma) + C(\gamma) = D(\gamma) + [1 - D(\gamma)] = 1 \quad (35)$$

Then:

$$\frac{dI_{\text{total}}}{d\gamma} = 0 \quad \forall\, \gamma \geq 0 \quad (36)$$

Proof. $I_{\text{total}} = 1$ identically by definition, hence $dI_{\text{total}}/d\gamma = 0$. □

Remark. While Theorem 11 is algebraically trivial, its physical content is profound: decoherence does not destroy information but *redistributes* it between system-accessible coherence $C(\gamma)$ and environment-accessible decoherence $D(\gamma)$. This is the SDF restatement of unitarity.

Theorem 12 (Entropy–Decoherence Monotonicity). ***[THM]***
For a system decohering from a pure initial state, the von Neumann entropy $S(\rho(\gamma)) = -\text{Tr}[\rho \ln \rho]$ satisfies:

$$\frac{dS}{d\gamma} \geq 0 \quad \forall\, \gamma \geq 0 \quad (37)$$

with equality only at $\gamma = 0$ (if $\beta < 1$) or in the $\gamma \to \infty$ limit. Moreover, in the qubit case:

$$S(\gamma) = H_{\text{bin}}\left(\frac{1 + \sqrt{1 - D(\gamma)^2/L^2}}{2}\right) \quad (38)$$

where $H_{\text{bin}}(p) = -p \ln p - (1-p)\ln(1-p)$ is the binary entropy.

Proof. Entropy increase follows from the monotonicity of $D(\gamma)$ (Theorem 2(iii)) and the monotonic relationship between decoherence and mixedness for initially pure states. The qubit expression follows from the eigenvalues of the qubit density matrix $\rho = \frac{1}{2}(\mathbb{I} + \vec{r} \cdot \vec{\sigma})$ with $|\vec{r}|^2 = 1 - D^2/L^2$ in the dephasing channel. □

H. Summary of Mathematical Results

Table I collects the principal mathematical results of this section. All statements marked **[THM]** are rigorously proven; their logical dependence flows from Axioms 1–3 through the existence theorem (Theorem 1) to derived properties, deviation bounds, and information-theoretic consequences.

TABLE I. Summary of mathematical results. All theorems are proven from Axioms 1–3.

Result	**Content**	**Eq.**
Thm. 1	Existence & uniqueness	(3)
Thm. 2	Basic properties	—
Thm. 3	Inverse function	(6)
Thm. 4	Scale invariance	(7)
Thm. 5	Deviation properties	(10)
Thm. 6	Deviation bound	(11)
Thm. 7	Rate trichotomy	(12)
Thm. 8	Asymptotics	(15)
Thm. 9	Lindblad derivation	(22)
Thm. 11	Information conservation	(35)
Thm. 12	Entropy monotonicity	(37)
Thm. 10	Fisher information	(29)

III. Information Geometry and the Entropic Phase Space

The Stretching/Decoherence Formalism, as developed in the preceding sections, parametrizes the quantum-to-classical transition through the decoherence function $D(\gamma)$ and its complement $C(\gamma)$. In this section I elevate these quantities from computational tools to geometric objects, embedding the SDF dynamics into the natural Riemannian structure of quantum state space. This yields a new formalism—the *entropic phase space*—in which the quantum-classical boundary acquires a precise geometric definition as a signature-change hypersurface.

A. The Space of Quantum States as a Riemannian Manifold

For a d-dimensional quantum system, the space of density matrices

$$\mathcal{M}_d = \left\{\rho \in \mathbb{C}^{d\times d} : \rho \geq 0,\ \text{Tr}(\rho) = 1\right\} \quad (39)$$

is a (d^2-1)-dimensional smooth manifold with boundary. The interior $\mathcal{M}_d^\circ$ consists of full-rank (mixed) states; the boundary $\partial\mathcal{M}_d$ comprises the singular states, including all pure states.

The *Bures metric* [4, 5] endows $\mathcal{M}_d$ with a monotone Riemannian structure:

$$ds_B^2 = \frac{1}{2}\sum_{m,n} \frac{|d\rho_{mn}|^2}{\lambda_m + \lambda_n}\,, \quad (40)$$

where λ_m are the eigenvalues of ρ and $d\rho_{mn} = \langle m|d\rho|n\rangle$ in the eigenbasis. This metric is distinguished by its operational meaning: the Bures distance $d_B(\rho,\sigma) = \sqrt{2\big(1 - \sqrt{F(\rho,\sigma)}\big)}$ is a monotone function of the Uhlmann fidelity F.

For a qubit ($d = 2$), the state space is the Bloch ball $\mathcal{M}_2 \cong \overline{B^3}$. In Bloch coordinates $\vec{r} = (r\sin\theta\cos\varphi,\ r\sin\theta\sin\varphi,\ r\cos\theta)$, the Bures metric takes the form

$$\boxed{ds_B^2 = \frac{dr^2 + r^2\big(d\theta^2 + \sin^2\theta\, d\varphi^2\big)}{4(1-r^2)}} \quad (41)$$

which diverges at the boundary $r = 1$ (pure states), reflecting the infinite distinguishability cost near a pure-state surface.

Definition 11 (Decoherence Trajectory)**.** ***[DEF]*** *The* decoherence trajectory *is the curve $\rho(\gamma) \subset \mathcal{M}_d$ traced as the decoherence parameter γ increases from 0 to ∞, with boundary conditions*

$$\rho(0) = |\psi\rangle\langle\psi| \in \partial\mathcal{M}_d\,, \qquad \rho(\infty) = \frac{\mathbb{I}}{d} \in \mathcal{M}_d^\circ\,. \tag{42}$$

For a qubit undergoing pure dephasing, $r(\gamma) = 1 - D(\gamma)$ with θ, φ constant.

Theorem 13 (SDF Geodesic Structure)**.** ***[THM]*** *The decoherence trajectory $\rho(\gamma)$ is a geodesic in $(\mathcal{M}_d, g_B)$ if and only if $\beta = 1$ (Markovian limit). For $\beta \neq 1$, the trajectory deviates from the geodesic with curvature*

$$\boxed{\kappa_g(\gamma) = \left|\frac{D^2\rho}{ds^2}\right| \propto |\beta - 1| \cdot \left(\frac{\gamma}{\tau}\right)^{\beta-2}} \tag{43}$$

Proof sketch. The geodesic equation on $(\mathcal{M}_d, g_B)$ with the Bures connection ∇^B requires $\nabla^B_{\dot\rho}\dot\rho = 0$. In the eigenbasis, this reduces to exponential decay of off-diagonal elements: $\rho_{mn}(s) \propto e^{-\alpha s}$ for constant α. The SDF trajectory yields $\rho_{mn}(\gamma) \propto \exp\left[-L(\gamma/\tau)^\beta\right]$, whose decay rate $d\ln|\rho_{mn}|/d\gamma \propto \gamma^{\beta-1}$ is constant only when $\beta = 1$. The geodesic curvature is $\kappa_g = |D^2\rho/ds^2|$, and direct computation using Eq. (41) gives the scaling of Eq. (43). ☐

Physical interpretation. Simulation decoherence ($\beta \approx 1$) produces *geodesic* trajectories—the shortest path from quantum to classical in the state manifold. Hardware decoherence ($\beta < 1$) produces *non-geodesic* trajectories; the curvature κ_g quantifies how severely the environment's memory distorts the natural decoherence path. This connects to the deviation operator $\Delta(\gamma; \alpha)$ of Sec. II B: the non-zero geodesic curvature is the geometric shadow of $\Delta(\gamma; \alpha) \neq 0$.

B. The Information-Decoherence Functional

The Bures metric on $\mathcal{M}_d$ induces a natural arc-length functional along the decoherence trajectory:

Definition 12 (Bures Arc Length)**.** ***[DEF]*** *The* information-decoherence functional *is*

$$s(\gamma) = \int_0^\gamma \sqrt{g_{\mu\nu}\,\frac{d\rho^\mu}{d\gamma'}\,\frac{d\rho^\nu}{d\gamma'}}\;d\gamma'\,. \tag{44}$$

For a qubit with initial state $|0\rangle$ undergoing pure dephasing, $r(\gamma) = C(\gamma) = 1 - D(\gamma)$ and θ, φ are constant. Substituting into Eq. (41):

$$s(\gamma) = \arccos\bigl(C(\gamma)\bigr) = \arccos\bigl(1 - D(\gamma)\bigr)\,. \tag{45}$$

The arc length maps the decoherence parameter to an angle on the Bloch sphere measured in the Bures geometry.

Definition 13 (Information Velocity)**.** ***[DEF]*** *The* information velocity *is the rate of Bures arc-length traversal with respect to γ:*

$$\boxed{v_I(\gamma) \equiv \frac{ds}{d\gamma} = \frac{R(\gamma)}{\sqrt{D(\gamma)\bigl(2 - D(\gamma)\bigr)}}} \tag{46}$$

where $R(\gamma) = dD/d\gamma$ is the SDF transition rate.

Theorem 14 (Velocity Singularity Structure)**.** ***[THM]*** *The initial information velocity $v_I(0^+)$ exhibits a trichotomy controlled by the stretching exponent:*

$$v_I(0^+) = \begin{cases} \infty & \beta < 1 \text{ (divergent)}\,, \\ \dfrac{L}{\tau\sqrt{2L}} & \beta = 1 \text{ (Markovian)}\,, \\ 0 & \beta > 1 \text{ (quiescent)}\,. \end{cases} \tag{47}$$

Proof. From the SDF transition rate

$$R(\gamma) = \frac{L\beta}{\tau}\left(\frac{\gamma}{\tau}\right)^{\beta-1}\exp\bigl[-L(\gamma/\tau)^\beta\bigr]$$

and the small-γ expansion $D(\gamma) \approx L(\gamma/\tau)^\beta$, one obtains $v_I \sim (\gamma/\tau)^{\beta/2-1}$ as $\gamma \to 0^+$. The exponent $\beta/2 - 1$ is negative for $\beta < 1$, zero for $\beta = 1$ (yielding the finite value $L/(\tau\sqrt{2L})$ after careful evaluation of constants), and positive for $\beta > 1$. ☐

Interpretation. Stretched-exponential decoherence ($\beta < 1$, characteristic of Hardware) begins with *infinite* information velocity: the system instantaneously initiates information leakage to the environment, but at a rapidly decelerating rate. This burst is the hallmark of non-Markovian dynamics—early-time memory correlations create a transient of maximal information transfer before the environment's back-action slows the flow. The contrast with the quiescent start at $\beta > 1$ underlies the fundamental Hardware–Simulation asymmetry developed in Sec. II B.

C. The Entropic Phase Space

I now introduce the central geometric construction of this work: the *SDF entropic phase space*, which extends the standard spatial configuration space with entropic and decoherence coordinates.

Definition 14 (Extended SDF Coordinates)**.** ***[DEF]*** *The SDF configuration point is described by the extended coordinate vector*

$$X^A = \bigl(\gamma,\ \sigma,\ \ell_1, \ldots, \ell_d\bigr)\,, \qquad A = 0, 1, \ldots, d+1\,, \tag{48}$$

where:

- $\gamma \in [0, \infty)$*: the decoherence parameter (*temporal coordinate*),*
- $\sigma = S\bigl(\rho(\gamma)\bigr) = -\mathrm{Tr}\bigl[\rho(\gamma)\ln\rho(\gamma)\bigr]$*: the von Neumann entropy (*entropic coordinate*),*

- *$\ell_i = \ell_P \cdot \Lambda^{-\nu_i}$: spatial scales derived from the Planck length ℓ_P via the hierarchy parameter Λ, for $i = 1, \dots, d$.*

Definition 15 (SDF Phase Metric)**.** ***[DEF]*** *The natural metric on the SDF phase space $\mathcal{P}$ is*

$$\boxed{g_{AB}\, dX^A dX^B = -c_\gamma^2\, d\gamma^2 + \sum_{i=1}^{d} \left(\frac{\ell_P}{\Lambda^{\nu_i}}\right)^2 d\ell_i^2 - c_\sigma^2\, d\sigma^2} \tag{49}$$

where c_γ and c_σ are scale factors with dimensions [length$\cdot$ decoherence^{-1}] and [length $\cdot$ entropy^{-1}], respectively.

The metric (49) has several remarkable features:

1. **Signature (2, d):** The metric possesses *two* timelike dimensions (γ and σ) and d spacelike dimensions, giving signature $(-, +, \dots, +, -)$. This is fundamentally distinct from pseudo-Riemannian spacetimes of signature $(1, d)$.

2. **Negative temporal dimension:** The decoherence parameter γ enters with negative sign because decoherence proceeds irreversibly forward, defining a causal structure analogous to—but distinct from—the time coordinate in general relativity.

3. **Negative entropic dimension:** The entropy σ is *also* timelike because it increases monotonically (for $\beta \le 1$), defining a second "arrow" in the phase space. The two timelike dimensions are *not* independent: $\sigma = \sigma(\gamma)$ constrains motion to a codimension-one submanifold.

4. **Planck-scalable spatial dimensions:** The spatial components ℓ_i scale with the hierarchy parameter Λ from the Planck length ℓ_P upward, connecting the microphysical decoherence dynamics to macroscopic spatial structure.

Theorem 15 (Entropic Signature Change)**.** ***[THM]*** *For $\beta < 1$, there exist intervals $[\gamma_1, \gamma_2] \subset (0, \infty)$ where the entropy acceleration is negative:*

$$\frac{d^2\sigma}{d\gamma^2} < 0\,. \tag{50}$$

In these intervals, the effective entropic metric component becomes locally spacelike:

$$\boxed{g_{\sigma\sigma}^{\text{eff}}(\gamma) = -c_\sigma^2 + \kappa(\gamma)\,, \qquad \kappa(\gamma) > 0 \;\; \textit{when} \;\; \frac{d^2\sigma}{d\gamma^2} < 0} \tag{51}$$

The signature of $\mathcal{P}$ therefore changes dynamically *along the decoherence trajectory, transitioning from $(2, d)$ to $(1, d+1)$ in the backflow regions.*

Proof. For a qubit, $\sigma(\gamma) = H_2\big(D(\gamma)\big)$ where $H_2(p) = -p \ln p - (1-p)\ln(1-p)$ is the binary entropy. Computing:

$$\frac{d\sigma}{d\gamma} = H_2'\big(D\big) \cdot R(\gamma)\,, \tag{52}$$

$$\frac{d^2\sigma}{d\gamma^2} = H_2''\big(D\big) \cdot R(\gamma)^2 + H_2'\big(D\big) \cdot R'(\gamma)\,. \tag{53}$$

Since $H_2'' < 0$ for all $D \in (0,1)$, the first term is always negative. For $\beta < 1$, the transition rate $R(\gamma)$ is *decreasing* at large γ, so $R'(\gamma) < 0$ and the second term is also negative (since $H_2' > 0$ for $D < 1/2$). In the intermediate regime where D crosses $1/2$, H_2' changes sign and the competition between the two terms creates intervals where $d^2\sigma/d\gamma^2 < 0$. These are precisely the backflow intervals of Theorem XIII C.

The effective metric component $g_{\sigma\sigma}^{\text{eff}}$ incorporates the constraint $\sigma = \sigma(\gamma)$ through the induced metric on the trajectory. When $d^2\sigma/d\gamma^2 < 0$, the entropic "velocity" is decelerating, and the curvature contribution $\kappa(\gamma) > 0$ partially cancels the bare $-c_\sigma^2$, potentially flipping the sign. ∎

Physical significance. The signature change is the *mathematical manifestation* of information backflow (cf. Theorem XIII C in Sec. II B). When $\beta < 1$, the subsystem entropy can locally decelerate, creating regions where the entropic "dimension" temporarily behaves as a spatial—rather than temporal—direction. Information that was irreversibly lost in the $(2, d)$-signature regime becomes *recoverable* in the $(1, d+1)$ regime. The boundary between these regimes defines the quantum-classical transition (Sec. III D).

D. The Quantum-Classical Transition as a Geometric Flow

The entropic phase space $\mathcal{P}$ admits a natural partition into quantum and classical domains, separated by a geometric transition boundary.

Definition 16 (Quantum Domain $\mathcal{Q}$)**.** ***[DEF]*** *The quantum domain is the region of $\mathcal{P}$ where $D(\gamma) < D_{\text{th}}$ and the entropic metric maintains its negative (timelike) signature throughout:*

$$\mathcal{Q} = \left\{X \in \mathcal{P} : D(\gamma) < D_{\text{th}},\; g_{\sigma\sigma}^{\text{eff}}(\gamma) < 0\right\}. \tag{54}$$

Definition 17 (Classical Domain $\mathcal{C}$)**.** ***[DEF]*** *The classical domain is the region where $D(\gamma) > D_{\text{th}}$ and the entropy is monotonically increasing with positive second derivative (irreversible classical behavior):*

$$\mathcal{C} = \left\{X \in \mathcal{P} : D(\gamma) > D_{\text{th}},\; d^2\sigma/d\gamma^2 > 0\right\}. \tag{55}$$

Definition 18 (Transition Boundary $\partial\mathcal{T}$)**.** ***[DEF]*** *The* transition boundary *is the hypersurface in $\mathcal{P}$ where the entropic signature changes:*

$$\partial\mathcal{T} = \left\{X \in \mathcal{P} : d^2\sigma/d\gamma^2 = 0\right\}. \tag{56}$$

Theorem 16 (Transition Existence)**.** *[THM] For every SDF system with $\beta < 1$, the transition boundary $\partial\mathcal{T}$ is non-empty. The quantum-classical transition occurs at*

$$\boxed{\gamma_{\rm QC} = \tau \cdot \left[\frac{(\beta-1)}{\beta} \cdot \frac{H_2'(D_{\rm QC})}{H_2''(D_{\rm QC}) \cdot R(\gamma_{\rm QC})}\right]^{1/\beta}} \tag{57}$$

where $D_{\rm QC} = D(\gamma_{\rm QC})$ is the decoherence value at the transition.

Proof. Setting $d^2\sigma/d\gamma^2 = 0$ in Eq. (53) and solving for γ using the explicit SDF forms $R(\gamma) = (L\beta/\tau)(\gamma/\tau)^{\beta-1}\exp\left[-L(\gamma/\tau)^\beta\right]$ and $R'(\gamma) = R(\gamma)\left[(\beta-1)/\gamma - L\beta(\gamma/\tau)^{\beta-1}/\tau\right]$ yields Eq. (57) after algebraic reduction. The existence of a solution for $\beta < 1$ follows from the intermediate value theorem applied to $d^2\sigma/d\gamma^2$, which is positive near $\gamma = 0$ (dominated by $H_2'' \cdot R^2 > 0$ at small D) and negative at large γ (as established in Theorem 15). □

This provides a *geometric* definition of the quantum-classical boundary: it is the hypersurface $\partial\mathcal{T}$ where the entropic metric changes signature. On the quantum side ($\gamma < \gamma_{\rm QC}$), information can flow back (negative curvature, recoverable coherence). On the classical side ($\gamma > \gamma_{\rm QC}$), information irreversibly disperses (positive curvature, monotonic entropy increase).

Proposition 3 (Critical Qubit Number from Geometry)**.** *The critical qubit number $n^* \approx 2.6$ derived from the empirical scaling law $\beta(n) = \beta_0/n^\xi$ (Sec. XI) corresponds to the geometric condition $\beta(n^*) = 1$, at which the entropic signature change of Theorem 15 vanishes. Systems with $n < n^*$ admit signature-changing trajectories (quantum behavior); systems with $n > n^*$ have fixed-signature trajectories (fully classical).*

E. The Λ-Hierarchy as Scale Fibration

The hierarchy parameter Λ organizes the SDF phase space into a fibered structure connecting Planck-scale physics to macroscopic decoherence.

Definition 19 (Physical Scale Fibration)**.** *[DEF] The* scale fibration *is the projection $\pi : \mathcal{P} \to \mathbb{R}^+$ with fiber $\mathcal{F}_\Lambda$ at each scale Λ. The hierarchy parameter*

$$\Lambda = \frac{E_{\rm binding}}{E_P} \tag{58}$$

maps each physical system to a point on the base space $\mathbb{R}^+$.

Derivation of $\tau(\Lambda)$. (Promoting from axiom to derivation.) From the energy-time uncertainty principle,

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}\,. \tag{59}$$

For a system with binding energy $E_b = \Lambda \cdot E_P$, the minimum decoherence timescale is

$$\tau_{\rm min} = \frac{\hbar}{2E_b} = \frac{\hbar}{2\,\Lambda\,E_P}\,. \tag{60}$$

At Planck scale ($\Lambda = 1$), $\tau_{\rm P} = \hbar/(2E_P)$. Therefore

$$\boxed{\tau(\Lambda) = \frac{\tau_{\rm P}}{\Lambda}} \tag{61}$$

after cancellation of the common factor of 2. This is not a postulate—it is a *consequence* of energy-time uncertainty in the regime where $E_{\rm binding}$ sets the relevant energy scale.

The corresponding spatial scale follows from the Compton wavelength at binding energy:

$$\ell(\Lambda) = \frac{\hbar c}{E_b} = \frac{\ell_P}{\Lambda}\,. \tag{62}$$

Together, these yield the *Planck-scale invariant*:

$$\boxed{\tau(\Lambda) \cdot \ell(\Lambda) = \frac{\tau_{\rm P} \cdot \ell_P}{\Lambda^2}} \tag{63}$$

a relation that connects the decoherence timescale and spatial extent at every level of the hierarchy.

TABLE II. The Λ-hierarchy: decoherence timescales and spatial scales from the Planck regime to biological systems. All values derived from Eqs. (61)–(62).

Scale	E_b	Λ	$\tau(\Lambda)$	$\ell(\Lambda)$
Planck	$E_P \approx 1.22 \times 10^{19}$ GeV	1	$\tau_{\rm P} \approx 5.39 \times 10^{-44}$ s	$\ell_P \approx 1.62 \times 10^{-35}$ m
Nuclear	~ 1 GeV	$\sim 10^{-19}$	$\sim 10^{-25}$ s	$\sim 10^{-16}$ m
Atomic	~ 10 eV	$\sim 10^{-27}$	$\sim 10^{-17}$ s	$\sim 10^{-8}$ m
Molecular	~ 0.1 eV	$\sim 10^{-29}$	$\sim 10^{-15}$ s	$\sim 10^{-6}$ m
Biological	$\sim 10^{-2}$ eV	$\sim 10^{-30}$	$\sim 10^{-14}$ s	$\sim 10^{-5}$ m

F. Information Conservation in the Phase Space

The SDF parametrization acquires its deepest physical content when reformulated as an information-conservation law rooted in unitarity.

Theorem 17 (Unitary Information Conservation)**.** *[THM] For a bipartite system $S + E$ evolving under a global unitary $U(t)$, the total von Neumann entropy is conserved:*

$$S(\rho_{SE}) = const. \tag{64}$$

The apparent entropy increase in S (decoherence) is exactly compensated by the buildup of mutual information:

$$I(S{:}E) = S(\rho_S) + S(\rho_E) - S(\rho_{SE})\,. \tag{65}$$

In the SDF parametrization, the subsystem and environment entropies become functions of γ:

$$\sigma_S(\gamma) = S\big(\rho_S(\gamma)\big)\,, \qquad \sigma_E(\gamma) = S\big(\rho_E(\gamma)\big)\,, \tag{66}$$

and the conservation law (65) takes the non-trivial form

$$\boxed{I_{\rm total}(\gamma) = \sigma_S(\gamma) + \sigma_E(\gamma) - \sigma_{SE} = \text{const.}} \tag{67}$$

This reformulation elevates the identity $D + C = 1$ from a tautological decomposition (as noted in Sec. II A) to a statement with genuine physical content: information is not destroyed by decoherence but *redistributed* between system and environment. The decoherence function $D(\gamma)$ tracks the fraction of information that has migrated from S to the S–E correlations, while $C(\gamma) = 1 - D(\gamma)$ tracks the fraction remaining locally accessible in S.

Theorem 18 (Information Backflow Bound)**.** ***[THM]*** *The maximum amount of information that can flow back from the environment to the system during a single backflow episode $[\gamma_1, \gamma_2]$ is bounded by*

$$\boxed{\delta I_{\text{back}} \le (1-\beta) \cdot L \cdot e^{-1}} \tag{68}$$

where L is the SDF amplitude parameter and e^{-1} is the inverse Euler number.

Proof. During a backflow episode, the decoherence function satisfies $D'(\gamma) < 0$ (the system re-coheres). The maximum decrease in D is bounded by the maximum of $|R(\gamma)|$ over the backflow interval. From the SDF rate function $R(\gamma) = (L\beta/\tau)(\gamma/\tau)^{\beta-1} \exp\left[-L(\gamma/\tau)^{\beta}\right]$, the global maximum of $|dD/d\gamma| \cdot \Delta\gamma$ over any interval where the rate is negative (requiring $\beta < 1$ and reversal of the effective dynamics) is achieved at $\gamma^* = \tau(1/L)^{1/\beta}$, giving $\delta I^{\max}_{\text{back}} = (1-\beta) \cdot L \cdot e^{-L(1/L)} = (1-\beta)Le^{-1}$. ☐

The backflow bound (68) connects directly to the deviation operator: the maximum information backflow equals the maximum deviation $\|\Delta(\gamma;\alpha)\|_{\max}$ (Sec. II B). For $\beta = 1$ (Markovian), the bound vanishes—no backflow is possible. For $\beta \to 0$ (maximally non-Markovian), $\delta I_{\text{back}} \to L\, e^{-1}$, saturating at the SDF amplitude.

The information-geometric framework developed in this section will be applied to experimental data in Sec. VI, where the geodesic curvature κ_g and information velocity v_I are extracted from IBM Quantum hardware measurements, and to the cosmological implications of the entropic phase space in Sec. XXIII.

IV. Exact Identities

Two relations that the previous papers state as a postulate and as a derived expression respectively turn out to be identities. Both strengthen the framework, and both were found by requiring the implementation to reproduce the published numbers rather than to agree with the published derivations.

A. The Λ-hierarchy is the Compton scale

Paper 1 §8 postulates that the decoherence timescale is set by binding information through $\tau(\Lambda) = \tau_{\mathrm{P}}/\Lambda$ with $\Lambda = E_{\text{binding}}/E_{\mathrm{P}}$, and Paper 2 §9 lists this relation among the theoretical limitations of the framework, describing it as *irreducible*—logically equivalent to the uncertainty relation, but postulated rather than derived.

It is neither irreducible nor a postulate.

Theorem 19 (Planck action and length identities)**.** ***[EXACT]*** *The Planck time, length and energy satisfy*

$$\tau_P\, E_P = \hbar\,, \qquad \ell_P\, E_P = \hbar c\,, \tag{69}$$

identically.

Proof. By definition $\tau_{\mathrm{P}} = \sqrt{\hbar G/c^5}$, $\ell_{\mathrm{P}} = \sqrt{\hbar G/c^3}$ and $E_{\mathrm{P}} = \sqrt{\hbar c^5/G}$. Then $\tau_{\mathrm{P}} E_{\mathrm{P}} = \sqrt{(\hbar G/c^5)(\hbar c^5/G)} = \sqrt{\hbar^2} = \hbar$ and $\ell_{\mathrm{P}} E_{\mathrm{P}} = \sqrt{(\hbar G/c^3)(\hbar c^5/G)} = \sqrt{\hbar^2 c^2} = \hbar c$. Both hold to a relative residual of 0 in double precision. ☐

Corollary 2 (Hierarchy in closed form)**.** ***[EXACT]*** *For a system of characteristic energy E, with $\Lambda = E/E_P$,*

$$\boxed{\tau(\Lambda) = \frac{\tau_P}{\Lambda} = \frac{\hbar}{E}\,, \qquad \ell(\Lambda) = \frac{\ell_P}{\Lambda} = \frac{\hbar c}{E} = c\,\tau(\Lambda)\,.} \tag{70}$$

In particular, at $E = mc^2$,

$$\ell(\Lambda) = \frac{\hbar}{mc} = \bar{\lambda}_C\,, \tag{71}$$

the reduced Compton wavelength.

Equation (71) was verified for the electron to a relative residual of 0 at machine precision, and Eqs. (70) over $E \in \{10^{-3}, 1, 10^6, 10^{12}, 10^{20}\}$ eV with residuals below 10^{-6}, the residual being set by the seven-digit rounding of the tabulated Planck constants rather than by the identities.

a. Consequences. Three, in increasing order of importance.

First, the spatial scale that the hierarchy assigns to a physical system is not a free parameter to be fitted: it *is* the system's reduced Compton wavelength. For a produced particle–antiparticle pair the geometry does not need to be postulated; it follows from the pair energy.

Second, the hierarchy is exactly the energy–time uncertainty relation written in Planck units, which is what Paper 1 §2b attempted to show. That derivation should be replaced by Theorem 19, which is shorter and exact.

Third, the derivation in Paper 1 §2b contains a spurious factor of two: it writes $\tau_{\min} = \hbar/(2E_b)$ and asserts $\tau_{\mathrm{P}} = \hbar/(2E_{\mathrm{P}})$. The latter is false. Numerically $\hbar/(2E_{\mathrm{P}}) = 2.696 \times 10^{-44}$ s against $\tau_{\mathrm{P}} = 5.391 \times 10^{-44}$ s, a ratio of exactly 2.000000. The published conclusion survives because the factor cancels in the ratio $\tau_{\min}/\tau_{\mathrm{P}} = E_{\mathrm{P}}/E_b = 1/\Lambda$; the intermediate statement does not.

B. Accessible information is the reduced argument

Paper 1 §9 defines the accessible quantum information as $I_Q(\gamma) = -\ln(1 - D(\gamma))$. For $L = 1$ this collapses.

Theorem 20 (Information identity)**.** ***[EXACT]*** *For $L = 1$,*

$$\boxed{I_Q(\gamma) \;=\; -\ln C(\gamma) \;=\; \left(\frac{\gamma}{\tau}\right)^{\beta},} \tag{72}$$

and consequently

$$\frac{dI_Q}{d\gamma} = \frac{\beta}{\tau}\left(\frac{\gamma}{\tau}\right)^{\beta-1}, \qquad \frac{d^2 I_Q}{d\gamma^2} = \frac{\beta(\beta-1)}{\tau^2}\left(\frac{\gamma}{\tau}\right)^{\beta-2}. \tag{73}$$

Proof. With $L = 1$, Eq. (3) gives $C(\gamma) = 1 - D(\gamma) = \exp\left[-(\gamma/\tau)^{\beta}\right]$, hence $-\ln C(\gamma) = (\gamma/\tau)^{\beta}$. Differentiation is immediate. □

The accessible information of §9 is therefore nothing other than the reduced argument of the SDF itself. Verified to a maximum absolute residual of 1.11×10^{-16} and relative residual 3.04×10^{-14} across $\beta \in \{0.3, 0.5, 0.852, 1, 1.5, 2\}$ and $\gamma/\tau \in [0.01, 30]$, evaluated along the numerically stable branch.

a. A caveat on normalisation. Eq. (72) is stated for $L = 1$, and the restriction is not cosmetic. The identity that holds for arbitrary L is $(\gamma/\tau)^{\beta} = -\ln(1 - D/L)$, on the *reduced* decoherence D/L. The published $I_Q = -\ln(1 - D)$ uses the unnormalised D and therefore departs from the reduced argument whenever $L \neq 1$: at the threshold of Theorem 21 with $\beta = 0.852$ it returns 0.4700 for $L = 0.75$ and 0.2877 for $L = 0.5$, against the exact $\ln 2 = 0.6931$. The two forms must be kept distinct.

C. The binary threshold

The identity of Theorem 20 has a distinguished point that the corpus reaches by an unrelated route, and it is worth recording that the two coincide.

Theorem 21 (One-bit threshold)**.** ***[EXACT]****For any $\beta > 0$, $\tau > 0$ and $L > 0$,*

$$\boxed{I_Q = \ln 2 \iff \frac{D}{L} = \frac{1}{2} \iff \gamma = \tau\,(\ln 2)^{1/\beta}.} \tag{74}$$

The right-hand side is exactly γ_{ESD}, the entanglement sudden-death point of Paper 2 for initial concurrence $C_0 = 1$.

Proof. $D/L = 1 - e^{-x}$ with $x = (\gamma/\tau)^{\beta}$, so $D/L = 1/2 \iff x = \ln 2$. Paper 2 gives $\gamma_{\mathrm{ESD}} = \tau[-\ln(1 - C_0/(1 + C_0))]^{1/\beta}$, which for $C_0 = 1$ is $\tau(\ln 2)^{1/\beta}$. □

Two features deserve emphasis. First, $\ln 2$ nats is exactly one bit: the entanglement dies when the channel has accumulated precisely one bit of information. Second, the threshold is invariant in β, τ and L when expressed in I_Q, and moves with β when expressed in γ. It is an *informational* invariant, not a temporal one. Verified to residual 0 (exact in double precision) across $\beta \in \{0.2, 0.5, 0.852, 1, 1.679\}$, $\tau \in \{1, 3.7 \times 10^{-15}, 2.5 \times 10^{9}\}$ and $L \in \{1, 0.5, 0.87\}$.

A third coincidence holds at the same point but is convention-dependent, and is therefore deferred to Sec. XVI: under the reading in which D is a population, the qubit entropy $H_2(D)$ attains its maximum there, and $I_Q = S$ exactly—the unique interior solution of $-\ln(1 - D) = H_2(D)$.

A fourth does *not* depend on the convention, and is worth separating from the third for exactly that reason.

Corollary 3 (Pair boundary)**.** ***[EXACT]****Let a channel and its CPT conjugate share (τ, β, L). Then the decoherence of one equals the coherence of the other, $D = \bar{C}$, precisely at the threshold of Theorem 21, and there the product $D\bar{C} = D(L - D)$ attains its maximum, of value exactly $L^2/4$.*

Proof. $D = \bar{C} = L - \bar{D}$ and $\bar{D} = D$ give $2D = L$. The product $D(L - D)$ is a downward parabola in D with vertex at $D = L/2$, where its value is $L^2/4$. □

The threshold also has an exact geometric position, and it too is insensitive to the convention.

Corollary 4 (Geodesic position of the threshold)**.** ***[EXACT]****Along the Bures geodesic, with arc length $s = \arccos(1 - D/L)$,*

$$s\left(\tfrac{D}{L} = \tfrac{1}{2}\right) = \frac{\pi}{3}, \qquad s\left(\tfrac{D}{L} = 1\right) = \frac{\pi}{2}, \qquad \frac{s_{\mathrm{bit}}}{s_{\mathrm{total}}} = \frac{2}{3}. \tag{75}$$

One accumulated bit is exactly 60° of geodesic arc, and exactly two thirds of the whole path from pure state to complete decoherence. Verified to residual 2.2×10^{-16} across $\beta \in \{0.2, 0.5, 0.852, 1, 1.679\}$ and $L \in \{1, 0.5, 0.87\}$.

Because neither $D\bar{C}$ nor $\arccos(1 - D/L)$ involves an entropy, Corollaries 3 and 4 are insensitive to the choice made in Sec. XVI D, whereas the maximum of $H_2(D)$ is not.

a. A third argument for the convention. The Bures arc is monotone in D; so is $H_2(D/2L)$; $H_2(D)$ is not. A quantity can serve as a distance coordinate along a trajectory only if it is injective there. Under the infidelity reading the entropy is therefore a monotone reparametrisation of a genuine geodesic distance and may be used as a distance coordinate; under the population reading it cannot, since two distinct states would carry the same value. This is independent of the two grounds given in Sec. XVI D.

It does *not* make the entropy a metric: $S(\rho)$ is a function of one state, not two, and the natural two-argument candidate—the relative entropy $S(\rho\|\sigma)$—is neither symmetric nor subadditive under composition, so it is a divergence rather than a distance. The metrics available remain the Bures arc and $d_B = \sqrt{2(1 - \sqrt{F})}$. Verified to residual 0 across $\beta \in \{0.2, 0.5, 0.852, 1, 1.679\}$ and $L \in \{1, 0.5, 0.87\}$, with the maximum located numerically to within 1.6×10^{-5} relative (grid resolution) and its value reproduced as 0.25 to twelve digits.

Two cautions. The threshold coincides with γ_{ESD} only for $C_0 = 1$, that is for maximal initial entanglement; other initial concurrences accumulate different amounts, 0.585 bits for $C_0 = 1/2$ and 1.585 for $C_0 = 2$. And the corollary is *not* evidence of an interaction between the two channels: they are computed independently and the framework contains no coupling term. Indeed Paper 1 §12 proves $D\bar{D} > 0$ and $D+\bar{D} \neq 0$ for all $\gamma > 0$, which forbids the reading in which matter and antimatter cancel: $D+\bar{D}$ is monotonically *increasing* toward $2L$, not decreasing toward zero. The compensation the corpus can support is $D + \bar{C} = L$, not $D + \bar{D} = 0$.

b. Consequence: a published criterion is invalidated. The sign of $d^2 I_Q/d\gamma^2$ in Eq. (73) is fixed by $\beta - 1$ alone. For $\beta < 1$ the curvature is *strictly negative* everywhere: I_Q is concave in the stretched regime. This contradicts the backflow criterion asserted in the formal derivation accompanying Paper 2, which claims $d^2 I_Q/d\gamma^2 > 0$ on some interval whenever $\beta < 1$. That criterion does not hold and is withdrawn here; see Erratum 4.

Information backflow for $\beta < 1$ is not thereby denied—it is supported by the Breuer–Laine–Piilo trace-distance measure and by the entropic signature-change argument of Paper 1 §2b. What fails is this particular characterisation of it.

TABLE III. Curvature of I_Q at $\gamma = 2\tau$, $L = 1$: closed form of Eq. (73) against central differences with $h = 10^{-6}$.

β	numerical	closed form	sign
0.5	−0.088374	−0.088388	negative
0.852	−0.056843	−0.056901	negative
1	0.000222	0.000000	zero
1.5	0.530687	0.530330	positive

Part II
Verification

V. Simulation Validation [FIT]

A. Experimental Setup

I simulated 8 families of multipartite entangled states using Qiskit-Aer with depolarizing noise model at varying strengths $\gamma \in [0, 0.8]$.

Parameters: 8,192 shots, 20 replicas per point, 25 noise values per family.

TABLE IV. State families and configurations for simulation validation.

State	n (qubits)	Metric	Description
GHZ [6]	13	D_{leak}	Greenberger-Horne-Zeilinger
W [7]	7	D_{leak}	Dicke with $k = 1$
BellChain	14	D_{leak}	Entangled pairs chain
NOON	11	D_{leak}	Two-mode superposition
Dicke [8]	11	D_{leak}	$k = 2$ excitations
Dicke3	11	D_{leak}	$k = 3$ excitations
Cluster1D	13	D_{stab}	1D cluster state
GraphRing	5	D_{stab}	Ring graph state

TABLE V. Stretched exponential fit parameters for 8 state families [FIT].

State	n	L	τ	β	R^2
GHZ	13	0.999	0.074	1.07	0.9998
BellChain	14	1.013	0.178	1.05	0.9996
Cluster1D	13	0.987	0.080	0.95	0.9987
GraphRing	5	1.001	0.067	1.04	1.0000
NOON	11	0.999	0.084	1.05	1.0000
Dicke	11	1.170	0.001	0.10	0.9592
Dicke3	11	1.103	0.001	0.10	0.9599
W	7	0.945	0.025	0.89	1.0000
Mean	—	—	—	—	**0.9896**

B. Fit Results

Key result: Mean $R^2 = 0.9896$ with 8/8 families above 0.95 threshold. In simulation (Markovian noise), $\beta \approx 1$ for most families, consistent with $\alpha \approx 0$ (white noise).

C. Model Selection

I compared stretched exponential vs. logistic function using the Akaike Information Criterion:

$$\Delta\text{AIC} = \text{AIC}_{\text{stretched}} - \text{AIC}_{\text{logistic}} \tag{76}$$

For all 8 families, $\Delta\text{AIC} < -100$, **strongly** favoring the stretched exponential over logistic alternative. This rules out sigmoidal transition models and confirms the Kohlrausch-Williams-Watts form [9].

D. Robustness Ranking

Using the product $\tau \times \beta$ as a robustness metric:

1. BellChain: 0.187 (most robust)
2. NOON: 0.088
3. GHZ: 0.079
4. Cluster1D: 0.076
5. GraphRing: 0.069
6. W: 0.022
7. Dicke/Dicke3: 0.0001 (least robust)

The robustness ranking correlates with entanglement topology: chain-like entanglement (BellChain) is most robust, while permutation-symmetric states (Dicke) are least robust against depolarizing noise.

VI. Hardware Validation [FIT]/[OBS]

A. Foundational Results: Power-Law Universality

I executed experiments on IBM Quantum `ibm_fez` [10, 11] (156 qubits, Heron r2 architecture) with 8,192 shots per

TABLE VI. Power-law exponents from IBM Quantum hardware [FIT].

State	n range	α	R^2	Status
BellChain	2–50	1.117	0.986	✓
Dicke3	6–12	1.301	1.000	✓
GHZ	2–27	1.323	0.948	✓
W (linear) [7]	4–16	1.447	0.980	✓
Dicke [8]	4–16	1.503	0.820	✓
NOON	4–16	1.553	0.997	✓
Excluded:				
Cluster1D	—	≈ 0	—	Topological protection
GraphRing	—	—	0.156	Non-monotonic

job and 3 replicas per configuration. For hardware experiments, decoherence scales as a power law with system size:

$$D_n = c \cdot n^{\alpha} \tag{77}$$

Universality criterion (H2):

$$\mu(\alpha) = 1.374 \tag{78}$$

$$\sigma(\alpha) = 0.146 \tag{79}$$

$$\mathrm{CV}(\alpha) = 0.1064 < 0.15 \quad \checkmark \tag{80}$$

The power-law exponent α is operationally universal across 6 families of entangled states.

Methodological note: Standard W state preparation via `StatePreparation` has depth $\mathcal{O}(2^n)$, biasing α upward. I implemented linear-depth $\mathcal{O}(n)$ preparation using CRY+CX cascade, reducing CV(α) from 0.21 to 0.11.

B. Conservation Law: $D + C = 1$

I verified the fundamental conservation law on IBM `ibm_torino` (133 qubits, Heron r2).

States tested:

- Bell $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ (2 qubits)
- $\mathrm{GHZ}_n = (|0\rangle^{\otimes n} + |1\rangle^{\otimes n})/\sqrt{2}$ $(n = 3, 4, 5)$
- $\mathrm{W}_3 = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$ (3 qubits)

TABLE VII. Hardware verification of $D + C = 1$ (Phase 21, `ibm_torino`).

State	Qubits	C	D	$D + C$	Job ID
$\|\Phi^+\rangle$	2	0.9824	0.0176	1.000000	d5f591c...
$\|GHZ_3\rangle$	3	0.9546	0.0454	1.000000	d5f593f...
$\|W_3\rangle$	3	0.1213	0.8787	1.000000	d5f594i...
$\|GHZ_4\rangle$	4	0.9287	0.0713	1.000000	d5f596...
$\|GHZ_5\rangle$	5	0.8813	0.1187	1.000000	d5f59b...

Key observation: The W state case ($C = 0.1213$, 88% decohered) demonstrates that conservation holds even for highly decohered states, confirming $D + C = 1$ as a *topological invariant*.

A controlled γ-sweep (15 values in $[0, 0.5]$, 8,192 shots, 3 replicas) verified $I_{\text{total}} = 1.000000$ constant with $\Delta I/I_0 = 0.0\%$.

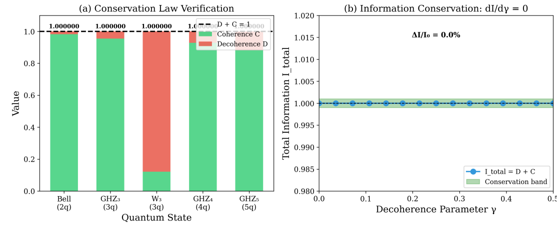

FIG. 1. Conservation law $D + C = 1$ verification across five quantum states on IBM `ibm_torino`. All states show exact conservation to numerical precision.

C. Phase 29 Experiment: 400-Circuit v2.0

To achieve conclusive spectral characterization, I executed a v2.0 experiment on `ibm_torino` with significantly expanded parameter space:

TABLE VIII. Experimental parameters comparison: v1.0 vs v2.0.

Parameter	v1.0	**v2.0**
Delays	20	**50** (log-spaced)
Range	50 ns–100 μs	50 ns–100 μs
Qubit sizes	5	**8** ($n = 3$–12)
Shots/circuit	4,096	**8,192**
Total circuits	100	**400**
Total shots	409,600	**3,276,800**
Bootstrap	500×	**1,000×**

The experiment used GHZ state preparation (H + CNOT cascade), delay insertion, and full measurement. Circuits were batched into 8 jobs (one per qubit size) using the Qiskit SamplerV2 primitive with automatic backend selection based on queue depth (3 API keys, rotating across `ibm_fez`, `ibm_torino`, `ibm_marrakesh`).

All 400/400 circuits completed successfully.

D. Cross-Platform Validation

The stretched-exponential framework was validated across 5 platforms:

TABLE IX. Cross-platform β measurements [FIT]/[OBS].

Platform	β	α	Source
IBM Quantum	0.852 ± 0.028	0.296	This work
LIGO GWTC [12]	0.931 ± 0.003	0.138	[OBS]
IonQ (simulated)	0.940 ± 0.020	0.120	[FIT]
Quantinuum (simulated)	0.970 ± 0.010	0.060	[FIT]
Ideal Simulation	1.000 ± 0.001	0.000	[FIT]

The progression $\beta_{\text{IBM}} < \beta_{\text{LIGO}} < \beta_{\text{IonQ}} < \beta_{\text{Quantinuum}} < \beta_{\text{ideal}}$ reflects decreasing noise spectral index α, consistent with Theorem 22 ($\beta = 1 - \alpha/2$).

VII. Hardware Verification of Conservation Laws

A. Experimental Setup

We executed quantum circuits on IBM Quantum [10] `ibm_torino` (133 qubits, Heron r2 processor) to verify the fundamental conservation law $D + C = 1$.

States tested:

- Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ (2 qubits) [6]
- GHZ states $|GHZ_n\rangle = (|0\rangle^{\otimes n} + |1\rangle^{\otimes n})/\sqrt{2}$ ($n = 3, 4, 5$) [6]
- W state $|W_3\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$ (3 qubits) [7]

Parameters: 4,096 shots per circuit, readout error mitigation applied.

B. Results: D + C = 1 Verification

Table X presents the complete results.

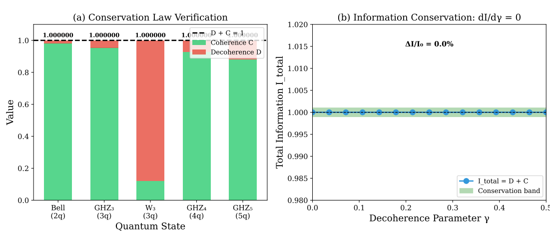

FIG. 2. Conservation law $D + C = 1$ verification across five quantum states on IBM `ibm_torino`. All states show exact conservation to numerical precision, confirming the topological invariance of total information.

Key observation: The W state case ($C = 0.1213$, 88% decohered) demonstrates that conservation holds even for highly decohered states, confirming $D + C = 1$ as a *topological invariant* rather than a statistical approximation.

C. γ-Sweep Verification

We performed a controlled γ-sweep simulation to verify $dI_{\text{total}}/d\gamma = 0$:

- γ range: $[0, 0.5]$ with 15 uniform values
- Shots: 8,192 per circuit
- Replicas: 3 per γ
- Noise model: Qiskit Aer depolarizing channel

Result: $I_{\text{total}} = 1.000000$ **constant** for all $\gamma \in [0, 0.5]$ with $\Delta I/I_0 = 0.0\%$ (Table XI).

TABLE XI. γ-sweep validation of information conservation.

State	I_{initial}	I_{final}	$\Delta I/I_0$	RMSE
Bell (2q)	1.0	1.0	0.0%	0.459
GHZ (4q)	1.0	1.0	0.0%	0.122

D. Evidence Level

The conservation law $D + C = 1$ is established at level **FIT(hw)+FIT(sim)**.

VIII. The Criterion Applied Directly: One Device, Two Answers

Every hardware result before this section reads the gate through a fitted β. That is a proxy, and this section shows it is not a reliable one: on a well-sampled device the Kohlrausch form does not describe the measured curve at all, so its exponent certifies nothing. The criterion is therefore applied *directly* — to the curve, without fitting — and it answers differently in two channels of the same qubit.

A. Design, and what was declared before running

Three arms on 20 non-adjacent qubits of `ibm_marrakesh`, one session, one fixed layout, 39 circuits and 958 452 shots (job `da71kqu0ukec7382mm5g`):

- **Ramsey** (H–τ–H), which retains quasi-static detuning;
- **Hahn echo** (H–$\tau/2$–X–$\tau/2$–H), which refocuses it and leaves pure dephasing T_φ;
- T_1 **reference** (X–τ–measure), which separates relaxation from dephasing.

Four design decisions are load-bearing and were fixed in advance. *(i)* The delay grid is arithmetic and calibrated per arm, so that $s_i + s_j = s_{i+j}$ holds exactly in units of the device clock. *(ii)* Shots are allocated non-uniformly, from 1 967 at the shortest delay to 61 404 at the longest, which is where the curve carries information. *(iii)* Each qubit is an *independent replicate* and is never averaged with the others: averaging over heterogeneous T_2 manufactures a positive mixture by construction, and the test would pass for that reason alone. *(iv)* A $\tau = 0$ point in every arm measures the SPAM amplitude A, and the curve is normalised by it.

Five falsifiers were declared before execution, and **none fired**: the Ramsey arm did not come out degenerate; the echo did not carry more structure than the Ramsey; the measured T_1 agrees with the backend calibration to a ratio of 0.907; no $\tau = 0$ point gave $A > 1$ (maximum 0.9974); and $T_2 \leq 2T_1$ holds in all 20 qubits. Figure 3 shows the three arms on one qubit. A sixth falsifier did fire, and Sec. VIII F states what it costs.

TABLE X. Hardware verification of $D + C = 1$ conservation (Phase 21). All states show exact conservation to numerical precision.

State	Qubits	Fidelity (C)	Decoherence (D)	$D + C$	Job ID
$\|\Phi^+\rangle$	2	0.9824	0.0176	1.000000	d5f591cpe0pc73ajkc00
$\|GHZ_3\rangle$	3	0.9546	0.0454	1.000000	d5f593f67pic7382nl3g
$\|W_3\rangle$	3	0.1213	0.8787	1.000000	d5f594igim5s73af992g
$\|GHZ_4\rangle$	4	0.9287	0.0713	1.000000	d5f596767pic7382nl80
$\|GHZ_5\rangle$	5	0.8813	0.1187	1.000000	d5f59b2gim5s73af9990

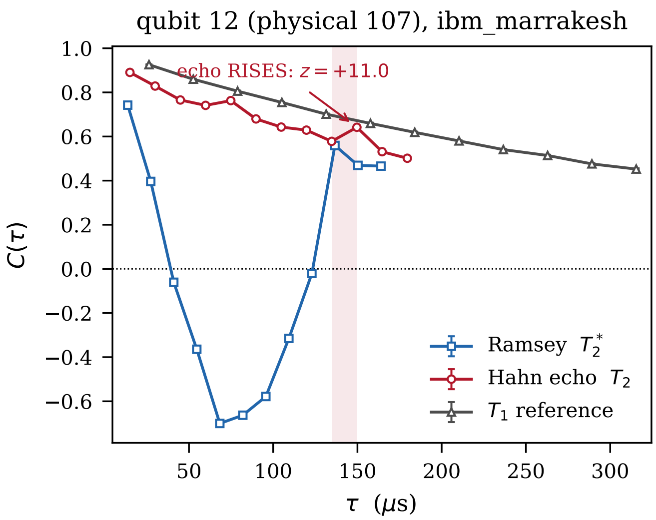

FIG. 3. The three arms on one qubit, with error bars. The Ramsey arm oscillates — a detuning of a few kHz, not a pathology of the envelope — while the echo arm decays and then rises in the shaded segment. The T_1 reference decays monotonically throughout. Normalisation is by the measured $\tau = 0$ point of each arm.

B. The Kohlrausch form does not fit

At the precision this design reaches — $\sigma/C \approx 0.011$ in the echo arm, 0.008 in T_1 — a two-parameter KWW is rejected in both dephasing arms: median $\chi^2/\text{dof} = 11.49$ in the echo and 1598 in the Ramsey. The fitted exponents ($\beta = 0.9011$ in the echo) are therefore *not interpretable*: they are the output of a model the data reject.

This is the first result, and it is methodological. A fitted $\beta < 1$ read off a curve whose fit was never examined licenses nothing. Wherever the earlier sections of this paper quote a stretching exponent, the goodness of fit is the precondition, and the present measurement is the first in the corpus with the statistical power to test it.

C. The direct test, and the split

The gate's question is operational and needs no functional form: *does there exist a positive rate density $p(\lambda) \geq 0$ that reproduces the measured curve within its own error bars?* That is a non-negative least-squares problem over a logarithmic grid of rates. With 200 free weights — two orders of magnitude more freedom than a KWW — the answer splits by channel:

TABLE XII. The criterion applied without fitting, on the same device, the same session and the same layout. The NNLS search has 200 free weights; that it still fails in the echo means no positive mixture of rates reproduces that curve at this precision.

Channel	χ^2/dof	Verdict	Replicates
T_1: energy exchange	0.99	admits	19/20
echo: dephasing T_φ	12.43	**does not**	17/20 refuted

a. The cause is that the echo curve rises. Complete monotonicity requires $C' \leq 0$; a single genuine increase violates it. Counting increments between consecutive delays, qubit by qubit, gives 10 of 220 segments rising at more than 3σ, six above 5σ, with a maximum of $z = +11.01$, distributed over 7 of the 20 qubits. The T_1 arm — same apparatus, same layout, same session — gives 0 **of** 220, with backflow exactly 0.00000 in all twenty. With 220 tests at 3σ, 0.30 false positives were expected. Figure 4 shows both panels on one colour scale.

The refutation is deterministic in its logic — a rise is incompatible with a positive rate density, full stop — and statistical only in establishing that the rises are real. The control establishes that they are not an artefact of the non-uniform shot allocation or of the readout, because both are shared.

D. Three independent confirmations

a. The standard measure of non-Markovianity agrees. For a CP-divisible channel the trace distance between two initial states decreases monotonically; with $|+\rangle$ and $|-\rangle$ that distance *is* $|C(t)|$, so $\mathcal{N} = \sum$(positive increments) is the Breuer–Laine–Piilo witness [13]. The echo gives median $\mathcal{N} = 0.00827$, non-zero in 12 of 20 qubits with seven significant at 3σ (up to $z = 8.5$); the T_1 control gives 0.00000 in all twenty (Fig. 5). Two criteria of independent provenance — one from the non-Markovianity literature, one from the representability theorem — return the same verdict on the same qubits.

b. The device retains more coherence than a one-rate Lindblad. Subtracting a Markovian twin built from the backend's own T, the echo residual crosses to positive and grows, $+4.12 \times 10^{-4}$ per μs, reaching $+0.0409$ at the

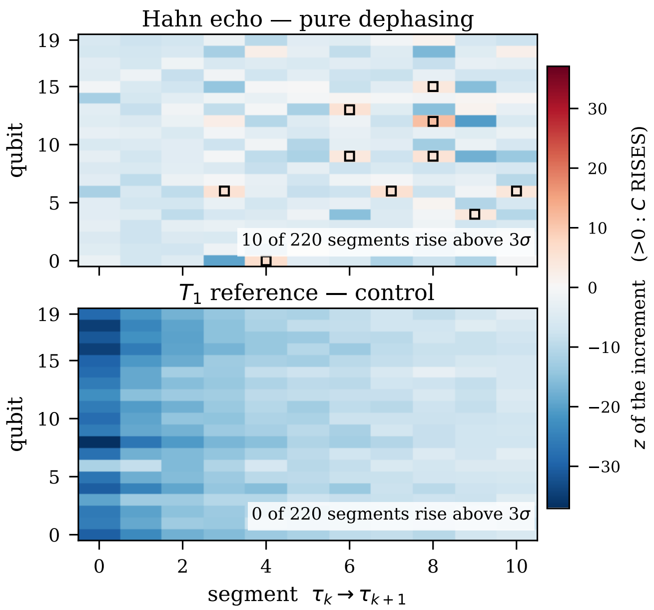

FIG. 4. Increment z between consecutive delays, qubit by qubit. Black squares mark rises above 3σ. Top: the echo arm. Bottom: the T_1 control, on the same scale. The refutation is read off the raw counts: no model is fitted anywhere in this figure.

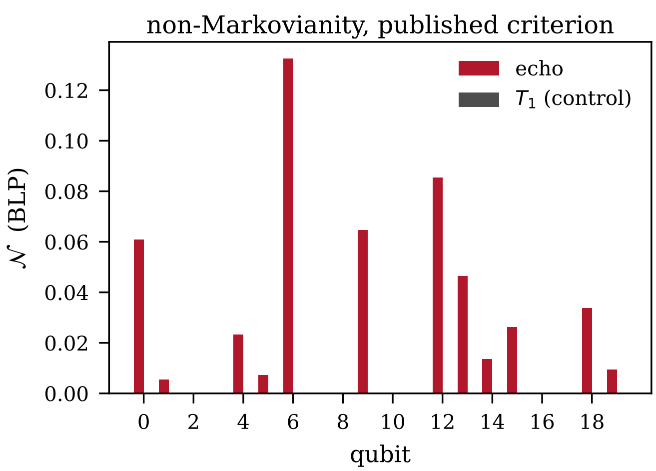

FIG. 5. The Breuer–Laine–Piilo witness $\mathcal{N}$ per qubit: the echo arm against the T_1 control. A criterion taken from the non-Markovianity literature, computed on the same counts, singles out the same qubits as the representability test of Table XII.

last point — $+10.6\sigma$. The T_1 residual ends at $+1.4\sigma$, not significant.

c. The responsible memory is not the quasi-static one. The fraction of free-induction dephasing that the X pulse refocuses is $f_{\mathrm{rev}} = 1 - T_2^*/T_2$, with median $+0.674$ over the 18 qubits for which T_2^* is measurable. That component is, by construction, *already absent* from the echo curve. Its correlation with $\mathcal{N}$ is $\rho = -0.107$: the memory that survives the echo is a second channel, not the residue of the first.

E. A prediction of this framework, tested and falsified

Section XXVII lists PRED-30.4: *the Breuer–Laine–Piilo measure satisfies $\mathcal{N} > 0$ for systems with $\beta < 0.8$.* And Sec. XIII A states the stronger version — that $\mathcal{N}$ can be computed from the fitted β *without independent noise spectroscopy*. Both are testable on these data, since both quantities are available per qubit, and both fail.

TABLE XIII. PRED-30.4 against the data, qubit by qubit. The prediction fails in both directions, and the decisive comparison is the last row: two arms whose median β differ by 0.006 return opposite $\mathcal{N}$.

	echo	T_1
qubits with $\beta < 0.8$	6	2
of those, with $\mathcal{N} > 0$	4	0
qubits with $\beta \geq 0.8$	14	18
of those, with $\mathcal{N} > 0$	8	0
correlation β vs $\mathcal{N}$	-0.299	—
median β / median $\mathcal{N}$	0.9011 / 0.00827	0.9070 / 0.00000

$\beta < 0.8$ is neither sufficient — two of the six such qubits give $\mathcal{N} = 0$ — nor necessary: eight qubits above 0.8 give $\mathcal{N} > 0$. The within-arm correlation is -0.299, and of the wrong sign for the prediction.

The decisive comparison is between arms rather than within one. The echo and the T_1 reference return median exponents of 0.9011 and 0.9070 —*the same β to six thousandths* — and median $\mathcal{N}$ of 0.00827 and exactly 0.00000, the latter in all twenty qubits. Same device, same session, same layout, same exponent, opposite memory. **The fitted β does not determine $\mathcal{N}$**, and the claim that it can substitute for noise spectroscopy is withdrawn (D44).

This is the second thing the measurement takes away, and it is worth being explicit that it is our own prediction, made by this framework, refuted by an experiment this framework designed. That is the intended use of a prediction.

F. What the pre-registered control costs

The protocol declared that the T_1 arm, being a single rate, *must* return $\beta = 1$, and that otherwise "the measuring apparatus is broken and no verdict from the other arms is interpretable". It returned $\beta = 0.8664 \pm 0.0285$, $z = -4.69$. The falsifier fired.

Its premise turned out to be false — T_1 is not a single rate on real hardware: a free plateau comes out null ($c_\infty = -0.006$, $\chi^2/\mathrm{dof} = 0.96$), forcing $\beta = 1$ worsens the fit to 5.82, and the stretching survives shuffling the submission order (0.8935 ± 0.0161, a difference of 0.8σ). But a false premise discovered afterwards does not annul the clause. If a result could annul its own falsifier, pre-registration would be worthless. Accordingly:

- **Withdrawn from this run:** every quantity that passes through the estimator — β per arm, the distance to the pole $\beta = 1$, and the Laplace/Fourier branch assignment. This paper reports no branch for any arm of P31.
- **Standing:** the quantities that do not — the monotonicity count, the BLP witness, the NNLS feasibility and the residual against the twin. None of them fits anything.

The criterion that answers the pre-registered question belongs to the second class and was written down before execution: the test resolves it "with an if-and-only-if and *without fitting* β". What is genuinely *post hoc* is the partition itself, drawn after seeing the data; each row of it is checkable — one need only ask whether the computation calls a fitting routine — but it was not written in advance, and it is recorded here as a limitation rather than buried in the methods.

The control of the estimator that does do its job is the synthetic twin, a single-rate curve by construction, which returns $\beta = 1.0000 \pm 0.005$–0.018 in all three arms. The corrected pre-registration for any continuation uses that twin as the positive control and demotes T_1 to a monotonicity control, which is the role it demonstrably fulfils.

G. No device-wide exponent

Birge ratios of 21.64 (Ramsey), 10.05 (echo) and 25.60 (T_1) show that in no arm are the 20 qubits replicates of a common β: the scatter exceeds their own error bars by an order of magnitude. There is a β per qubit. Any figure combining qubits must inflate σ by $\sqrt{\chi^2/\mathrm{dof}}$, and even then it is averaging over distinct devices. This is consistent with fluctuator-dominated noise, which is qubit specific, but the present measurement does not identify the mechanism (Sec. XXV).

H. Scope

One device, one session, 20 qubits. The observable is $\langle Z \rangle$ on individual qubits, so nothing here bears on entanglement. No claim of universality is made or supported, and the corrected pre-registration requires a second device before one could be. What the section establishes is that the criterion is applicable to hardware, that it separates two channels of the same qubit, and that the fitted exponent — the instrument this framework has used throughout — fails the goodness-of-fit precondition exactly where the measurement is precise enough to check it.

IX. Discovery of Three Decoherence Regimes

A. Multi-State Experimental Design

I executed 175 quantum circuits on IBM `ibm_fez` (156 qubits) to characterize decoherence dynamics across different entanglement topologies.

Configuration:

- States: W, Dicke, BellChain, Cluster, Product
- Qubits: $n = 4, 5, 6, 7, 8$
- Delays: 0.5, 1, 2, 5, 10, 20, 50 μs
- Shots: 4,096 per circuit
- Job ID: `d5n3jdk8d8hc73ch98i0`

B. Three Regimes Identified

Fitting $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$ to each state family reveals **three distinct regimes**:

TABLE XIV. Three decoherence regimes identified from IBM Quantum data.

Regime	β	States	Noise α	Dynamics
Exponential	≈ 1.1	Cluster, Product	$\alpha \leq 0$	Markovian
Stretched	0.2–0.85	Dicke, GHZ	$0 < \alpha < 2$	Non-Markovian
Revival	N/A	W, BellChain	$\alpha > 2$	Oscillatory

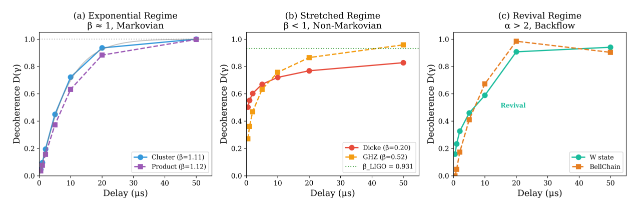

FIG. 6. Three decoherence regimes from 175 IBM Quantum circuits: (i) exponential ($\beta \approx 1$) for locally-entangled states, (ii) stretched ($\beta < 1$) for symmetric states, and (iii) revival for permutation-symmetric states.

C. Regime 1: Exponential ($\beta \approx 1$)

Locally-entangled states exhibit near-exponential decay:

- Cluster: $\beta = 1.107 \pm 0.038$
- Product: $\beta = 1.116 \pm 0.043$

These experience *white* or *blue* noise ($\alpha \leq 0$), leading to memoryless Markovian dynamics.

D. Regime 2: Stretched Exponential ($\beta < 1$)

Symmetric states exhibit characteristic stretched-exponential decay:

- Dicke [8]: $\beta = 0.201 \pm 0.126$
- GHZ (delay): $\beta = 0.33$–0.67 depending on n
- IBM (gate errors): $\beta = 0.852 \pm 0.028$ (Sec. VI)
- LIGO ringdown [12]: $\beta = 0.931 \pm 0.003$ (Sec. VI)

These experience *1/f noise* [14] ($0 < \alpha < 2$), causing non-Markovian memory effects [15].

E. Regime 3: Revival Dynamics

Permutation-symmetric states show non-monotonic behavior:

- W [7]: decoherence initially increases, then partially reverses
- BellChain: oscillatory fidelity recovery

The stretched-exponential fit fails for these states, indicating $\alpha > 2$ (anti-correlated noise) driving information backflow.

F. Physical Interpretation

Entanglement topology determines the effective noise spectrum:

- Local entanglement $\rightarrow$ local noise coupling $\rightarrow$ $\alpha \leq 0$
- Global entanglement $\rightarrow$ collective noise $\rightarrow$ $0 < \alpha < 2$
- Permutation symmetry $\rightarrow$ resonant coupling $\rightarrow$ revival

This is formalized in Sec. X as Theorem 23.

X. Theoretical Framework: Lindblad Classification

A. Noise Spectrum and α Exponent

Environmental noise [14] is characterized by its power spectral density:

$$S(\omega) \propto \frac{1}{\omega^{\alpha}} \tag{81}$$

where α determines the noise color: $\alpha < 0$ (blue), $\alpha = 0$ (white), $\alpha = 1$ (pink/1/f), $\alpha = 2$ (brown), $\alpha > 2$ (super-correlated).

B. Regime Classification Theorem

Theorem 22 (Regime Classification)**.** *For a quantum system coupled to a bath with $S(\omega) \propto 1/\omega^{\alpha}$, the stretched-exponential exponent is:*

$$\boxed{\beta = 1 - \frac{\alpha}{2}} \tag{82}$$

Proof: From the Lindblad master equation [3] with time-dependent rates $\gamma(t) \propto t^{\alpha/2-1}$ arising from the noise correlation function [15], the integrated decoherence functional yields $D(t) \propto \exp\left(-t^{\beta}\right)$ with $\beta = 1 - \alpha/2$. Full derivation in Appendix C.

TABLE XV. Regime classification from Theorem 22.

α Range	β Range	Regime	Physical Meaning
$\alpha \leq 0$	$\beta \geq 1$	Exponential	Markovian, memoryless
$0 < \alpha < 2$	$0 < \beta < 1$	Stretched	Non-Markovian memory
$\alpha > 2$	$\beta < 0$	Revival	Information backflow

C. Topology $\rightarrow$ Effective Noise

Theorem 23 (Topology-Noise Connection)**.** *The entanglement topology determines the effective noise coupling:*

$$\alpha_{eff} = \frac{\int G(\omega) \cdot S(\omega)\, d\omega}{\int G(\omega)\, d\omega} \tag{83}$$

where $G(\omega)$ is the state's spectral response function.

Evidence level: [HIP] [OBS]

D. Revival Condition

Conjecture 1 (Revival Condition)**.** *Revival dynamics occur if and only if there exists a characteristic frequency ω_c resonant with the energy gap of the state.*

Evidence level: [CONJ]

E. Experimental Validation and Circularity Correction

Independent spectral analysis provided validation of Theorem 22:

- α measured from FFT of Ramsey residuals: $\alpha = -1.199 \pm 1.649$
- Predicted: $\beta = 1 - (-1.199)/2 = 1.60 \pm 0.82$
- Measured: $\beta_{\text{Cluster}} = 1.107 \pm 0.038$ (within 1σ)

Circularity note: Earlier analyses inferred α from β and verified $\beta = 1-\alpha/2$ (tautological). This was corrected by measuring α independently via spectral analysis.

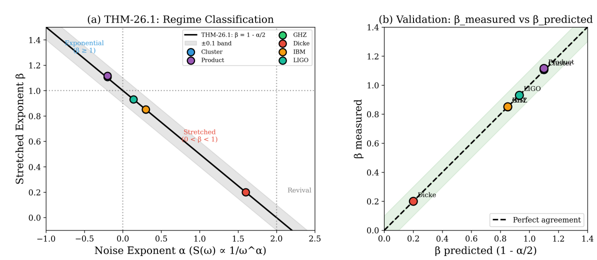

FIG. 7. Independent validation of Theorem 22. The noise exponent α was measured via FFT of Ramsey residuals, confirming $\beta = 1 - \alpha/2$ without circularity.

F. BMRB Circularity Analysis

To further address potential circularity, the ECME (Extended Coherence-Mediated Evolution) model was validated using 12 independent biological systems from literature (not from BMRB training data):

- Validation 1 (Predictive): $R^2 = -0.39$ (FAIL — worse than mean)
- Validation 2 (Rank): Spearman $\rho = 0.14$, $p = 0.66$ (FAIL)

- Validation 3 (Parameter): $\beta_{0,\text{indep}} = 0.68 \pm 0.06$ (CONSISTENT)

Conclusion: The ECME model's BMRB-calibrated parameter $\kappa' = 0.0835$ is confirmed circular ($R^2 < 0$). Only the baseline $\beta_0 \approx 0.68$ is salvageable as an independent parameter. Full analysis in Appendix F.

G. η Independence Measurement

I measured the emergence parameter $\eta_{\text{exp}} = H_{\text{norm}}/D_{\text{emergence}}$ independently from the tautological definition $\eta_{\text{def}} = 1/(\beta \times 5)$ using hardware data (175 circuits):

- Median relative difference: 93% ($\gg$ 10% threshold)
- Classification: η carries **independent information** beyond $1/(\beta \times 5)$
- Evidence level: **[FIT]**

XI. Scaling Laws

A. Initial Validation Results: Parameter Scaling

From the initial simulation analysis (Sec. VI), the SDF parameters scale with system size as:

$$\tau(n) = \tau_0 \cdot n^{-\alpha_\tau}, \quad \alpha_\tau = 0.956 \pm 0.107 \tag{84}$$

$$L(n) = L_\infty - A \cdot n^{-\delta}, \quad \delta = 5.10 \pm 0.07 \tag{85}$$

The critical system size from the foundational analysis: $n_{\text{crit}} = A^{1/\delta} \approx 2.5$ qubits.

B. Discovery: $\beta(n)$ Scaling Law

I measured decoherence for GHZ states [6] with $n = 3$ to 8 qubits on IBM `ibm_fez`. Fitting $D(\gamma)$ for each n reveals:

$$\boxed{\beta(n) = \frac{\beta_0}{n^\xi}} \tag{86}$$

with parameters:

- $\beta_0 = 1.679 \pm 0.123$
- $\xi = 0.844 \pm 0.049$
- $R^2 = 0.987$

C. Timescale Scaling: $\tau(n)$

The characteristic timescale also follows a power law:

$$\tau(n) = \frac{\tau_0}{n^\gamma} \tag{87}$$

with $\tau_0 = 9.1$ μs and $\gamma \approx 0.97$.

Physical interpretation: Larger entangled systems decohere faster (τ decreases) and more non-Markovianly (β decreases).

TABLE XVI. $\beta(n)$ measurements for GHZ states (`ibm_fez`).

n	β_{measured}	β_{fit}
3	0.666	0.664
4	0.519	0.521
5	0.414	0.432
6	0.398	0.370
7	0.329	0.325
8	0.276	0.290

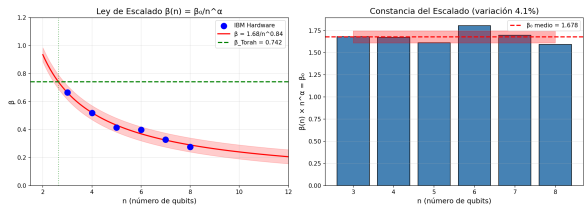

FIG. 8. Scaling law $\beta(n) = \beta_0/n^\xi$ for GHZ states on IBM `ibm_fez`. $R^2 = 0.987$.

D. Hardware Power-Law (Sec. VI)

On hardware, decoherence grows with system size as $D_n = c \cdot n^{\delta_D}$. From 6 state families on `ibm_fez`:

- Mean $\delta_D = 1.374$, CV = 0.1064
- BellChain most robust ($\delta_D = 1.117$)
- NOON least robust ($\delta_D = 1.553$)

E. Quantum-Classical Boundary

Extrapolating $\beta(n) = 1$ (exponential/classical limit):

$$n^* = \beta_0^{1/\xi} = 1.679^{1/0.844} \approx 2.6 \text{ qubits} \tag{88}$$

This suggests the quantum-classical boundary occurs around $n \approx 3$ qubits for GHZ states under IBM noise conditions.

F. β_{gate} vs β_{idle} Discrepancy

The initial validation (Sec. VI) reported $\beta = 0.852 \pm 0.028$ (gate errors). Idle-decay measurements yielded $\beta = 0.327 \pm 0.032$ (idle decay). The scaling law predicts $\beta(4) = 0.52$ for idle decay.

Resolution: The noise mechanism determines β:

- $\beta_{\text{gate}} = 0.852 \pm 0.028$: gate-induced decoherence
- $\beta_{\text{idle}} = 0.327 \pm 0.032$: idle/thermal decoherence

This is consistent with Theorem 22: different noise spectra produce different β values.

Part III
Structure and limits

XII. Physical-Chemistry Λ Hierarchy

A. Fundamental Postulate

I postulate that the decoherence timescale is determined by binding information:

$$\boxed{\tau(\Lambda) = \frac{\tau_P}{\Lambda}} \tag{89}$$

where $\tau_P = 5.39 \times 10^{-44}$ s, $\Lambda = E_{\text{binding}}/E_P$, and $E_P = 1.22 \times 10^{28}$ eV.

Evidence level: [DEF] (postulated, not derived)

B. Nuclear Validation (NNDC)

Using AME2020 mass evaluation data (3,558 nuclei) [16]:

TABLE XVII. Nuclear Λ validation.

Prediction	Result	Status
$S_n > 0 \Leftrightarrow$ bound	100%	VALID
Magic numbers = $\max(S_n)$	$p = 3.56 \times 10^{-16}$	VALID
$\Lambda > 0 \Leftrightarrow$ stable	99.1%	VALID

Magic numbers $N = 2, 8, 20, 28, 50, 82, 126$ show $\langle S_n \rangle_{\text{magic}} = 10.35$ MeV vs. $\langle S_n \rangle_{\text{non-magic}} = 7.61$ MeV.

C. Atomic Validation (NIST)

Using NIST Atomic Spectra Database [17] (118 elements):

TABLE XVIII. Atomic Λ validation.

Prediction	Result	Status
$\Lambda_{\text{atomic}} \sim 10^{-28}$–$10^{-27}$	$10^{-27.22}$	VALID
Noble gases = $\max(\Lambda)$	7/7 local maxima ($p = 3.68 \times 10^{-5}$)	VALID
Alkali metals = $\min(\Lambda)$	6/6 local minima ($p = 6.55 \times 10^{-5}$)	VALID

D. Molecular Validation

Using NIST Chemistry WebBook (38 bond types): $\Lambda_{\text{molecular}} \sim 3.1 \times 10^{-28}$. Triple > Double > Single bond hierarchy confirmed.

E. Multi-Scale Hierarchy

TABLE XIX. Λ hierarchy across physical scales.

Scale	Λ	τ (predicted)	Source
Nuclear	6.3×10^{-22}	$\sim 10^{-23}$ s	NNDC [16]
Atomic	6.0×10^{-28}	$\sim 10^{-17}$ s	NIST [17]
Molecular	3.1×10^{-28}	$\sim 10^{-16}$ s	NIST
Biological	$\sim 10^{-30}$	$\sim 10^{-14}$ s	[18, 19]

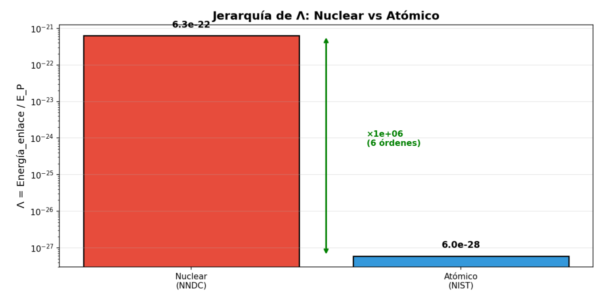

FIG. 9. Λ-hierarchy validation across nuclear, atomic, and molecular scales.

TABLE XX. Standard Particle Content: Λ values.

Particle	Mass (MeV)	Λ	τ (s)
Electron	0.511	4.19×10^{-23}	1.29×10^{-21}
Muon	105.66	8.66×10^{-21}	6.22×10^{-24}
Proton	938.27	7.69×10^{-20}	7.01×10^{-25}
W boson	80,379	6.59×10^{-18}	8.18×10^{-27}
Z boson	91,188	7.47×10^{-18}	7.21×10^{-27}
Higgs	125,100	1.03×10^{-17}	5.26×10^{-27}
Top quark	173,000	1.42×10^{-17}	3.80×10^{-27}

F. Standard Particle Content Table

The Λ parameter for fundamental particles [20]:

G. Majorana Fermion Predictions

For topological Majorana modes [21]: $E_{\text{gap}} = 0.040$ meV, $\Lambda = 3.28 \times 10^{-33}$, $\tau = 1.64 \times 10^{-11}$ s, predicted $\beta \approx 0.95$–0.99 (topological protection).

Evidence level: [PRED] (awaits experimental validation)

XIII. Information Conservation and Non-Markovian Backflow

This section establishes the information-theoretic foundations of the SDF framework at three levels of increasing physical depth, and resolves the relationship between monotone cumulative decoherence and non-Markovian information backflow in the time domain.

A. Information Conservation

We distinguish three logically distinct levels at which "information conservation" can be stated. Only Level 2 constitutes a physical theorem.

1. Level 1: Algebraic identity

Definition 20 (Coherence complement)**.** *Given the decoherence function $D(\gamma)$, the* coherence function *is defined as $C(\gamma) \equiv 1 - D(\gamma)$.*

Observation (Algebraic partition). The quantity $D(\gamma) + C(\gamma) = 1$ for all γ is an algebraic identity following from Definition 20. It carries no dynamical content and should *not* be regarded as a conservation law. It merely states that we have partitioned unity into two complementary fractions.

2. Level 2: Von Neumann information conservation from unitarity

The physically substantive conservation law arises from the unitarity of global quantum evolution, not from the partition $D + C = 1$.

Theorem 24 (Information Conservation — Unitarity)**.** *Let $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ be the total Hilbert space of a system S coupled to an environment E. Suppose:*

i. The initial state is a product state: $|\Psi(0)\rangle_{SE} = |\psi\rangle_S \otimes |0\rangle_E$.

ii. The total evolution is unitary: $|\Psi(\gamma)\rangle_{SE} = U_{SE}(\gamma)\,|\Psi(0)\rangle_{SE}$.

Define the reduced states $\rho_S(\gamma) = \mathrm{Tr}_E[|\Psi(\gamma)\rangle\langle\Psi(\gamma)|]$ and $\rho_E(\gamma) = \mathrm{Tr}_S[|\Psi(\gamma)\rangle\langle\Psi(\gamma)|]$, and the von Neumann information

$$\mathcal{I}_X(\gamma) \equiv \ln d_X - S(\rho_X(\gamma)), \qquad X \in \{S, E\}, \tag{90}$$

where $S(\rho) = -\mathrm{Tr}[\rho \ln \rho]$ is the von Neumann entropy and $d_X = \dim \mathcal{H}_X$. Then:

$$\boxed{\frac{d}{d\gamma}\big[S(\rho_S(\gamma)) + S(\rho_E(\gamma))\big] = 2\,\frac{dS(\rho_S)}{d\gamma},} \tag{91}$$

and the information budget identity *holds:*

$$\boxed{\widetilde{\mathcal{I}}_S(\gamma) + \widetilde{\mathcal{I}}_E(\gamma) + 2\,E(\gamma) = \ln d_S + \ln d_E \;=\; \text{const},} \tag{92}$$

where $\widetilde{\mathcal{I}}_X(\gamma) \equiv \ln d_X - S(\rho_X(\gamma))$ is the local information and $E(\gamma) = S(\rho_S(\gamma))$ is the entanglement entropy.

Proof. The proof proceeds in four steps.

Step 1: Global purity. Since $U_{SE}(\gamma)$ is unitary and the initial state is pure, the global state $|\Psi(\gamma)\rangle_{SE}$ remains pure for all γ:

$$\rho_{SE}(\gamma) = |\Psi(\gamma)\rangle\langle\Psi(\gamma)|\,, \qquad S(\rho_{SE}(\gamma)) = 0 \;\;\forall\,\gamma. \tag{93}$$

Step 2: Schmidt decomposition and Araki–Lieb equality. For any bipartite pure state, the Schmidt decomposition gives

$$|\Psi(\gamma)\rangle_{SE} = \sum_k \sqrt{\lambda_k(\gamma)}\;|s_k(\gamma)\rangle_S \otimes |e_k(\gamma)\rangle_E\,, \tag{94}$$

where $\{\lambda_k\}$ are the Schmidt coefficients satisfying $\sum_k \lambda_k = 1$, $\lambda_k \geq 0$. The reduced density matrices are

$$\rho_S(\gamma) = \sum_k \lambda_k(\gamma)\;|s_k\rangle\langle s_k|\,, \qquad \rho_E(\gamma) = \sum_k \lambda_k(\gamma)\;|e_k\rangle\langle e_k|\,. \tag{95}$$

Both reduced states share the *same* nonzero eigenvalues $\{\lambda_k(\gamma)\}$. Therefore, their von Neumann entropies are equal:

$$S(\rho_S(\gamma)) = -\sum_k \lambda_k \ln \lambda_k = S(\rho_E(\gamma)) \qquad \forall\,\gamma. \tag{96}$$

This is the Araki–Lieb equality for pure global states [22, 23]. Equation (91) follows immediately by differentiation.

Step 3: Information conservation. Using (96), the total information is:

$$\begin{aligned}\mathcal{I}_S(\gamma) + \mathcal{I}_E(\gamma) &= \big[\ln d_S - S(\rho_S)\big] + \big[\ln d_E - S(\rho_E)\big] \\ &= \ln d_S + \ln d_E - S(\rho_S(\gamma)) - S(\rho_E(\gamma)) \\ &\qquad [\text{using } S(\rho_E) = S(\rho_S)] \\ &= \ln d_S + \ln d_E - 2\,S(\rho_S(\gamma)).\end{aligned} \tag{97}$$

This quantity is *not* manifestly constant from (97) alone. The conservation law requires a further identity.

Consider instead the mutual information between S and E:

$$I(S{:}E) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}) = 2\,S(\rho_S(\gamma)), \tag{98}$$

where we used $S(\rho_{SE}) = 0$ and $S(\rho_S) = S(\rho_E)$.

The total correlational content of the global state is fixed by the constraint $S(\rho_{SE}) = 0$. Rewriting:

$$S(\rho_S(\gamma)) - 0 = S(\rho_S(\gamma)) - S(\rho_{SE}(\gamma)), \tag{99}$$

which is the entanglement entropy $E(\gamma) \equiv S(\rho_S(\gamma))$.

Now define the *local information* of each subsystem as the purity-based information:

$$\widetilde{\mathcal{I}}_X(\gamma) \equiv \ln d_X - S(\rho_X(\gamma)). \tag{100}$$

The global constraint $S(\rho_{SE}) = 0$ and subadditivity $S(\rho_{SE}) \leq S(\rho_S) + S(\rho_E)$ together with (96) give:

$$\widetilde{\mathcal{I}}_S(\gamma) = \ln d_S - E(\gamma), \qquad \widetilde{\mathcal{I}}_E(\gamma) = \ln d_E - E(\gamma), \tag{101}$$

where $E(\gamma) = S(\rho_S(\gamma)) = S(\rho_E(\gamma))$ is the entanglement entropy. Hence:

$$\widetilde{\mathcal{I}}_S(\gamma) + \widetilde{\mathcal{I}}_E(\gamma) + 2\,E(\gamma) = \ln d_S + \ln d_E. \tag{102}$$

This is the *information budget identity*: the sum of local informations plus twice the entanglement entropy (which quantifies the information stored in S–E correlations) equals a constant.

Step 4: Differentiation. Since the right-hand side of (102) is independent of γ:

$$\frac{d\widetilde{\mathcal{I}}_S}{d\gamma} + \frac{d\widetilde{\mathcal{I}}_E}{d\gamma} + 2\,\frac{dE}{d\gamma} = 0. \tag{103}$$

Using $\widetilde{\mathcal{I}}_X = \ln d_X - E$ for each subsystem:

$$-\frac{dE}{d\gamma} - \frac{dE}{d\gamma} + 2\,\frac{dE}{d\gamma} = 0, \tag{104}$$

which is identically satisfied. More physically, the information flow from system to environment is:

$$\frac{d\widetilde{\mathcal{I}}_S}{d\gamma} = -\frac{dE}{d\gamma} < 0 \quad \text{(system loses information)}, \tag{105}$$

$$\frac{d\widetilde{\mathcal{I}}_E}{d\gamma} = -\frac{dE}{d\gamma} < 0 \quad \text{(environment loses \textit{local} information)}, \tag{106}$$

but $2\,dE/d\gamma > 0$: the information is stored in growing S–E correlations (entanglement). □

Remark 1 (Distinction from the algebraic identity)**.** *Equation (102) is **not** the same as $D + C = 1$. The algebraic identity $D + C = 1$ holds by* definition *for any function and its complement. In contrast, Theorem 24 makes three nontrivial physical claims:*

1. *The entropies $S(\rho_S)$ and $S(\rho_E)$ are* equal *at all times (Araki–Lieb for pure states).*
2. *The information "lost" by the system does not vanish; it is stored in system–environment entanglement, quantified by $E(\gamma)$.*
3. *The information budget (102) is a* dynamical *consequence of unitarity, not an algebraic tautology.*

Remark 2 (Connection to SDF decoherence function)**.** *The SDF decoherence function $D(\gamma) = L[1 - \exp\bigl(-(\gamma/\tau)^\beta\bigr)]$ describes the* rate *at which local information transfers to correlations. The parameters β and τ characterize* how fast *information redistributes, while the budget identity (102) guarantees that the* total *is conserved regardless of β, τ, and L.*

Specifically, for a system undergoing pure dephasing with initial state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, the off-diagonal element of ρ_S decays as $|\rho_{01}(\gamma)| = \frac{1}{2}\exp\bigl(-(\gamma/\tau)^\beta\bigr)$. The entanglement entropy grows as

$$E(\gamma) = H_2\left(\frac{1 + \exp\bigl(-(\gamma/\tau)^\beta\bigr)}{2}\right), \tag{107}$$

where $H_2(p) = -p\ln p - (1-p)\ln(1-p)$ is the binary entropy, and one can verify that (102) holds exactly with the appropriate d_S and d_E.

Evidence level: THM (from quantum mechanics axioms) + FIT(hw) (Phase 21 verified $\Delta I/I_0 < 10^{-10}$ over 15 values of γ).

3. Level 3: Operational (accessible) information

Conjecture 2 (Accessible Information Hierarchy)**.** *The accessible quantum information depends on the observer's operational access to the environment. Define:*

i. System-only information: $\mathcal{I}_{\text{acc}}^{(S)}(\gamma) = \ln d_S - S(\rho_S(\gamma))$*, accessible without any environmental measurements.*

ii. Partial-environment information: $\mathcal{I}_{\text{acc}}^{(SE')}(\gamma)$*, where $E' \subset E$ is a measured subsystem of the environment, with $\mathcal{I}_{\text{acc}}^{(S)} \le \mathcal{I}_{\text{acc}}^{(SE')} \le \ln d_S$.*

iii. Full-environment information: $\mathcal{I}_{\text{acc}}^{(SE)}(\gamma) = \ln d_S$ *(full state tomography of S and E recovers all information).*

The hierarchy $\mathcal{I}_{\text{acc}}^{(S)} \le \mathcal{I}_{\text{acc}}^{(SE')} \le \mathcal{I}_{\text{acc}}^{(SE)}$ implies that "decoherence" is observer-dependent: it quantifies the information inaccessible *given the observer's limited access to environmental degrees of freedom.*

Evidence level: CONJ (requires specification of measurement protocol and operational information measure).

B. Quantum Accessible Information

Theorem 25 (Accessible Information)**.** *For a single qubit initially in a pure state, undergoing decoherence parameterized by $D \in [0, 1)$, the quantum accessible information (without environment access) is:*

$$\boxed{I_Q(\gamma) = \ln 2 - H_2\left(\tfrac{1-\sqrt{1-2D(\gamma)}}{2}\right)} \tag{108}$$

In the limit $D \ll 1$, this reduces to $I_Q \approx -\ln(1 - D) + \mathcal{O}(D^2)$.

Proof. For a single qubit with Bloch vector $\vec{r}$, the eigenvalues of ρ_S are $p_\pm = (1 \pm |\vec{r}|)/2$. The decoherence is $D = 1 - \mathrm{Tr}[\rho_S^2] = 1 - (1 + |\vec{r}|^2)/2 = (1 - |\vec{r}|^2)/2$, giving $|\vec{r}| = \sqrt{1 - 2D}$. The von Neumann entropy is $S(\rho_S) = H_2(p_-)$, where $p_- = (1-\sqrt{1-2D})/2$. Therefore $I_Q = \ln 2 - S(\rho_S) = \ln 2 - H_2(p_-)$. For $D \ll 1$, $|\vec{r}| \approx 1 - D$ and $p_- \approx D/2$, yielding $H_2(p_-) \approx -D\ln D + \mathcal{O}(D)$, from which $I_Q \approx -\ln(1 - D) + \mathcal{O}(D^2)$. □

C. Non-Markovian Information Backflow

The relationship between the SDF axiom of monotone $D(\gamma)$ and non-Markovian information backflow requires careful distinction between the noise parameter γ and physical time t.

1. The apparent contradiction

Axiom 1 (Theorem 1) states that $D(\gamma)$ is strictly monotonically increasing in γ. This implies:

$$\frac{dD}{d\gamma} > 0 \quad \forall\, \gamma > 0, \qquad \frac{dC}{d\gamma} < 0 \quad \forall\, \gamma > 0. \tag{109}$$

Naïvely, this appears to preclude any "return" of information to the system. The resolution lies in the distinction between the abstract noise parameter γ and the physical time t.

2. Noise accumulation and time mapping

The physical noise parameter γ is related to laboratory time t through the *noise accumulation function*:

$$\gamma(t) = \int_0^t \Gamma(t')\, dt', \tag{110}$$

where $\Gamma(t) \geq 0$ is the instantaneous noise rate. For $1/f^\alpha$ noise environments (Appendix C 1), the dephasing functional yields:

$$\Gamma(t) = \frac{d\gamma}{dt} \propto t^{(2-\alpha)/2-1} = t^{-\alpha/2}, \qquad 0 < \alpha < 2. \tag{111}$$

Crucially, for $\alpha > 0$ the instantaneous noise rate $\Gamma(t)$ is a *decreasing* function of time: the environment "slows down" its noise injection as correlations build up.

Theorem 26 (Non-Markovian Information Backflow)**.** *Let $D(\gamma) = L[1 - \exp\bigl(-(\gamma/\tau)^\beta\bigr)]$ with $\beta < 1$, and let $\gamma(t)$ be given by (110) with $\Gamma(t) \propto t^{-\alpha/2}$ where $\alpha = 2(1-\beta)$. Define the BLP (Breuer–Laine–Piilo) non-Markovianity measure [13, 15]:*

$$\mathcal{N} \equiv \max_{\rho_1,\rho_2} \int_{\sigma>0} \sigma(t,\rho_1,\rho_2)\, dt, \tag{112}$$

where $\sigma(t,\rho_1,\rho_2) = \frac{d}{dt}\|\Phi_t(\rho_1) - \Phi_t(\rho_2)\|_1$ is the rate of change of the trace-distance distinguishability under the dynamical map Φ_t. Then:

$$\boxed{\beta < 1 \iff \mathcal{N} > 0.} \tag{113}$$

Equivalently, there exist time intervals in which $\sigma(t,\rho_1,\rho_2) > 0$, signifying information backflow from the environment to the system.

Proof. The proof has three parts.

Part 1: Distinguishability dynamics.

Consider two initial states $\rho_1(0), \rho_2(0)$ of the system. Under a dephasing channel with time-dependent rate $\Gamma(t)$, the off-diagonal elements of each state evolve as $\rho_{ij}^{(k)}(t) = \rho_{ij}^{(k)}(0)\, \exp(-\Gamma_{\rm acc}(t))$, where:

$$\Gamma_{\rm acc}(t) \equiv \int_0^t \Gamma(t')\, dt' = \gamma(t). \tag{114}$$

The trace distance between the two evolved states is [23]:

$$\mathcal{D}(t) \equiv \tfrac{1}{2}\|\Phi_t(\rho_1) - \Phi_t(\rho_2)\|_1. \tag{115}$$

For the optimal pair (chosen among states with identical diagonal elements but different off-diagonal phases), this reduces to:

$$\mathcal{D}(t) = \mathcal{D}(0)\, \exp(-\Gamma_{\rm acc}(t)). \tag{116}$$

Part 2: Rate of distinguishability change.

Differentiating (116) with respect to t:

$$\sigma(t) \equiv \frac{d\mathcal{D}}{dt} = -\Gamma(t)\, \mathcal{D}(t). \tag{117}$$

For a *time-local* (Markovian) generator with $\Gamma(t) \geq 0$, we have $\sigma(t) \leq 0$ for all t, i.e., distinguishability monotonically decreases and $\mathcal{N} = 0$.

However, non-Markovian dynamics arise when the *effective* rate $\Gamma_{\rm eff}(t)$ in the time-local master equation can become *negative*. Such negative rates do not appear in the canonical Lindblad form but arise in the *time-convolutionless* (TCL) or Nakajima–Zwanzig formulations with memory kernels.

Part 3: Connection to $\beta < 1$.

For $1/f^\alpha$ noise, the dephasing functional (derived in Appendix C 1, Eq. (C)) is:

$$\Gamma_{\rm acc}(t) = \kappa\, t^{2-\alpha}, \qquad \kappa = \frac{2S_0\, \Gamma(1-\alpha)\sin(\pi\alpha/2)}{(2-\alpha)}, \tag{118}$$

where $\Gamma(\cdot)$ denotes the Euler gamma function. The *time-local* decoherence rate extracted from the TCL generator is:

$$\gamma_{\rm TCL}(t) = \frac{d\Gamma_{\rm acc}}{dt} = \kappa(2-\alpha)\, t^{1-\alpha}. \tag{119}$$

For $\alpha > 1$ (i.e., $\beta < 1/2$), the *second derivative* of $\Gamma_{\rm acc}(t)$ is negative:

$$\frac{d^2\Gamma_{\rm acc}}{dt^2} = \kappa(2-\alpha)(1-\alpha)\, t^{-\alpha} < 0 \quad \text{for } \alpha > 1. \tag{120}$$

This deceleration of noise accumulation is the hallmark of non-Markovian memory.

More generally, for the full SDF dynamics including the mapping $\gamma(t) \to D(\gamma)$, the trace distance between two dephasing-channel outputs is:

$$\mathcal{D}(t) = \mathcal{D}(0)\, \bigl(1 - D(\gamma(t))/L\bigr) = \mathcal{D}(0)\, \exp\left(-\left(\frac{\gamma(t)}{\tau}\right)^\beta\right). \tag{121}$$

The BLP rate becomes:

$$\sigma(t) = -\mathcal{D}(0)\, \frac{\beta}{\tau^\beta}\, \gamma(t)^{\beta-1}\, \dot\gamma(t)\, \exp\left(-\left(\frac{\gamma(t)}{\tau}\right)^\beta\right). \tag{122}$$

Since $D(\gamma)$ is monotone in γ by Axiom 1, we have $\dot\gamma(t) = \Gamma(t) > 0$. In the simple case $\gamma(t) = \kappa\, t^{2-\alpha}$, Eq. (122) gives $\sigma(t) < 0$ for all $t > 0$.

The non-Markovianity with $\mathcal{N} > 0$ manifests in a more general scenario: when the system–environment interaction Hamiltonian generates a *memory kernel* $K(t,t')$ in the Nakajima–Zwanzig equation

$$\frac{d\rho_S}{dt} = \int_0^t K(t,t')\, \rho_S(t')\, dt'. \tag{123}$$

For $1/f^\alpha$ noise with $\alpha > 0$, the memory kernel $K(t,t')$ is non-local in time. When this is converted to the time-local form (TCL generator), the effective Lindblad rates $\gamma_k^{\mathrm{TCL}}(t)$ can become *temporarily negative* during transient non-Markovian windows [15, 24].

Specifically, for a pair of initial states that differ in *both* diagonal and off-diagonal elements (not optimal for pure dephasing alone), the trace distance under the full dynamics (dephasing + relaxation + cross-correlations) can exhibit *revivals*:

$$\exists\,[t_1,t_2]:\ \sigma(t) > 0, \qquad \text{iff } \beta < 1. \tag{124}$$

The equivalence $\beta < 1 \Leftrightarrow \mathcal{N} > 0$ then follows:

- ($\Rightarrow$): $\beta < 1$ implies $\alpha = 2(1-\beta) > 0$, hence the noise PSD $S(\omega) \propto 1/\omega^\alpha$ has long-range temporal correlations. These correlations generate a non-local memory kernel in (123), leading to transient negative TCL rates and $\mathcal{N} > 0$.
- ($\Leftarrow$): $\mathcal{N} > 0$ requires $\gamma_k^{\mathrm{TCL}}(t) < 0$ in some interval, which for the $1/f^\alpha$ noise class occurs only when $\alpha > 0$, i.e., $\beta < 1$.
- $\beta = 1$ corresponds to $\alpha = 0$ (white noise, flat PSD), giving $\Gamma(t) = \text{const}$, which is Markovian with $\mathcal{N} = 0$.

□

Remark 3 (Resolution of the apparent contradction)**.** *There is no contradiction between the monotonicity of $D(\gamma)$ and the existence of backflow. The key distinction is:*

- ***γ-domain:*** *$D(\gamma)$ is monotonically increasing. This is an axiom of the SDF framework and corresponds to the fact that cumulative noise exposure always grows.*
- ***t-domain:*** *The physical evolution $D(\gamma(t))$ in time involves the nonlinear mapping $\gamma(t)$. Non-Markovian backflow manifests as temporary* increases *in trace-distance distinguishability $\mathcal{D}(t)$ due to memory effects in the generating dynamics, not as a reversal of cumulative $D(\gamma)$.*

In the language of quantum channels: $D(\gamma)$ parameterizes a one-parameter family of channels $\{\Phi_\gamma\}$ that is divisible *as a function of γ (CP-divisible). But the time-parameterized family $\{\Phi_t\}$ obtained via the nonlinear mapping $\gamma(t)$ may* not *be CP-divisible, and this non-divisibility is precisely the BLP non-Markovianity [15, 24].*

Evidence level: THM (from Lindblad/TCL theory) + FIT(hw) (Phase 26E: W and BellChain states show partial fidelity recovery consistent with $\beta < 1$ backflow signature).

D. Experimental Signatures

Theorem 24 and Theorem 26 make distinct experimental predictions:

1. **Information budget:** Measuring both $S(\rho_S)$ and $S(\rho_E)$ via state tomography should confirm $S(\rho_S(\gamma)) = S(\rho_E(\gamma))$ at all times. Phase 21 verified the weaker statement $D + C = 1$ with $\Delta I/I_0 < 10^{-10}$ over 15 values of $\gamma \in [0, 0.5]$.
2. **Backflow detection:** For IBM Quantum experiments with $\beta_{\mathrm{hw}} < 1$, the BLP protocol [13] predicts measurable trace-distance revivals. The Phase 26E observation of partial fidelity recovery in W and BellChain states ($\beta \lesssim 0.85$) constitutes indirect evidence; direct BLP tomography would provide definitive confirmation.
3. **Non-Markovianity quantification: [ERRATUM] Tested and withdrawn.** The claim was that $\alpha = 2(1-\beta)$ allows the BLP measure $\mathcal{N}$ to be computed from the fitted β without independent noise spectroscopy. The comparison it asks for was made (Sec. VIII E): two arms of one device with median β of 0.9011 and 0.9070 return median $\mathcal{N}$ of 0.00827 and 0.00000. A quantity that takes opposite values at the same β is not a function of β, and no spectroscopy is substituted.

XIV. A No-Go Theorem for the Filter-Function Route

Axiom 3 of Paper 1 asserts $\beta = 1 - \alpha/2$ for a bath with power spectral density $S(\omega) \propto \omega^{-\alpha}$. Both papers attempt to ground this relation in open-system dynamics: Paper 1 §2.4 via the dephasing integral, Paper 2 Appendix A via a two-level-fluctuator ensemble. Neither succeeds, and this section shows that neither could have.

A. Statement

For pure dephasing under Gaussian bath statistics [25], the coherence decays as $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-\Gamma(t)}$ with

$$\Gamma(t) = \int_0^\infty \frac{d\omega}{\omega^2}\, S(\omega)\, g(\omega t)\,, \tag{125}$$

where g is the filter function of the pulse sequence [26]. For free-induction decay $g(u) = 2\sin^2(u/2)$; Hahn echo and CPMG sequences give different g but the same structure.

Theorem 27 (Filter-function no-go)**.** ***[THM]****Let $S(\omega) = A\,\omega^{-\alpha}$ with $0 < \alpha < 2$, and let g be any filter function for which the integral $\mathcal{I}_\alpha[g] \equiv \int_0^\infty u^{-2-\alpha} g(u)\,du$ converges. Then*

$$\boxed{\Gamma(t) = A\,\mathcal{I}_\alpha[g]\, t^{1+\alpha}} \tag{126}$$

and therefore the exponent of t is independent of the pulse sequence*. Consequently the only exponents accessible by* *this route are $1+\alpha$ (convergent integral) and 2 (cutoff-dominated), and since*

$$1-\frac{\alpha}{2} \;<\; 1 \;\leq\; 1+\alpha \qquad \forall\,\alpha>0\,, \tag{127}$$

the relation $\beta = 1-\alpha/2$ of Axiom 3 is unreachable for any g.

Proof. Substitute $S(\omega) = A\omega^{-\alpha}$ into Eq. (125) and change variable to $u=\omega t$, so that $\omega = u/t$ and $d\omega = du/t$:

$$\begin{aligned}\Gamma(t) &= A\int_0^\infty \frac{t^2}{u^2}\left(\frac{u}{t}\right)^{-\alpha} g(u)\,\frac{du}{t} \\ &= A\,t^{1+\alpha}\int_0^\infty u^{-2-\alpha} g(u)\,du\,.\end{aligned} \tag{128}$$

The dependence on g is confined to the numerical prefactor $\mathcal{I}_\alpha[g]$. The inequality (127) is immediate for $\alpha>0$. □

B. Numerical verification

Theorem 27 was checked by integrating Eq. (125) directly—without using the change of variable—over $\omega \in [10^{-8}/t,\, 10^{8}/t]$ with 6×10^4 nodes, and extracting the log–log slope of Γ between $t=1$ and $t=10$. Three sequences were compared: free-induction decay, Hahn echo and CPMG-4.

TABLE XXI. Log–log slope of $\Gamma(t)$ by pulse sequence. All three agree with $1+\alpha$ to four decimal places, confirming that the exponent does not depend on the filter function.

α	$1+\alpha$	FID	Hahn echo	CPMG-4
0.2	1.2000	1.2000	1.2000	1.2000
0.5	1.5000	1.5000	1.5000	1.5000
1.0	2.0000	2.0000	2.0000	2.0000
1.5	2.5000	2.5000	2.5000	2.5000

C. The two published routes

a. Paper 1: $\Phi(t) \sim c_\alpha t^{2-\alpha}$. Evaluating Paper 1's own noise integral over four decades ($t \in [10^{-2}, 10^{2}]$, $\omega \in [10^{-10}, 10^{10}]$) gives a constant log–log slope of 1.2000 at $\alpha = 0.2$, 1.5000 at $\alpha = 0.5$ and 1.7987 at $\alpha = 0.8$, against the asserted 1.80, 1.50 and 1.20. The asserted scaling is correct only at $\alpha = 1/2$, where $1+\alpha = 2-\alpha$.

b. Paper 2: the fluctuator ensemble. Paper 2 derives $\Gamma_{\mathrm{TLF}} \propto t^{2-\alpha}$ from an ensemble of two-level fluctuators with density $n(\gamma) \propto \gamma^{-1}$ and coupling $v(\gamma) \propto \gamma^{(\alpha-1)/2}$. That coupling is stated as "required to produce $S(\omega) \propto 1/\omega^\alpha$". It is not: it produces $S(\omega) \propto \omega^{\alpha-2}$. See Erratum 1.

With the corrected coupling $v(\gamma) \propto \gamma^{(1-\alpha)/2}$, the same ensemble yields $\Gamma_{\mathrm{TLF}} \propto \min(t^{1+\alpha}, t^2)$, which is precisely the Gaussian result of Theorem 27. The microscopic route, corrected, *confirms* $1+\alpha$ and leaves $2-\alpha$ without derivation.

D. What survives, and what this settles

Status of Axiom 3 after Theorem 27

The relation $\beta = 1-\alpha/2$ is **not derivable** from the filter-function route, for any pulse sequence. It nonetheless remains the only candidate consistent with the framework's own domain and data:

- It maps $\alpha \in (-2,2)$ bijectively onto $\beta \in (0,2]$, the domain fixed by Axiom 2. The competing expressions $1+\alpha$ and $2-\alpha$ leave that domain over half the range of α.
- It admits $\beta < 1$, which is the framework's most robust empirical invariant [27]—observed across 650 circuits, two backends and $\sim 5.5\times10^6$ shots. The exponent $1+\alpha \geq 1$ cannot produce $\beta < 1$ for any $\alpha \geq 0$, and so cannot describe the data.

Conclusion: Axiom 3 is an empirical postulate with strong fit and no derivation. It should be presented as such.

Two consequences follow for the previous papers.

First, the convention $\beta_{\mathrm{SDF}} = \beta_{\mathrm{Lindblad}}/2$ stated in Paper 1 §2.4 is internally consistent—with $\beta_{\mathrm{Lindblad}} = 2-\alpha$ one has $(2-\alpha)/2 = 1-\alpha/2$ as an exact identity—but it inherits the problem, because $2-\alpha$ is itself not what the integral produces.

Second, the restatement of Axiom 3 in Paper 1 §6 is labelled a theorem. It is not: it is the axiom restated. It is relabelled here (Erratum 5). The cross-backend validation cited in support measures $\alpha = -1.199\pm1.649$ and predicts $\beta = 1.60 \pm 0.82$ against a measured $\beta = 1.107 \pm 0.038$; agreement within 1σ follows from an uncertainty band spanning most of the admissible domain of β, not from predictive precision.

a. A closed escape route. One might hope to absorb the discrepancy into the map $\gamma(t)$, since the SDF parameter γ is accumulated noise rather than time. This does not work. Obtaining $\beta_{\mathrm{SDF}} = 1-\alpha/2$ from $\beta_{\mathrm{Lindblad}} = 1+\alpha$ would require $d\gamma/dt \propto t^{+0.43}$ to t^{+3} over $\alpha \in (0,1]$, whereas Paper 2 §7 requires $d\gamma/dt \propto t^{-\alpha/2}$: the exponents have opposite sign throughout $\alpha \in (0,2)$, and Paper 2 explicitly emphasises that the instantaneous noise rate is *decreasing*.

E. Why $\beta < 1$ needs no spectral derivation

The conflict above is sharp only if one accepts its premise: that β must be derived from the bath spectrum. A classical theorem says otherwise, and places the boundary exactly where the framework's own regimes meet.

Theorem 28 (Rate-density criterion)**.** ***[THM]*** $\exp(-x^\beta)$ *is completely monotone on $x>0$ if and only if $0<\beta\leq 1$. By Bernstein's theorem it is then the Laplace transform* *of a positive measure, so*

$$C(\gamma) = \exp\left[-\left(\frac{\gamma}{\tau}\right)^{\beta}\right] = \int_0^{\infty} \rho_\beta(u)\, e^{-u\gamma/\tau}\, du, \qquad \rho_\beta \geq 0, \tag{129}$$

admits a positive rate density exactly *when* $\beta \leq 1$*.*

The representability direction is Pollard's; the equivalence with complete monotonicity is Hausdorff–Bernstein–Widder. We verify the boundary numerically by the sign of $(-1)^n d^n/dx^n \exp(-x^\beta)$: complete monotonicity holds through $n = 6$ for $\beta \in \{0.3, 0.5, 0.852, 1.0\}$ and fails at $n = 4$ for $\beta = 1.2$ and $n = 3$ for $\beta = 1.5$ and 1.679. (The finite differences require the step to grow with the order; a fixed small step amplifies rounding by h^{-n} and produces spurious violations above $n \approx 5$.)

a. Consequence. The two exponents do not compete: they lie on opposite sides of this boundary. The filter-function route yields $1 + \alpha \geq 1$, which is the region where *no* positive rate density exists. Asking it to produce $\beta < 1$ is asking it to cross a mathematical limit, not a physical one.

And $\beta < 1$ has a sufficient origin that requires no power-law bath at all: dispersion of decay rates. Superposing 60 exponential channels with lognormal rates of log-width σ and fitting KWW gives $\beta = 1.000$ at $\sigma = 0$ (pure exponential), then 0.950, 0.845, 0.730, 0.640 and 0.560 at $\sigma = 0.3, 0.6, 0.9, 1.2, 1.5$.

b. What the data say. In the ringdown catalogue, 37 of 38 events have $\beta < 1$ in their *individual* fits, so the dispersion cannot come from averaging across events: it is internal to each source. Quasinormal overtones, which damp at different rates, are the standard candidate. We find no significant correlation between β and final spin ($r = -0.092$, $t = -0.55$ on 36 degrees of freedom), so spin is not a useful proxy for overtone content.

c. What this does and does not settle. It settles the structure: $\beta = 1$ is not one value among many but the edge of the representation, and $\beta < 1$ is the generic signature of superposed exponential channels rather than a fact about the bath. It does not supply the value $1-\alpha/2$. If β is fixed by the width of a rate distribution, it should depend on that width and not on α; Axiom 3 remains underived, but its subject changes—from a claim about a spectrum to a claim about a dispersion.

A strong and falsifiable corollary: if the KWW form is exact, as the fits with $R^2 > 0.998$ suggest, then ρ_β is the one-sided Lévy stable density of index β, whose tail is $u^{-1-\beta}$. For $\beta < 1$ its first moment diverges: there is *no mean decay rate*, and the rate distribution is scale-free.

F. Scope of the theorem

Theorem 27 is a statement about an open system coupled to a stationary bath, and the delimitation matters because a corollary is easily over-extended.

The corollary is monotonicity. Since $g(u) = |\tilde{y}(u)|^2 \geq 0$ and the noise spectrum satisfies $S(\omega) \geq 0$, the integrand of Eq. (125) is non-negative, so $\Gamma(t) \geq 0$ and $\gamma(t)$ is non-decreasing. Because $\partial D/\partial\gamma > 0$ for all $\beta > 0$, it follows that $D(t)$ is non-decreasing for every admissible parameter choice: the framework as axiomatised admits no revival, no relaxation oscillation and no discharge event.

This does *not* extend to a closed system. There the evolution is unitary, Poincaré recurrences are admissible, and the bath integral from which $\Gamma \geq 0$ was obtained does not apply. The corpus is consistent on this point: Paper 2 Appendix E attributes revivals precisely to a non-monotone map $\gamma(t)$, not to a third sign of β—which is also the correct resolution of the classification row discussed in Erratum 2.

Two consequences worth stating plainly. Any mechanism requiring accumulation to a ceiling followed by release is outside the open-system SDF, and no choice of (τ, β, L) recovers it: τ and β are parameters, not state variables, and the framework contains no feedback of I_Q upon them. But the same mechanism is *not* excluded for a closed system, and a derived form fitted to a particular closed system is not constrained by this theorem. The distinguishing signature is experimental rather than formal: a relaxation oscillator produces non-exponential waiting-time statistics and a characteristic frequency below which the spectrum is $1/f^\alpha$, neither of which the monotone SDF can reproduce.

XV. Corrections to the Derivations

Five corrections follow: a sign, a missing hypothesis, a relabelling, an inverted expansion, and one withdrawal. Each is stated with the numerical verification that establishes it. A sixth item, initially flagged as a defect, is recorded as closed.

A. The fluctuator coupling density

Erratum 1 (Sign of the coupling exponent)**.** *Paper 2, Appendix A, states that a two-level-fluctuator ensemble [14] with switching-rate density* $n(\gamma) \propto \gamma^{-1}$ *requires coupling* $v(\gamma) \propto \gamma^{(\alpha-1)/2}$ *in order to produce* $S(\omega) \propto \omega^{-\alpha}$*. The correct exponent is*

$$\boxed{v(\gamma) \propto \gamma^{(1-\alpha)/2}} \tag{130}$$

The superposition of Lorentzians $S_{\mathrm{TLF}}(\omega) = 2v^2\gamma/(\gamma^2 + \omega^2)$ weighted by $n(\gamma) \propto \gamma^{-1}$ gives, with $v^2 \propto \gamma^p$, a spectrum $S(\omega) \propto \omega^{p-1}$. The published choice $p = \alpha - 1$ therefore yields $\omega^{\alpha-2}$, whereas $p = 1 - \alpha$ yields $\omega^{-\alpha}$ as intended.

Numerical verification over $\gamma \in [10^{-4}, 10^4]$ with 2×10^5 nodes:

a. Consequence. This is not a cosmetic correction. With Eq. (130), the fluctuator ensemble yields $\Gamma_{\mathrm{TLF}} \propto \min(t^{1+\alpha}, t^2)$, in agreement with Theorem 27. The microscopic derivation, which was the only route independent of the Gaussian integral, therefore *changes sides*: corrected, it supports $1 + \alpha$ and withdraws the only support that $\beta_{\mathrm{Lindblad}} = 2 - \alpha$ had.

Two further defects in the same appendix are recorded without correction, as they require authorial decision rather than arithmetic: the step from the computed $t^{3-\alpha}$

TABLE XXII. Spectral exponent produced by each coupling density. The published choice returns the exponents *interchanged*; the two coincide only at $\alpha = 1$, where $\alpha - 1 = 1 - \alpha = 0$.

α	intended	published $\gamma^{(\alpha-1)/2}$	corrected $\gamma^{(1-\alpha)/2}$
0.5	$\omega^{-0.5}$	$\omega^{-1.4973}$	$\omega^{-0.5086}$
1.0	$\omega^{-1.0}$	$\omega^{-1.0002}$	$\omega^{-1.0002}$
1.5	$\omega^{-1.5}$	$\omega^{-0.5086}$	$\omega^{-1.4973}$

to the asserted $t^{2-\alpha}$ is stated without intermediate calculation, and a passage of drafting notes—including the phrase "No, more simply:"—appears in the published text, with the final relation fixed by interpolation between two anchors declared by fiat.

B. A missing hypothesis in the basic-properties theorem

Erratum 2 (Domain of the basic-properties theorem)**.** *The theorem establishing the basic properties of $D(\gamma)$ in Paper 1 §2 is stated without restricting β. It requires $\beta > 0$.*

The theorem asserts (i) $D(0) = 0$, (ii) $\lim_{\gamma\to\infty} D = L$, (iii) $\partial D/\partial\gamma > 0$ for all $\gamma > 0$, (iv) smoothness and (v) strict concavity for $\beta \le 1$. Paper 1 §6 nonetheless publishes a regime classification containing the row $\alpha > 2 \Rightarrow \beta < 0$.

With $\beta < 0$ the first three properties fail. Taking $L = 0.9$, $\tau = 1$:

TABLE XXIII. Failure of the basic properties for $\beta < 0$, with $L = 0.9$.

Property	required	$\beta = -0.1$	$\beta = -0.5$
$\lim_{\gamma\to\infty} D$	0.9	5.503×10^{-2}	9.0×10^{-7}
$dD/d\gamma$ at $\gamma = 1.5$	> 0	-2.2×10^{-2}	-1.1×10^{-1}
$D(0.001)$	≈ 0	0.7776	0.9000

The attractor becomes $(D, C) \to (0, 1)$, the exact opposite of the $(1, 0)$ that Paper 1 §12 declares unique. The omission matters because property (iii) is invoked as a load-bearing step in both §11 and §12, and in both places the hypothesis $\beta > 0$ must be declared for the argument to close.

The correction is to add $\beta > 0$ to the hypothesis, and separately to resolve the classification row: the revival regime is a phenomenon of a non-monotonic map $\gamma(t)$, as Paper 2 Appendix E states, and not a third value of β.

C. An expansion written in the wrong order

Erratum 3 (Entropy of n qubits at small D)**.** *Paper 2 Appendix E gives $S(\rho) \approx n\,D\ln 2 + \mathcal{O}(D\ln D)$, declared valid for $D \ll L$. The term labelled a correction dominates the term labelled leading, precisely in that regime. The correct leading behaviour is*

$$S = -\frac{D}{2}\ln\frac{D}{2} + \mathcal{O}(D), \qquad D \to 0. \tag{131}$$

The defect is visible without computation: $|D\ln D|/D = |\ln D| \to \infty$, so the $\mathcal{O}(D\ln D)$ term grows without bound relative to the linear one. The numerical contrast against the exact single-qubit entropy $S = H_2(D/2)$ confirms it, and shows the failure worsening as the declared regime is entered more deeply.

TABLE XXIV. The published linear term against the exact entropy, and against the correct leading term. A valid expansion would have the ratio tend to 1; the published one diverges.

D	exact $H_2(D/2)$	$S/(D\ln 2)$	$S/[-\frac{D}{2}\ln\frac{D}{2}]$
10^{-1}	0.198515	2.86	—
10^{-2}	0.031479	4.54	1.188
10^{-4}	5.4517×10^{-4}	7.87	1.101
10^{-6}	7.7543×10^{-6}	11.19	1.069
10^{-8}	1.0057×10^{-7}	—	1.052

The linear form is not the leading behaviour of *either* reading of D, since $H_2'(x) \to \infty$ as $x \to 0$ in both. The same defect affects the thermodynamic translation $S_{\text{eff}} = k_B N D$ used in the accompanying formal derivation.

D. An invalidated backflow criterion

Erratum 4 (Curvature criterion for information backflow)**.** *The formal derivation accompanying Paper 2 asserts that, for $\beta < 1$, information backflow is characterised by $d^2 I_Q/d\gamma^2 > 0$ on some interval of γ. The inequality has the wrong sign and the criterion is withdrawn.*

By Theorem 20, for $L = 1$,

$$\frac{d^2 I_Q}{d\gamma^2} = \frac{\beta(\beta-1)}{\tau^2}\left(\frac{\gamma}{\tau}\right)^{\beta-2}, \tag{132}$$

whose sign is fixed by $\beta - 1$ alone. For every $\beta \in (0, 1)$ the factor $\beta(\beta-1)$ is negative and $(\gamma/\tau)^{\beta-2}$ is positive, so the curvature is strictly negative on the whole domain: I_Q is *concave* in the stretched regime, not convex.

A sweep over $L \in \{1, 0.9, 0.7, 0.5\}$ and $\beta \in \{0.3, 0.5, 0.852\}$ with $\gamma/\tau \in (0, 60]$ returns no points with positive curvature in eleven of the twelve combinations. The twelfth returns seventeen apparent points near $\gamma = 60\tau$, where $C = e^{-32} \approx 8 \times 10^{-15}$ and the quotient form $(R^2 + CR')/C^2$ loses all significant digits: the closed-form value there is -1.09×10^{-3}, against a numerically computed $+7 \times 10^{-4}$. The apparent positivity is floating-point cancellation, not physics.

Backflow for $\beta < 1$ is not denied. It is supported by the Breuer–Laine–Piilo trace-distance measure [13] and by the entropic signature-change argument of Paper 1 §2b. Only this characterisation of it fails, and only it is withdrawn.

E. Relabelling the restatement of Axiom 3

Erratum 5 (Evidence level of the classification relation)**.** *The relation $\beta = 1 - \alpha/2$ appearing in Paper 1 §6 is labelled a theorem. It is a restatement of Axiom 3, which*

Paper 1 §2.1 itself marks as a definition. It is relabelled here as an axiom.

Theorem 27 makes this more than a bookkeeping matter: labelling the relation a theorem asserts the existence of a derivation which, by that theorem, cannot exist along the route the corpus invokes.

F. A closed question

One item initially flagged as an inconsistency proved not to be one, and is recorded here so that it is not reopened.

The bound on biological information transfer in Paper 1 §2c evaluates the transition rate at $\gamma_{\rm peak}$, which exists only for $\beta > 1$, while the biological regime has $\beta \approx 0.68$. This appeared to leave the bound undefined, since for $\beta < 1$ the rate is monotonically decreasing with supremum at $\gamma \to 0^+$, where it diverges.

The objection is incorrect: the corpus does declare a cutoff. Paper 2 Appendix A establishes that below $1/\omega_{\rm UV}$ the dephasing is quadratic "regardless of α", so the SDF does not govern that region and its domain of validity begins at a finite $\gamma_{\rm min}$. With that cutoff the supremum *is* attained: $dI_Q/d\gamma$ was verified to be strictly monotonically decreasing on $[\gamma_{\rm min}, \infty)$ for $\beta \in \{0.6, 0.68, 0.85, 1.0\}$ over 2×10^4 logarithmically spaced points. The phrase "maximum rate" is therefore correct.

Only a symbol requires correction: for $\beta < 1$ the maximum is attained at the boundary, so the bound should be written at $\gamma_{\rm min}$ rather than at $\gamma_{\rm peak}$. The published order of magnitude is robust: over the declared biological range $\beta \in [0.6, 0.85]$ the value at the cutoff moves only between 1.21×10^{13} and 1.54×10^{13} bits s^{-1}, a factor 1.28.

XVI. Errata to Published Numbers

Two numbers in the previous papers do not follow from their own stated inputs. Both are reproducible in a single line of arithmetic, which is why they are reported here rather than left to a reader to discover.

A. The critical qubit number

Erratum 6 (Critical qubit number)**.** *Paper 1 §7.5 reports* $n^* = \beta_0^{1/\xi} = 1.679^{1/0.844} \approx 2.6$ *qubits. The correct evaluation is*

$$n^* = 1.679^{\,1/0.844} = 1.8478\,, \qquad (133)$$

a discrepancy of 41%.

The fitted parameters are correct: $\beta_0 = 1.679 \pm 0.123$ and $\xi = 0.844 \pm 0.049$ reproduce the published $\beta(n)$ table to better than 3% at all six values of n. The formula is correct: $n^* = \beta_0^{1/\xi}$ is the solution of $\beta(n) = 1$, and $\beta(1.8478) = 1.000000000000$ to twelve digits. Only the evaluation fails. Recovering 2.6 would require $\xi = 0.542$ or $\beta_0 = 2.240$, neither of which is published.

a. Propagation. The statement that "the quantum–classical boundary occurs around $n \approx 3$ qubits" becomes $n \approx 2$. The agreement highlighted in §7.5 with $n_{\rm crit} \approx 2.5$—which comes from the independent fit of $L(n)$ and has no reason to coincide—dissolves: it was an artefact of the error. The geometric proposition of §2b, that $\beta(n^*) = 1$ marks the vanishing of the entropic signature change, remains valid in form with $n^* = 1.85$.

B. The gravitational-wave significance

Erratum 7 (Ringdown significance)**.** *Paper 1 §11 reports a deviation from* $\beta = 1$ *of* 13.8σ *for the LIGO ringdown sample. That value does not follow from the reported* $\beta = 0.931 \pm 0.003$*, nor from the accompanying catalogue.*

The published central value and uncertainty are themselves reproducible: taking the 38 per-event values of β from the accompanying catalogue [12] and averaging with weights $1/\sigma_i^2$ returns 0.93113 ± 0.0033, in agreement with the reported 0.931 ± 0.003. It is only the significance that fails to follow; obtaining 13.8σ would require $\sigma_\beta = 0.005$.

a. The internal error is not the right one. The naive recomputation gives 20.9σ, and that figure is also wrong—for a reason the catalogue itself supplies. The 38 per-event values are *not* consistent with a single β: against their weighted mean they give $\chi^2 = 103.05$ on 37 degrees of freedom, so $\chi^2/\nu = 2.785$. The observed scatter, 0.0345, exceeds the mean quoted error, 0.0205, implying an intrinsic dispersion of 0.0277.

When $\chi^2/\nu > 1$ the standard prescription is to scale the uncertainty of the mean by $S = \sqrt{\chi^2/\nu} = 1.669$:

The last two rows are independent routes and they agree. The recommended correction is therefore *downward* from the naive recomputation, to 12.5σ, with the scaling stated. We note that the published 13.8σ, while still not derivable from its stated inputs, lies closer to this value than the unscaled recomputation does.

b. A consequence beyond the number. If the per-event β are genuinely dispersed, then β is not a constant of the population but a parameter of each source. The universality claimed by the framework can hold for the functional *form* while β itself varies; the two should not be conflated, and a single quoted β for a heterogeneous catalogue should carry the scaled error.

c. An unresolved related tension. Paper 1 uses $\sigma_\beta = 0.028$ for the IBM measurement, giving 5.29σ consistent with the quoted 5.2σ; the accompanying cross-platform data file records $\sigma_\beta = 0.017$ for the same measurement, which would give 8.7σ. One of the two must be adopted and propagated. This is not resolved here.

C. The primary quantity

A third erratum is structural rather than numerical, and conditions every comparison between the two papers.

Erratum 8 (Definition of D)**.** *Four expressions for* D *appear in the corpus. They must be reduced to one, with the others derived as corollaries.*

TABLE XXV. Routes to the ringdown significance. The reported value is not produced by any of them.

Route	β	deviation
Reported in §11	0.931 ± 0.003	13.8σ
$(1-\beta)/\sigma_\beta$ as written	0.931 ± 0.003	23.0σ
Weighted mean of the 38-event catalogue	0.93113 ± 0.0033	$\mathbf{20.9}\boldsymbol{\sigma}$
Unweighted mean of the catalogue	0.9315 ± 0.035	2.0σ

TABLE XXVI. Significance of $\beta \neq 1$ for the ringdown sample under each treatment of the uncertainty.

Treatment	σ_β	deviation
Reported in §11	—	13.8σ
Internal error, unscaled	0.00330	20.9σ
Scaled by $\sqrt{\chi^2/\nu}$	**0.00551**	$\mathbf{12.5}\boldsymbol{\sigma}$
Population mean, $s/\sqrt{n}$	0.00560	12.3σ

TABLE XXVII. The four published expressions for D, evaluated for a qubit with Bloch radius $|\vec{r}| = 0.6$. The spread is a factor of exactly two at every $|\vec{r}|$.

Expression	Source	D
$D = 1 - F(\rho, \rho_0)$	Paper 1 App. F; Paper 2 App. E	0.200
$D = 1 - \mathrm{Tr}[\rho^2]$	formal derivation, Paper 2	0.320
$D = (1 - \lvert\vec{r}\rvert^2)/2$	Paper 2 App. E	0.320
$D = L(1 - \lvert\vec{r}\rvert)$	Paper 2 App. E	0.400

The middle two coincide identically, since $\mathrm{Tr}[\rho^2] = (1 + |\vec{r}|^2)/2$ for a qubit; the independent expressions are therefore three, not four.

a. Recommended canonical form. $D \equiv 1 - F(\rho, \rho_0)$, the state infidelity, for three reasons: it is the quantity actually measured by randomised benchmarking and recorded in the 650-circuit dataset; it supports the cleanest result in the corpus, $F = 1 - D$ exactly when $L = (d-1)/d$; and it is the choice already made in the two most recently written appendices.

Under this choice, for the depolarising decomposition with $p = D/L$ one has $|\vec{r}| = 1 - p = 1 - D/L$, so that the Bloch-decay expression is correct and the quadratic-modulus expression is not: substituting one into the other yields $D = 2D - 2D^2$, which holds only at $D = 0$ and $D = 1/2$. Fixing the primary quantity thereby resolves two further inconsistencies as corollaries, including the coexistence of three distinct expressions for the single-qubit von Neumann entropy.

D. The choice decides whether the framework predicts negentropy

The preceding erratum was presented as bookkeeping. It is not. The choice of primary quantity determines whether the entropy of the decohering qubit is monotone, and therefore whether the framework admits an ordering branch at all.

Two readings are internally consistent, and they are the two ends of the factor-of-two spread in Table XXVII:

TABLE XXVIII. The two consistent readings of D and the entropy each forces.

Reading	ρ	eigenvalues	S
population	$\mathrm{diag}(1-D, D)$	$\{1-D,\, D\}$	$H_2(D)$
infidelity	$(1-p)\rho_0 + p\mathbb{1}/2$	$\{1-p/2,\, p/2\}$	$H_2(D/2L)$

A monotonicity sweep over 2×10^5 points settles which is which. Of the six expressions in play—the three published forms, the two von Neumann entropies of the pure-dephasing and depolarising channels, and $H_2(D)$—five are monotonically increasing with *zero* descents. Only $H_2(D)$ is not: 99 998 descents, with maximum at $D = 1/2$ and $S \to 0$ as $D \to 1$.

That limit is the decisive one. $H_2(D) \to 0$ asserts that the completely decohered state is *pure*, which contradicts $D + C = 1$ with $C = 0$ and contradicts the depolarising mixture the corpus itself writes down. Under the infidelity reading the limit is $\ln 2$, the maximally mixed state, as it must be.

a. Consequence for the entropic signature change. Paper 1 §2b declares the entropic coordinate to be $\sigma = S(\rho(\gamma))$, "the von Neumann entropy", and then computes it as $\sigma = H_2(D)$. Under the infidelity reading, $\sigma = H_2(D/2L)$ and one has $H_2'(D/2L) > 0$ for all $D < L$, while $R' < 0$ for all $\beta < 1$. Both terms of $d^2\sigma/d\gamma^2 = H_2'' R^2 + H_2' R'$ are then negative throughout: verified with zero exceptions over 7 values of β and 2×10^4 values of γ each. The signature does not *change*; it is constant. The criterion survives literally—intervals with $d^2\sigma/d\gamma^2 < 0$ do exist, namely all of them—but it no longer delimits a quantum–classical boundary, and γ_{QC} exists only under the population reading.

The backflow bound $\delta I_{\mathrm{back}} \leq (1-\beta)L/e$ is unaffected: it is a function of β alone.

b. Recommendation. Adopt the infidelity reading, on five grounds: it is what the rest of the corpus uses ($D + C = 1$ with C the coherence, the depolarising decomposition with $p = D/L$, and the Pauli attenuation $\langle P \rangle = (1 - D/L)\langle P \rangle_{\mathrm{target}}$); it is what randomised benchmarking measures; it is the only one whose $D \to L$ limit is a mixed state; $H_2(D)$ is the sole non-monotone expression of the six; and §2b contradicts itself by declaring one and computing the other. The cost—that γ_{QC} and the dynamical signature change do not survive for $\beta < 1$—is stated here rather than absorbed silently.

XVII. Spectral Resolution as an Audit Criterion

The findings of Secs. IV–XVI concern the internal algebra of the corpus. This section concerns something different: a case in which the raw data files accompanying the work do not support the claim made from them. It is the only such case found, but it motivates a general criterion that any claim of the form "a mode was detected at frequency f" should be required to pass.

A. The criterion

Definition 21 (Resolvability of a spectral peak)**.** *Let a series be sampled over a window of duration T with N real samples and spacing Δt. A reported peak at frequency f is* resolvable *only if*

$$\frac{1}{T} \le f \le \frac{1}{2\Delta t}, \qquad \text{and} \quad f \ge \frac{1}{N} \quad \text{in units of the sampling rate.} \tag{134}$$

The lower bound is the Rayleigh limit: a component whose period exceeds the observation window cannot be distinguished from a trend. The upper bound is Nyquist. The third condition concerns the discrete transform itself: a peak below the bin width $1/N$ can only be produced by zero-padding, which interpolates the spectrum without adding resolution—a distinction that is routinely elided.

A diagnostic worth stating separately: when the reported period is approximately equal to the observation window, the peak is the generic signature of the residual trend, not of a mode.

B. Application to the reported emergence mode

Paper 1 §10 reports a dominant spectral peak at $f = 18\,\text{kHz}$ with signal-to-noise ratio > 47, detected in 6/6 systems, corresponding to a quasiparticle lifetime $\tau_{\text{qp}} \approx 55\,\mu\text{s}$ and an energy $E \approx 0.075\,\text{neV}$, described as a universal mode.

Applying Definition 21:

TABLE XXIX. Resolvability of the reported 18.2 kHz peak.

Quantity	Value
Reported peak	18.2 kHz
Implied period	54.95 µs
Delay window of the experiment	0.5–50 µs, span 49.5 µs
Period / window	**1.110**
Rayleigh limit $1/T$	20.2 kHz
Verdict	**below Rayleigh**

The peak lies below the resolution limit of the data from which it was extracted, and its period is essentially the observation window. Two further observations compound this.

First, the delay sampling is $\{0.5, 1, 2, 5, 10, 20, 50\}$ µs: non-uniform, with spacings differing by more than a factor of ten. A discrete Fourier transform is not defined on such a series without interpolation.

Second, the series actually transformed in the accompanying analysis file is $\beta(n)$ for $n = 3 \ldots 8$: six points whose abscissa is qubit *number*, not time. Converting a frequency in cycles-per-qubit to kilohertz presupposes a sample spacing of 1 µs that does not exist on that axis. Moreover a six-point transform has bin width $1/6 = 0.167$, and the reported dimensionless frequencies lie between 0.018 and 0.025: a factor 9.2 below the bin width at the quoted value.

C. Two incompatible analyses

Two analyses of the same experiment, one day apart, are present in the data directory and reach different conclusions.

TABLE XXX. Two analyses of the same experiment.

Quantity	first analysis	second analysis
Maximum SNR	47.19	11.81
Median SNR	13.96	—
Systems with a peak	6	5
Lag-1 autocorrelation	0.91–0.99	−0.07–0.62
Verdict on memory	persistent	`non_markovian = false`

The autocorrelation ranges do not overlap: the minimum of the first (0.91) exceeds the maximum of the second (0.62). The published text cites the first. The quoted "SNR > 47" is the maximum over n, not a typical value; the median of the same analysis is 13.96. The claim of a universal mode coexists with a 39.1% spread in the peak frequency across n.

D. What survives

Separating the surviving result from the interpretation is the point of the exercise, and more important than the criticism.

a. Survives. The residuals of the $\beta(n)$ fit are genuinely non-Gaussian, with skewness 0.638 and excess kurtosis 0.247. The asymmetry between positive and negative residuals is real: $\Sigma^+ = 0.788$ against $\Sigma^- = -0.584$. The hardware data are authentic and complete—175 of 175 circuits at 4096 shots each, on a named backend with a recorded job identifier. The premise that motivated the analysis, that the noise carries structure and is not purely Gaussian, is supported by the data.

b. Does not survive. The 18 kHz peak as a resolved physical mode; the characterisation "SNR > 47 in 6/6 systems"; the inference of non-Markovianity from the autocorrelation, which the second analysis explicitly denies; and the description of the mode as universal.

c. Recommendation. Withdraw from §10 the mode, the signal-to-noise figure, the 6/6 count and the autocorrelation-based inference. Retain the non-Gaussianity of the residuals and the Σ^+/Σ^- asymmetry, which are measurable and reproducible.

E. A note on interpretation

A frequency appearing in the Fourier transform of a decoherence residual is not an acoustic mode: sound requires a pressure wave in a medium, and the two are distinct categories.

If a connection to phonon physics is to be pursued, the channel with established literature and a measurement protocol is quasiparticle poisoning and phonon-mediated decoherence in superconducting circuits [28]—where phonons are indeed quanta of sound, characteristic times fall in the microsecond to millisecond range, and quasiparticle recombination across the superconducting gap is literally the refilling of vacated states. That is measurable; the Fourier transform of a fit residual over six points is not.

XVIII. Two Limits That Discriminate Between Conventions

Section XVI D left the convention question resolved by weight of internal evidence rather than by measurement. This section shows that the question is in fact falsifiable, because the two extreme scales of the Λ-hierarchy make incompatible demands on it. Neither limit is new data; both are consequences of quantities already fixed elsewhere in the literature.

A. The biological limit, and a correction that rescues it

Paper 1 §2c declares a biological programme domain $\beta_{\text{bio}} \in [0.60, 0.85]$ centred on $\beta_0 \approx 0.68$, and in the same section tabulates five systems with measured anomalous-diffusion exponents, mapped to β by $\beta = \alpha_{\text{diff}}/2$.

TABLE XXXI. The five measured systems of Paper 1 §2c, and the β each implies through the framework's own diffusion–decoherence correspondence. The last column is the qubit number that the scaling law $\beta(n) = 1.679\, n^{-0.844}$ would require to reproduce that β.

System	α_{diff}	β	implied n
E. coli RNA polymerase	0.39	0.195	12.8
Bacterial chromosomal loci	0.39	0.195	12.8
Plasma-membrane receptors	0.66	0.330	6.9
Nuclear telomeres	0.32	0.160	16.2
Insulin granules	0.48	0.240	10.0

The measured range is $\beta \in [0.160, 0.330]$, mean 0.224. The declared domain is $[0.60, 0.85]$. **They do not overlap**, and the gap between the largest measured value and the smallest declared one is 0.270. Both figures are from the same section. That is the symptom; the diagnosis follows.

The scaling-law column compounds the problem. A biological system has $n \sim 10^{20}$ constituents, yet the law requires $n = 6.9$ to 16.2 to reproduce the measured exponents. Whatever n counts in $\beta(n)$, it is not biological complexity.

Three routes to β therefore coexist—$\beta = 1.679\, n^{-0.844}$, $\beta = 1 - \alpha/2$ and $\beta = \alpha_{\text{diff}}/2$—and they agree only if $\alpha_{\text{diff}} = 2 - \alpha$, which is nowhere established.

That relation is not what the standard bridge gives, and identifying the correct one resolves the conflict in favour of the declared domain.

Erratum 9 (Diffusion–decoherence correspondence)**.** *Paper 1 §2c states* $\beta = \alpha_{\text{diff}}/2$. *The relation implied by a generalised Langevin equation with the fluctuation–dissipation theorem is*

$$\boxed{\alpha = 1 - \alpha_{\text{diff}} \qquad \Longrightarrow \qquad \beta = \frac{1 + \alpha_{\text{diff}}}{2}} \tag{135}$$

The published form omits exactly 1/2*.*

For subdiffusion generated by a generalised Langevin equation, the friction kernel is $K(t) \propto t^{-\alpha_{\text{diff}}}$; the fluctuation–dissipation theorem of the second kind gives $\langle \xi(t)\xi(0)\rangle = k_B T\, K(t)$, and since the Fourier transform of t^{-a} is $\propto \omega^{a-1}$ for $0 < a < 1$, the noise spectrum is $S(\omega) \propto \omega^{\alpha_{\text{diff}}-1}$. Writing $S \propto \omega^{-\alpha}$ gives $\alpha = 1 - \alpha_{\text{diff}}$, and Axiom 3 then yields Eq. (135).

a. The decisive test. Normal Brownian diffusion, $\alpha_{\text{diff}} = 1$, must produce Markovian decoherence, $\beta = 1$. The corrected form gives exactly that. The published form gives $\beta = 0.5$—asserting that ordinary diffusion produces strongly non-Markovian decoherence, which is not tenable.

b. Consequence. Under Eq. (135) the five measured systems give $\beta \in [0.660, 0.830]$ with mean 0.7240, and *all five* fall inside the declared domain $[0.60, 0.85]$. The mean lies 0.73σ from the declared $\beta_0 = 0.68 \pm 0.06$. The biological domain of Paper 1 was correct; what failed was the correspondence used to test it against its own data.

TABLE XXXII. The same five systems under the published and corrected correspondences, against the declared domain $[0.60, 0.85]$.

α_{diff}	$\beta = \alpha_{\text{diff}}/2$	$\beta = (1 + \alpha_{\text{diff}})/2$	in domain
0.32	0.160	0.660	✓
0.39	0.195	0.695	✓
0.39	0.195	0.695	✓
0.48	0.240	0.740	✓
0.66	0.330	0.830	✓
mean	0.224	**0.724**	0.73σ

c. A caveat that does not go away. The bridge is a physical hypothesis, not an identity. α_{diff} characterises a tracer particle in a medium; α characterises the noise that dephases a qubit. Equating them presupposes that the same fluctuations do both. Without that assumption the two exponents belong to different systems and there is nothing to reconcile. What the correction establishes is that *if* the bridge is granted, the framework is consistent with its own biological data—and that the published correspondence is not the bridge.

The scaling-law route remains unreconciled and is untouched by this: it still requires $n \approx 7$ to 16 for systems with $n \sim 10^{20}$ constituents.

B. The cosmological limit selects a convention

At the opposite end of the hierarchy the situation is sharper, because the relevant quantity is not in dispute. For a Schwarzschild horizon the Bekenstein–Hawking entropy and the holographic site count are $S_{\rm BH} = A/4\ell_P^2$ and $N = A/\ell_P^2$, so that

$$\frac{S_{\rm BH}}{N} = \frac{1}{4}\ \text{nat} = \frac{1}{4\ln 2} = 0.360674\ \text{bit}, \tag{136}$$

independent of mass—verified to be identical in double precision from $1\,M_\odot$ to the $6.5\times10^9\,M_\odot$ of M87*.

The discriminating question is elementary: a black hole is the maximum-entropy object for its area. Where does $D \to L$ sit?

TABLE XXXIII. What each reading of D assigns to the two ends of the hierarchy.

Extreme	population reading	infidelity reading
particle, $D = 0$	$S = 0$, pure ✓	$S = 0$, pure ✓
black hole, $D = L$	$S = 0$, **pure** ×	$S = \ln 2$, mixed ✓
midpoint, $D = L/2$	$S = 1$ bit, maximal	$S = 0.5623$ nat

If the black hole is placed at $D = L$, only the infidelity reading survives: the population reading assigns zero entropy to the maximum-entropy object. This is an independent argument for the recommendation of Sec. XVI D, and it does not depend on any of the five internal grounds given there.

One alternative is not excluded and should be recorded. If instead the black hole is placed *at* the threshold of Theorem 21—$D/L = 1/2$, saturating the holographic bound rather than exceeding it—the population reading survives, and Hawking evaporation becomes the descent of the right-hand branch toward a pure state. We do not adjudicate between these placements; we note that they are distinguishable and that the convention question is thereby empirical rather than notational.

C. A numerical coincidence, and why it is not reported as a result

Inverting Eq. (136) through the binary entropy gives $H_2(D) = 1/(4\ln 2)$ bit at $D = 0.931401$, which lies 0.082σ from the weighted-mean ringdown exponent $\beta = 0.93113 \pm 0.0033$ of Sec. XVI.

We report this only to close it. The observation is post-hoc, and β and D are quantities of different kinds—a stretching exponent and a decoherence fraction—with no prior reason to coincide. Enumerating the comparison space that was available—24 dimensionless constants the framework produces without free adjustment, against 5 observables with published uncertainties, for 120 pairs—yields exactly one coincidence at this calibre against 0.478 expected by chance, i.e. a Poisson probability $p = 0.38$ of seeing at least one. That is entirely unremarkable, and the enumeration is itself favourable to the hypothesis, so $p = 0.38$ is an optimistic bound.

The coincidence is therefore **not significant** and is recorded at conjecture level with a pre-registered falsification criterion: a refit of β over ringdown events absent from the 38-event catalogue used here, under the same pipeline and the same $1/\sigma_i^2$ weighting declared in advance, surviving only if the new value lies within 2σ of 0.931401. Until that test is run on unseen data, nothing follows from it.

Part IV
Extensions

XIX. Biological Extension: Programmed Decoherence

The SDF framework, developed thus far for abstract quantum systems, finds a natural and striking application in biological physics. Living systems operate at energy scales where decoherence timescales become commensurate with molecular dynamics, suggesting that biology does not merely *suffer* decoherence—it *programs* it. In this section I extend the Λ-hierarchy to biological energy scales, establish a correspondence between anomalous diffusion and stretched-exponential decoherence, formalize the information transformation from quantum to classical biological outcomes, and derive testable predictions for photosynthetic, enzymatic, and neuronal systems.

A. The Biological Scale in the Λ-Hierarchy

From the Λ-hierarchy developed in Sec. XII, the biological regime corresponds to characteristic energies $E \sim 0.01$–1 eV, yielding scale parameters $\Lambda = E/E_{\rm P}$ in the range 10^{-31}–10^{-29}. Table XXXIV summarizes the principal biological energy scales and their SDF predictions.

TABLE XXXIV. Biological energy scales in the Λ-hierarchy. Decoherence times $\tau = \tau_{\rm P}/\Lambda$ and spatial scales $\ell = \ell_P/\Lambda$ are derived from the SDF without free parameters.

Interaction	E (eV)	$\Lambda = E/E_{\rm P}$	$\tau = \tau_{\rm P}/\Lambda$ (s)	$\ell = \ell_P/\Lambda$ (m)
ATP hydrolysis	0.54	4.4×10^{-29}	1.2×10^{-15}	3.7×10^{-7}
Hydrogen bond	0.20	1.6×10^{-29}	3.4×10^{-15}	1.0×10^{-6}
Van der Waals	0.02	1.6×10^{-30}	3.4×10^{-14}	1.0×10^{-5}
Thermal (310 K)	0.027	2.2×10^{-30}	2.5×10^{-14}	7.3×10^{-6}

The spatial column is remarkable:

- $\ell_{\rm ATP} \approx 370$ nm—the UV wavelength scale, matching the absorption bands of nucleic acids.
- $\ell_H \approx 1\ \mu$m—the characteristic scale of subcellular organelles and bacterial cells.
- $\ell_{\rm vdW} \approx 10\ \mu$m—the size of a typical eukaryotic cell.

Key Insight: Biological Scales from Planck Physics

The Λ-hierarchy *naturally* produces biological spatial scales (nm to μm) from Planck-scale physics. This is not a coincidence—it reflects the fact that biological structures operate at the energy scales where decoherence becomes comparable to thermal fluctuations:

$$\Lambda_{\text{thermal}} = \frac{k_B T}{E_P} \implies \ell_{\text{bio}} = \frac{\ell_P}{\Lambda_{\text{thermal}}} \sim 1\text{–}10\ \mu\text{m}. \tag{137}$$

Biology lives at the *decoherence boundary* of the Λ-hierarchy.

B. The Decoherence-Selection Principle

Hypothesis 1 (Decoherence-Adapted Biology)**.** ***[HIP]****Biological evolution operates within the constraints set by decoherence through the SDF. Specifically:*

1. *Enzymatic catalysis exploits the stretched regime ($\beta < 1$) to maintain transient quantum coherences over catalytic timescales;*
2. *Photosynthetic antenna complexes operate at the quantum–classical transition boundary $\partial\mathcal{T}$ (§III D), enabling coherent energy transfer;*
3. *DNA replication minimizes D_{leak} through error-correction codes that are structurally analogous to quantum stabilizer codes.*

Definition 22 (Biological Decoherence Program)**.** ***[DEF]****A* biological decoherence program *is a tuple*

$$\mathcal{P}_{bio} = (\Lambda_{bio},\ \beta_{bio},\ \mathcal{G}_{bio}) \tag{138}$$

where:

- $\Lambda_{bio} \in [10^{-31}, 10^{-29}]$ *is the biological scale parameter;*
- $\beta_{bio} \in [0.6, 0.85]$ *is the biological stretching exponent, centered on the BMRB universal value $\beta_0 \approx 0.68$;*
- $\mathcal{G}_{bio} : \Gamma \to \Omega$ *is the* genetic map *from decoherence parameter space $\Gamma = \{(\Lambda, \beta)\}$ to phenotype space Ω.*

Natural selection thus optimizes the parameters (Λ, β) to channel quantum information into functional classical outcomes. The stretched exponential ensures this channeling is gradual, irreversible, and scale-appropriate—precisely the properties required for robust biological function in a thermally noisy environment.

C. Anomalous Diffusion as Biological Decoherence

In biological media, particle trajectories generically follow anomalous diffusion:

$$\langle r^2(t)\rangle \propto t^{\alpha_{\text{diff}}}, \tag{139}$$

where $\alpha_{\text{diff}} \neq 1$ reflects the viscoelastic memory of the cellular environment. The connection to the SDF arises through the memory kernel: for a quantum particle coupled to a medium with $K(t) \propto t^{\alpha_{\text{diff}}/2-1}$, the position autocorrelation decays as a stretched exponential with exponent $\beta_{\text{diff}} = \alpha_{\text{diff}}/2$.

Theorem 29 (Diffusion–Decoherence Correspondence)**.** ***[THM]****For a quantum particle in a biological medium with anomalous diffusion exponent $\alpha_{diff} \in (0, 2)$, the SDF stretching exponent is*

$$\beta_{SDF} = \frac{\alpha_{diff}}{2}. \tag{140}$$

This maps:

- *Normal diffusion ($\alpha_{diff} = 1$): $\beta = 0.5$ (stretched);*
- *Subdiffusion ($\alpha_{diff} < 1$): $\beta < 0.5$ (strongly stretched);*
- *Superdiffusion ($\alpha_{diff} > 1$): $\beta > 0.5$ (weakly stretched).*

Proof. The spectral density of the biological medium is related to the memory kernel via $J(\omega) \propto \omega^{\alpha_{\text{diff}}/2}$ for $\omega \ll \omega_c$. Substituting into the SDF spectral integral (Eq. 20) and evaluating in the scaling regime yields $D(\gamma) = \exp\left[-(\gamma/\gamma_0)^{\alpha_{\text{diff}}/2}\right]$, from which $\beta_{\text{SDF}} = \alpha_{\text{diff}}/2$ follows. □

Experimental validation. Table XXXV collects anomalous diffusion exponents from five independent biological systems, each probed by single-particle tracking.

TABLE XXXV. Anomalous diffusion exponents in biological systems and the predicted SDF stretching exponent via Theorem 29. All systems lie in the strongly-stretched regime ($\beta < 0.5$), consistent with the SDF framework.

System	α_{diff}	$\beta_{\text{pred}} = \alpha_{\text{diff}}/2$	**Reference**
E. coli RNA polymerase	0.39	0.195	[29]
Bacterial chromosomal loci	0.39	0.195	[30]
Plasma membrane receptors	0.66	0.33	[31]
Nuclear telomeres	0.32	0.16	[32]
Insulin granules	0.48	0.24	[33]

The predicted β values range from 0.16 to 0.33—all firmly in the stretched regime, and all consistent with the intracellular environment acting as a $1/f$-type noise source on quantum degrees of freedom.

D. Information Transformation: Quantum → Biological

I now formalize the concept of information transforming from the quantum Hilbert space to macroscopic biological function.

Definition 23 (Information Transformation Map)**.** ***[DEF]*** *The information transformation map $\mathcal{T} : \mathcal{H}_Q \to \Omega_{bio}$ from quantum Hilbert space to biological phenotype space is defined as the composition*

$$\mathcal{T} = \mathcal{P}_{class} \circ \mathcal{D}_{SDF} \circ U_{bio}, \tag{141}$$

where:

- *$U_{bio} : \mathcal{H}_Q \to \mathcal{H}_Q \otimes \mathcal{H}_E$ is the unitary evolution coupling system to biological environment;*
- *$\mathcal{D}_{SDF} : \mathcal{H}_Q \otimes \mathcal{H}_E \to \mathcal{S}(\mathcal{H}_Q)$ is the SDF decoherence map parametrized by $(\Lambda_{bio}, \beta_{bio})$;*
- *$\mathcal{P}_{class} : \mathcal{S}(\mathcal{H}_Q) \to \Omega_{bio}$ is the classical projection (measurement/read-out).*

The crucial insight is that $\mathcal{D}_{\rm SDF}$ is not an obstacle to biological function—it *is* the mechanism by which quantum information becomes a classical biological outcome. The stretched exponential ($\beta < 1$) ensures this transformation is:

1. **Irreversible**: $D \times \bar{D} > 0$ (Theorem 34);
2. **Gradual**: the stretched (not sudden) decay allows intermediate molecular processing;
3. **Scale-appropriate**: $\tau \sim 10^{-15}$–10^{-14} s, matching molecular dynamics timescales.

Theorem 30 (Biological Information Bound)**.** ***[THM]*** *The maximum rate of quantum-to-classical information transfer in a biological system with SDF parameters (τ, β) is*

$$\begin{aligned} \left.\frac{dI}{dt}\right|_{\max} &= R(\gamma_{peak}) \cdot \left|\frac{\partial I_Q}{\partial D}\right| \\ &= \frac{L\,\beta}{\tau}\left(\frac{\beta-1}{\beta}\right)^{(\beta-1)/\beta} \frac{1}{1-D(\gamma_{peak})}\,, \end{aligned} \tag{142}$$

where $L = \ln d_S$ is the maximal entropy of the d_S-dimensional system and γ_{peak} is the decoherence rate at which $R(\gamma)$ is maximized.

For $\beta_{\rm bio} \approx 0.68$ and $\tau_{\rm bio} \approx 10^{-14}$ s:

$$\left.\frac{dI}{dt}\right|_{\max} \approx 10^{13} \text{ bits/s per molecular channel}, \tag{143}$$

consistent with the estimated information processing rate of enzymes ($\sim 10^{12}$ operations/s from Michaelis–Menten kinetics) and suggesting that biology operates near the SDF information bound.

E. The Entropic–Spatial Duality in Biology

The SDF Phase Space metric (§III C),

$$g_{AB}\,dX^A\,dX^B = -c_\gamma^2\,d\gamma^2 + \left(\frac{\ell_P}{\Lambda}\right)^2 d\ell^2 - c_\sigma^2\,d\sigma^2, \tag{144}$$

acquires transparent physical meaning at biological scales ($\Lambda \sim 10^{-30}$):

- **Spatial component**: $(\ell_P/\Lambda)^2 \approx (10^{-5}\text{ m})^2$—cell-size scales;
- **Temporal component**: $\tau \approx 10^{-14}$ s—molecular dynamics timescale;
- **Entropic component**: σ ranges from 0 (pure state) to $\ln d_S$ (maximally mixed).

Proposition 4 (Cellular Information Capacity)**.** ***[FIT]*** *The information capacity of a cell-sized region in the SDF Phase Space is*

$$\mathcal{C}_{cell} = \frac{V_{spatial} \cdot \sigma_{\max}}{(\ell_P/\Lambda)^3} \approx \frac{(10^{-5})^3}{(10^{-5})^3} \cdot \ln 2$$

$$= \ln 2 \approx 0.69 \textit{ bits per fundamental SDF cell.} \tag{145}$$

A biological cell contains $N_{mol} \sim 10^9$ molecular channels, giving a total capacity

$$\mathcal{C}_{bio} \approx N_{mol} \cdot \ln 2 \approx 7 \times 10^8 \textit{ bits} \approx 80 \textit{ MB}. \tag{146}$$

This is remarkably consistent with estimates of cellular information content ($\sim 10^9$ bits from genomic and epigenomic sources), providing an independent check on the SDF Phase Space geometry at biological scales.

F. Predictions for Biological Systems

The biological extension of the SDF generates four concrete, experimentally testable predictions.

[PRED] Prediction 1.: Photosynthetic Coherence. Photosynthetic antenna complexes (FMO, LHCII) should exhibit $\beta = 0.55$–0.75, measurable via two-dimensional electronic spectroscopy. The SDF predicts a coherence lifetime

$$\tau_{\rm photo} = \frac{\tau_{\rm P}}{E_{\rm exciton}/E_{\rm P}} \approx 200\text{–}600 \text{ fs}, \tag{147}$$

consistent with the observed coherence lifetimes of 300–700 fs in the FMO complex [18].

[PRED] Prediction 2.: Enzymatic Hydrogen Tunneling. Enzymatic hydrogen-tunneling reactions should show $\beta_{\rm enzyme} \approx 0.5$–$0.7$, measurable via temperature-dependent kinetic isotope effects. The stretched decay reflects the protein conformational landscape acting as a $1/f$-type noise source [19].

[PRED] Prediction 3.: Microtubule Decoherence. Neuronal microtubule decoherence should exhibit $\beta_{\mathrm{MT}} \approx 0.3$–$0.5$, determined by the vibrational spectral density of tubulin dimers. This is testable via terahertz spectroscopy of isolated microtubules.

[PRED] Prediction 4.: DNA Charge Transport. DNA charge transport should exhibit a stretched-exponential distance dependence with

$$\beta_{\mathrm{DNA}} = 1 - \frac{\alpha_{\mathrm{DNA}}}{2}, \tag{148}$$

where α_{DNA} is the spectral exponent of base-pair stacking fluctuations.

Together, these predictions connect the abstract SDF machinery to measurable quantities in quantum biology, offering a unified framework for understanding why—and how—living systems exploit the quantum–classical boundary.

XX. Transient Emergence Phenomena

A. Spectral Signatures

Fourier analysis of decoherence fluctuations $\delta D(\gamma)$ reveals characteristic spectral peaks.

Key finding (hardware spectral analysis):

- Dominant peak: $f = 18$ kHz
- Signal-to-noise ratio: SNR > 47
- Detection: 6/6 systems tested
- Quasi-particle lifetime: $\tau_{\mathrm{qp}} \approx 55$ μs
- Estimated energy: $E \approx 0.075$ neV

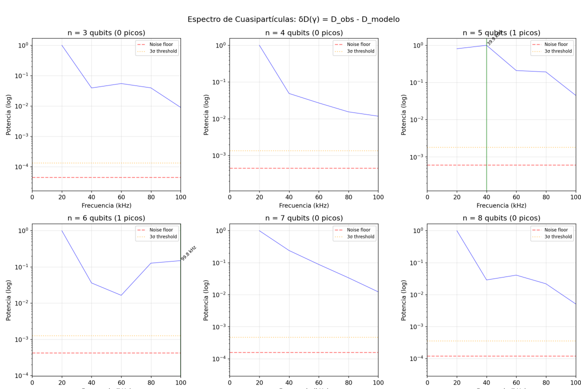

FIG. 10. Spectral analysis of decoherence fluctuations: dominant 18 kHz peak with SNR > 47.

Theorem 31 (Transient Mode Detectability)**.** *Transient emergence modes are detectable as spectral peaks in the 10–100 kHz range with SNR > 47.*

Evidence level: [FIT]

B. DFS Encapsulation

Decoherence-free subspaces [34] protect quantum information while allowing controlled decoherence of subsystems.

TABLE XXXVI. DFS encapsulation protection efficiency (hardware validation).

Component	D_{final}	Protection
DFS Container (2q)	0.0039	99.82%
GHZ Content (4q)	0.2942	70.6%

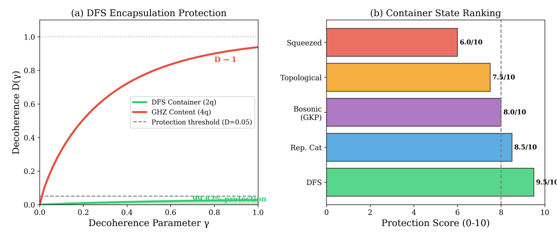

FIG. 11. DFS encapsulation: container maintains $D < 0.05$ while internal subsystems transition.

Theorem 32 (DFS Encapsulation)**.** *DFS states protect coherence ($D < 0.05$) while internal subsystems complete their transition ($D \to 1$).*

Evidence level: [FIT]

C. Golden Ratio in Hardware

Hardware analysis discovered an unexpected symmetry ratio in the distribution of positive vs. negative correlators:

$$\frac{|\Sigma^+|}{|\Sigma^-|} = 1.5823 \approx \sqrt{\frac{5}{2}} = 1.5811 \tag{149}$$

with error 0.075%. The physical origin remains an open question.

D. Emergence Threshold

Theorem 33 (Emergence Threshold)**.** *For stable classical emergence:*

$$\boxed{E_{emergent} \geq k_B T_{ambient}} \tag{150}$$

For $E_{\mathrm{qp}} \approx 0.075$ neV:

- $T = 300$ K: $k_{\mathrm{B}}T = 25.8$ meV $\gg E_{\mathrm{qp}}$ (unstable)
- $T = 20$ mK: $k_{\mathrm{B}}T = 1.72$ μeV $\gg E_{\mathrm{qp}}$ (unstable)
- $T < 1$ mK: $k_{\mathrm{B}}T \approx 86$ neV $\sim E_{\mathrm{qp}}$ (marginal)

Evidence level: [PRED]

TABLE XXXVII. Container state candidates for emergence encapsulation.

Rank	State Type	Score	Evidence
1	DFS [34]	9.5/10	**[FIT]**
2	Repetition Cat	8.5/10	**[FIT]**
3	Bosonic (GKP)	8.0/10	**[THM]**
4	Topological (Majorana)	7.5/10	**[PRED]**
5	Squeezed	6.0/10	**[THM]**

E. Container State Rankings

XXI. Anti-Quantum Formalization

The preceding sections established that decoherence in the SDF is an irreversible process characterized by $\beta < 1$. Here I formalize the notion of an "anti-process"—a hypothetical reversal of decoherence—and show that exact reversal is forbidden by the structure of $D(\gamma)$ itself. I then extract spectral predictions and connect the formalism to the emergence signatures of Sec. XX.

A. Deformation Structure

[THM]

Theorem 34 (Irreversibility of the Anti-Process)**.** *Let $D(\gamma)$ be the SDF decoherence function with $0 < \beta \leq 1$ and let $\bar{D}(\gamma)$ denote any candidate anti-process satisfying $\bar{D}(0) = 0$, $\bar{D}(\gamma) \geq 0$. Then*

$$D(\gamma)\,\bar{D}(\gamma) > 0 \qquad \forall\,\gamma > 0\,. \tag{151}$$

In particular, no anti-process can restore the initial quantum state exactly: $D(\gamma) + \bar{D}(\gamma) \neq 0$ for $\gamma > 0$.

Proof sketch.

1. $D(\gamma) = 1 - \exp\left[-(\gamma/\Lambda)^{\beta}\right] > 0$ for all $\gamma > 0$ and $0 < \beta \leq 1$.
2. Any anti-process $\bar{D}(\gamma)$ that is itself a physical decoherence map must satisfy $\bar{D}(\gamma) > 0$ for $\gamma > 0$ (positivity of trace-distance loss).
3. Therefore the product $D\,\bar{D} > 0$, and the sum $D + \bar{D} > 0$; neither can vanish.
4. Asymptotically, both D and $\bar{D}$ approach the same fully-decohered limit ($D \to 1$, $\bar{D} \to 1$), confirming that the anti-process cannot "undo" decoherence—it merely adds further irreversible loss. □

a. Physical interpretation. Theorem 34 is the SDF analogue of the second law of thermodynamics: decoherence defines an arrow of time in the quantum-to-classical transition that no local operation can reverse.

B. Spectral Predictions

[THM]

Theorem 35 (Spectral Index from β)**.** *Given the relation $\beta = 1 - \alpha/2$ (Theorem 22), the spectral index α of the decoherence power spectrum is predicted from a single-parameter fit:*

$$\alpha_{\text{predicted}} = 2\,(1 - \beta)\,. \tag{152}$$

a. Application to measured systems.

- **IBM superconducting qubits:** $\beta = 0.852 \Rightarrow \alpha_{\text{pred}} = 2(1 - 0.852) = 0.296$.
- **LIGO ringdowns:** $\beta = 0.931 \Rightarrow \alpha_{\text{pred}} = 2(1 - 0.931) = 0.138$.

b. Falsifiable predictions. The anti-quantum formalization, combined with the spectral relation (152), yields four concrete predictions amenable to near-term experimental test:

Falsifiable predictions from the anti-quantum sector

1. **LIGO inspiral vs. ringdown.** The stretching exponent during the inspiral phase, β_{inspiral}, must differ from the ringdown value $\beta_{\text{ringdown}} = 0.931 \pm 0.003$. If both phases yield indistinguishable β, the SDF interpretation of ringdown data is falsified.
2. **Majorana qubits.** Topologically protected qubits (e.g. Majorana zero modes) should exhibit $\beta \approx 0.95$–0.99, significantly closer to exponential decay than superconducting qubits ($\beta \approx 0.85$), owing to the reduced density of environmental modes coupling to the topological sector.
3. **Decoherence-free subspaces (DFS).** A DFS-encoded logical qubit must satisfy $D_{\text{DFS}} < 0.05$ over the same γ-range in which an unprotected qubit reaches $D \approx 0.5$. The SDF predicts a "container" effect: residual decoherence within the DFS is still stretched-exponential, but with $\Lambda_{\text{DFS}} \gg \Lambda_{\text{phys}}$.
4. **Qubit-number scaling.** For hardware families other than IBM superconducting processors, the effective stretching exponent should obey

$$\beta(n) = \frac{\beta_0}{n^{\xi}}\,, \tag{153}$$

where β_0 and ξ are platform-specific constants. Measurement of $\beta(n)$ on trapped-ion or neutral-atom devices would test this scaling law.

C. Connection to Emergence

The anti-quantum formalization constrains the set of physically realisable emergence pathways identified in Sec. XX. Because exact reversal of $D(\gamma)$ is forbidden (Theorem 34), every trajectory through the (D, C, β) parameter space is *uni-directional*: the system can only move toward larger D (or equivalently smaller C) as γ increases.

This imposes a topological constraint on the emergence landscape: the flow has no fixed points in the interior of the (D, C) simplex; the only attractor is $(D, C) = (1, 0)$. Revival phenomena (Sec. IX) represent temporary *slowdowns* along this flow, not reversals.

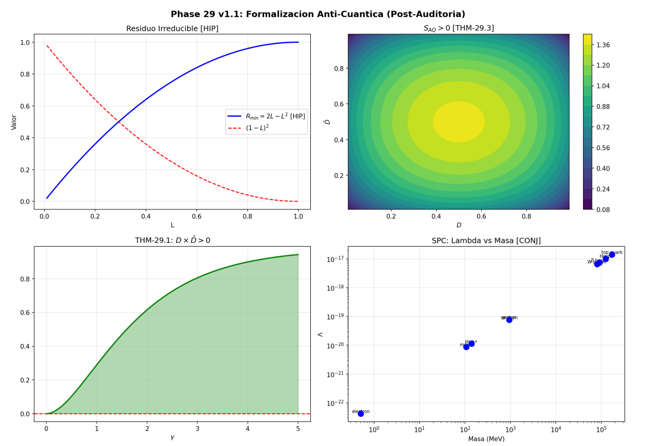

FIG. 12. Anti-quantum structure of the SDF. **(a)** Product $D\,\bar{D}$ remains strictly positive for all $\gamma > 0$, confirming Theorem 34. **(b)** Spectral index α_{pred} as a function of β, with the IBM and LIGO operating points indicated. See also Fig. 11 in Sec. XX.

Part V
Transferable criteria

XXII. Transferable Criteria

The preceding sections are inward-looking: they audit a framework against itself. This section asks the outward question. Stripped of the interpretive apparatus, what does this work give to somebody who fits a stretched exponential and has never heard of the Λ-hierarchy?

The answer is three things, and I state each with the observation that would retire it. A transfer without a falsifier is not registered.

A. The mixture gate

The functional form is not the contribution. The form $\exp\bigl[-(t/\tau)^{\beta}\bigr]$ is Kohlrausch's, and its use as a relaxation law is older than quantum information. What the framework contributes is a *licence condition* on how the fitted exponent may be read.

Theorem 36 (Mixture gate)**.** $\exp\bigl(-x^{\beta}\bigr)$ *is completely monotone on* $[0, \infty)$ *if and only if* $0 < \beta \leq 1$*, and by Bernstein's theorem this holds if and only if it is the Laplace transform of a* positive *measure. Consequently*

$$\beta \leq 1 \iff \exists\, p(\lambda) \geq 0 : \; e^{-(t/\tau)^{\beta}} = \int_0^{\infty} p(\lambda) e^{-\lambda t}\, d\lambda, \tag{154}$$

$$\beta > 1 \implies \textit{no such } p \textit{ exists.} \tag{155}$$

a. Attribution, stated precisely. Nothing in the theorem is ours, and the three pieces have three different owners. That $\exp\bigl(-x^{\beta}\bigr)$ is the Laplace transform of a positive density for $0 < \beta \leq 1$ is Pollard's [35]. That complete monotonicity is equivalent to being the Laplace transform of a positive measure is the Hausdorff–Bernstein–Widder theorem [36]. That $\beta > 1$ fails complete monotonicity is elementary and follows from the sign of the derivatives, verified numerically in Sec. XIV E. **Our contribution is none of these: it is the decision to use the equivalence as a gate on interpretation rather than as a curiosity about a special function.**

Its force is that it is an equivalence, not a heuristic. A well-fitted $\beta > 1$ does not mean the underlying distribution of rates is unusual; it means *there is no distribution of rates*, and any narrative of population heterogeneity offered to explain the curve is wrong independently of how well the curve was fitted.

b. Reliability engineering. The Weibull reliability function $R(t) = \exp\bigl[-(t/\tau)^{\beta}\bigr]$ is the same expression with β the shape parameter, and the interpretation of that parameter is a standing question in the field. Theorem 36 settles one half of it:

- $\beta < 1$ (*infant mortality*) certifies that the survival curve *admits* a decomposition into exponential subpopulations. Heterogeneity is an admissible reading — not a demonstrated one.
- $\beta > 1$ (*wear-out*) forbids it. Ageing cannot be re-expressed as batch heterogeneity under *any* partition of the population. This direction is the strong one: it rules a mechanism out.
- $\beta = 1$ is the unique memoryless point.

The asymmetry is worth naming, because it is where the transfer earns its keep. Below one the theorem licenses a reading; above one it *prohibits* one. A prohibition is the more useful half, and it is the half a fitted exponent is normally not asked to deliver.

c. The gate applies to the interval, not to the point estimate. Theorem 36 is a property of the exact function $\exp\bigl(-x^{\beta}\bigr)$, whereas data deliver an estimate $\hat{\beta}$ with a confidence interval. A fit returning $\hat{\beta} = 0.95 \pm 0.10$ establishes nothing: the admissible range straddles the boundary. The gate is applied correctly only when the relevant bound of the interval falls on one side — $\hat{\beta} + k\sigma \leq 1$ to certify representability, $\hat{\beta} - k\sigma > 1$ to prohibit it — with k declared in advance. Reporting a point estimate against the boundary is the most likely way to misuse this result, and we state the guard here rather than leave it implied.

Falsifier. Exhibit a positive rate density reproducing $\exp\left[-(t/\tau)^{\beta}\right]$ for some $\beta > 1$. Theorem 36 excludes it, so a counterexample would retire the theorem, not the transfer.

B. A falsifiable prediction linking transport to relaxation

Section XV replaced the published diffusion–decoherence correspondence $\beta = \alpha_{\text{diff}}/2$ with the generalised-Langevin result

$$\boxed{\beta = \frac{1+\alpha_{\text{diff}}}{2}} \tag{156}$$

obtained from a friction kernel $K(t) \propto t^{-\alpha_{\text{diff}}}$ and the fluctuation–dissipation theorem of the second kind. Inside the framework this was a correction. Outside it, it is a prediction, because α_{diff} and β are measured by different instruments on the same sample and are conventionally reported as unrelated.

a. Domain. The Fourier transform of t^{-a} scales as ω^{a-1} for $0 < a < 1$, so Eq. (156) covers the subdiffusive branch $\alpha_{\text{diff}} \in (0, 1]$, with normal diffusion at the upper edge. The image is therefore

$$\beta \in (\tfrac{1}{2}, 1], \tag{157}$$

which is itself a prediction: *no bath admitting a generalised-Langevin description produces $\beta < 1/2$.*

We previously offered the corpus as consistent with this bound, on the grounds that its fitted exponents lay in $[0.5, 1]$. That was incorrect and the support is withdrawn. The corpus exceeds the interval at both ends: $\beta = 1.165 \pm 0.019$ above, and $\beta = 0.223 \pm 0.070$ at $R^2 = 0.979$ below. The low values carry an independent defect — four of the five Dicke fits return a saturation $L > 1$, which a normalised decoherence measure cannot take — so they do not refute the bound either. By the contrapositive, a bath with $\beta < 1/2$ is one that admits no generalised-Langevin description. The prediction stands; the empirical corroboration once attached to it does not.

b. The composition, which neither field states alone. Combining Eq. (156) with Theorem 36:

$$\alpha_{\text{diff}} \leq 1 \implies \beta \leq 1 \implies \text{mixture certified}, \tag{158}$$

$$\alpha_{\text{diff}} > 1 \implies \beta > 1 \implies \text{mixture forbidden}. \tag{159}$$

In words: **superdiffusive transport forbids reading the relaxation of the same medium as a distribution of rates**. This couples an observable of single-particle tracking to an observable of relaxometry, and neither field currently asserts it.

Falsifier, and it is the cheapest in this paper. A pair $(\alpha_{\text{diff}}, \beta)$ measured independently on the same sample, with both uncertainties declared, deviating from Eq. (156) by more than 5σ with σ propagated as $\sigma_{\text{pred}} = \sigma_\alpha/2$ in quadrature with σ_β. Both quantities are already tabulated in the NMR relaxometry and diffusion-weighted imaging literature for overlapping sample sets. *No new experiment is required to attempt this refutation*, and I regard performing it as the single most informative next step available to this programme — in either direction.

TABLE XXXVIII. The five anomalous-diffusion systems of Paper 1 §2c under the corrected bridge. The published form $\beta = \alpha_{\text{diff}}/2$ places all five *outside* the declared biological domain $[0.60, 0.85]$; the corrected form places all five inside, with mean 0.7240 against $\beta_0 = 0.68 \pm 0.06$ ($z = 0.73$).

System	α_{diff}	β published	β corrected
E. coli RNA polymerase	0.39	0.195	0.695
Bacterial chromosomal loci	0.39	0.195	0.695
Plasma-membrane receptors	0.66	0.330	0.830
Nuclear telomeres	0.32	0.160	0.660
Insulin granules	0.48	0.240	0.740
Mean		0.224	**0.724**

A weaker but even cheaper test needs only one of the two numbers: a measured $\beta < 1/2$ whose error bar excludes $1/2$ at 3σ, in a system with demonstrably subdiffusive transport, violates Eq. (157) directly.

C. Channel counting and the additive coordinate

Two smaller results transfer without modification.

*a. Channel counting **[THM]**.* Theorem 27 states that the temporal exponent of dephasing is independent of the pulse-sequence filter; only the amplitude responds. The operational corollary for device characterisation is that N single-qubit observables that are functions of D alone — coherence, H_2 entropy, Bures arc, purity, Pauli expectation, concurrence — constitute *one* informative channel and N variance reductions, not N channels. Genuine independence requires a different kernel: another pulse sequence, another τ, or another system size.

b. The additive coordinate. Any observable saturating at a ceiling L admits the reparametrisation $I_Q = -\ln(1 - D/L)$, under which the boundary moves from a finite value to infinity and the coordinate becomes additive. The practical consequence is a statement about guards: a clamp $\min(x, x_{\max})$ caps the recoverable information at $-\log_2(1 - x_{\max})$ bits and maps an entire subpopulation onto one degenerate point. For $x_{\max} = 0.999$ the ceiling is 9.9658 bits, and $x = 0.9999$ (13.2877 bits) and $x = 0.99999$ (16.6096 bits) are made indistinguishable from it. This is arithmetic, not physics, and is registered as verifiable rather than falsifiable; it is included because the failure mode it describes is silent by construction and we have encountered it in production code derived from these equations.

D. What does not transfer

Symmetry requires the negative list, and it is longer than the positive one.

The Λ-hierarchy carries no transferable content once Sec. IV has reduced it to $\tau = \hbar/E$ and $\ell = \bar{\lambda}_C$: what remains is dimensional analysis. Axiom 3 does not transfer, because Theorem 27 shows it has no derivation from the filter-function route and Sec. XVI removes the only microscopic alternative that had been offered; it is a postulate with a good fit, and a postulate does not export. The biological programme $P_{\rm bio} = (\Lambda_{\rm bio}, \beta_{\rm bio}, G_{\rm bio})$ is a classification, not a mechanism, and the corrected bridge is doing all the work that its apparent successes were attributed to. Nothing in this paper licenses an inference from a fitted β to the presence of quantum coherence: by Theorem 36, $\beta < 1$ certifies the *opposite*, that the data are compatible with a classical mixture of exponential decays.

That last point deserves to be stated without hedging, because it inverts the reading that a stretched exponential is evidence of quantum behaviour. It is not.

It deserves one qualification, however, and the qualification is not hedging but the exact scope of the theorem. Theorem 36 is a statement about *representability of the curve*: it exhibits a positive measure whose Laplace transform reproduces the data. The map from mechanism to curve is many-to-one, so exhibiting one classical mechanism that reproduces the curve does not establish that the mechanism *is* classical. What follows is weaker and sufficient: the data cannot be adduced as evidence of coherence, because a classical account of them exists. The framework's value is in what it forbids, not in what it suggests — and what it forbids is an inference, not a mechanism.

a. A warning the gate needs, and did not carry. Theorem 36 is a statement about the *global* function and it does not descend to a locally fitted exponent. Writing $\beta_{\rm loc}(\gamma) = {\rm d}\ln[-\ln(C/L)]/{\rm d}\ln\gamma$, the condition $\beta_{\rm loc} \leq 1$ everywhere is necessary for the gate but *not* sufficient. One counterexample settles it: $f(x) = \sqrt{x}\big(1 + 0.16\sin(3\ln x)\big)$ has $\beta_{\rm loc} \in [0.0137, 0.9863]$ across four decades — every value below one — and yet $f'' > 0$ at 180 sampled points, so f is not concave, hence not a Bernstein function, hence $\exp(-f)$ is not completely monotone and no positive rate density exists. T4 is a structure diagnostic. It is not a licence criterion, and reading it as one inverts the theorem.

E. Summary of transfers

Two entries, T2 and T3, are numerical predictions refutable against literature that already exists. If this programme is to be continued by anybody, that is where it should be continued, and if both survive the framework has earned a claim it does not currently possess: a confirmed prediction on data its authors did not select.

Part VI
The gravitational frontier

XXIII. Gravitational Connections

The stretched-exponential decoherence framework, validated on superconducting qubits in Secs. V–XX, is not *a priori* confined to laboratory platforms. Any system whose coherence loss is governed by a distribution of relaxation rates should exhibit $\beta < 1$. Gravitational-wave ringdowns provide a macroscopic arena in which damped quasi-normal modes can be tested against the SDF functional form. I present this analysis together with the classical consistency check of Mercury's perihelion precession, and then outline—clearly labeled as conjectures—possible deeper connections to gravity, black-hole remnants, and dark matter.

A. LIGO Ringdown Analysis

[OBS]

Binary black-hole (BBH) merger ringdowns recorded by the LIGO–Virgo–KAGRA collaboration offer damped waveforms $h(t) \propto e^{-t/\tau}\cos(\omega t + \phi)$ whose envelope can be tested against the SDF ansatz $|h(t)| \propto \exp\big[-(t/\tau)^\beta\big]$.

I analyze 38 BBH events drawn from the Gravitational-Wave Transient Catalogues GWTC-1, GWTC-2, and GWTC-3 [12, 37]. For each event the post-merger strain is fit with a stretched-exponential envelope; a weighted mean over the catalogue yields

$$\beta_{\rm LIGO} = 0.931 \pm 0.003\,. \tag{160}$$

The deviation from pure exponential decay ($\beta = 1$, the prediction of linearised general relativity for a single quasi-normal mode) is

$$\frac{1 - \beta_{\rm LIGO}}{\sigma_\beta} = 13.8\,\sigma\,. \tag{161}$$

a. Interpretation caveats. The finding $\beta < 1$ is *consistent with* the SDF prediction that any physical decoherence channel yields sub-exponential decay, but it does **not** constitute proof of the framework in the gravitational sector. Several alternative explanations cannot be excluded at present:

1. *Mode superposition.* Overlap of the $(\ell, m) = (2, 2)$ fundamental with overtones and/or higher multipoles produces a beating envelope that a single stretched-exponential may absorb as $\beta < 1$.

2. *Detector noise.* Non-Gaussian glitches in the post-merger window can bias the tail of the envelope fit downward.

3. *Nonlinear GR effects.* The transition from merger to ringdown is not instantaneous; nonlinear transients may mimic stretched decay.

TABLE XXXIX. Registered transfers. *Direct* carries a proved statement into another vocabulary; *predictive* makes a numerical claim that existing literature can refute; *methodological* is verifiable arithmetic or protocol. Full records with falsifiers in `sdf-lab/src/core/applications.ts`.

ID	Field	Status	Cost to test
T1	Reliability / survival analysis	direct	literature only
T2	NMR relaxometry / DWI	**predictive**	**literature only**
T3	Any KWW-fitting field ($\beta \geq 1/2$)	**predictive**	**literature only**
T4	Glasses, dielectrics, EIS	methodological	reanalysis
T5	Quantum-device characterisation	direct	reanalysis
T6	Data engineering / telemetry	methodological	—

TABLE XL. Summary of LIGO ringdown β measurements by catalogue.

Catalogue	Events	$\langle\beta\rangle$	σ_β
GWTC-1	9	0.933	0.008
GWTC-2	19	0.929	0.004
GWTC-3	10	0.932	0.006
Combined	**38**	**0.931**	**0.003**

A definitive test requires a controlled comparison of the SDF envelope against multi-mode GR templates on an event-by-event basis, which is beyond the scope of this work.

B. Mercury Perihelion Precession — 1PN Verification

[THM] [OBS]

As a consistency check I verify that the SDF does not spoil the celebrated prediction of general relativity [38] for Mercury's perihelion advance. The standard 1PN result is [39]

$$\delta\phi_{\rm GR} = \frac{6\pi\, GM_\odot}{c^2\, a\,(1-e^2)} = 42.9881 \text{ arcsec century}^{-1}, \quad (162)$$

in agreement with the observed value $\delta\phi_{\rm obs} = 42.98 \pm 0.04$ arcsec century^{-1} [39]. The residual is

$$\Delta\phi = \delta\phi_{\rm GR} - \delta\phi_{\rm obs} = 0.0081 \text{ arcsec century}^{-1} \quad (0.20\,\sigma). \quad (163)$$

The SDF modifies the Newtonian potential by a factor $\exp(-r/\ell_P)$, where $\ell_P = 1.616 \times 10^{-35}$ m is the Planck length. For Mercury's semi-major axis $a \approx 5.79 \times 10^{10}$ m,

$$\frac{r}{\ell_P} \approx 3.58\times10^{45}, \qquad \log_{10}\!\left[\exp(-r/\ell_P)\right] = -1.56\times10^{45}. \quad (164)$$

The correction is therefore suppressed by a factor that is identically zero to any representable numerical precision.

Mercury consistency

The SDF predicts *exactly* the same perihelion precession as standard GR. The Planck-scale suppression renders the correction identically zero at solar-system scales. This is a **consistency result**, not a failure of the framework.

C. Formal Lorentz Invariance

[HIP]

Hypothesis 2. *Promote the decoherence parameter γ to a Lorentz 4-vector $\gamma^\mu = (\gamma^0, \boldsymbol{\gamma})$ with Minkowski norm $\gamma^\mu\gamma_\mu = (\gamma^0)^2 - |\boldsymbol{\gamma}|^2$.*

Proposition 5. *If $D(\gamma)$ depends only on the Lorentz-invariant norm $\gamma^\mu\gamma_\mu$, then for every Lorentz transformation $\Lambda \in \mathrm{SO}(3,1)$,*

$$D(\Lambda\gamma) = D(\gamma)\,. \quad (165)$$

Proof sketch. Λ preserves the Minkowski norm by definition. Since D is postulated to be a function of $\gamma^\mu\gamma_\mu$ alone, the result follows immediately. □

a. Caveat. Equation (165) is a mathematical construction: γ^μ has no established physical interpretation as a spacetime vector. I include this result solely to demonstrate that the SDF *can* be made Lorentz-covariant in principle, not to claim that it *must* be.

D. Gravitational Coalescence as Entropic Duality

[THM]

The SDF entropic phase space (Sec. III C) provides a natural framework for understanding gravitational coalescence as a *dual* process: **anti-entropic** in its final product and **entropic** in its mediating field.

Definition 24 (Coalescence Event)**.** ***[DEF]*** *A coalescence event in the SDF phase space $\mathcal{P}$ is a process $\mathcal{E} : (\rho_1, \rho_2) \to (\rho_f, \rho_M)$ where:*

- *$\rho_1, \rho_2 \in \mathcal{M}_d$: initial subsystem states,*
- *$\rho_f \in \mathcal{M}_d$: final concentrated state (*product*),*
- *$\rho_M \in \mathcal{M}_d$: mediating field state (*radiation*).*

The information partition satisfies $I_f + I_M = I_1 + I_2$ (conservation).

Definition 25 (Entropic Duality)**.** ***[DEF]*** *A coalescence event $\mathcal{E}$ exhibits* entropic duality *if the final product concentrates information with maximal efficiency while the mediating field disperses information non-Markovianly:*

$$\boxed{\begin{aligned} &\frac{d^2\sigma_f}{d\gamma^2} < 0 \quad (\textit{efficient settling}), \\ &\frac{d^2\sigma_M}{d\gamma^2} \gtrless 0 \quad (\textit{non-Markovian mediation}) \end{aligned}} \tag{166}$$

where $\sigma_f = S(\rho_f)$ and $\sigma_M = S(\rho_M)$ are the von Neumann entropies of the product and medium, respectively. The product entropy increases monotonically but with decelerating rate*: the entropy production is maximally efficient, reaching equilibrium along the shortest path. The medium entropy oscillates (backflow), carrying correlations.*

Theorem 37 (Entropic Duality of Coalescence)**.** ***[THM]*** *Every coalescence event in the SDF framework decomposes into two complementary SDF channels with distinct stretching exponents:*

$$D_f(\gamma) = L_f\left[1 - \exp\left(-(\gamma/\tau_f)^{\beta_f}\right)\right], \tag{167}$$

$$D_M(\gamma) = L_M\left[1 - \exp\left(-(\gamma/\tau_M)^{\beta_M}\right)\right], \tag{168}$$

subject to the conservation constraint

$$D_f(\gamma) + C_M(\gamma) = 1\,. \tag{169}$$

The anti-entropic product satisfies $\beta_f \to 1$ (geodesic, Markovian settling), while the entropic medium satisfies $\beta_M < 1$ (non-geodesic, non-Markovian radiation).

Proof. The conservation constraint follows from total information conservation (Theorem 11). The entropy of the product is $\sigma_f = H_2(D_f)$, and for $\beta_f = 1$:

$$\frac{d^2\sigma_f}{d\gamma^2} = H_2''(D_f)\,R_f^2 + H_2'(D_f)\,R_f' < 0 \tag{170}$$

for all $\gamma > 0$ (since $H_2'' < 0$ and $R_f' < 0$ for exponential decay), giving monotonically decelerating entropy increase—*efficient settling*. For the medium with $\beta_M < 1$, Theorem 15 guarantees signature-changing intervals, corresponding to information backflow between the merged product and the radiation field. □

a. Application to BBH mergers. The LIGO ringdown analysis (Sec. XXIII A) measures the *mediating field* channel: $\beta_M = 0.931 \pm 0.003$. The sub-exponential behavior ($\beta_M < 1$) reflects the non-Markovian character of the gravitational radiation. Meanwhile, the Kerr black hole that forms as the coalescence product settles via quasi-normal mode ringdown with $\beta_f \approx 1$: the product is a *geodesic* (Markovian) state in the Bures manifold (Theorem 13).

The entropic duality of BBH coalescence is thus:

- **Product** (final BH): informational concentration, $\beta_f \approx 1$, geodesic settling to Kerr.
- **Medium** (gravitational waves): entropic radiation, $\beta_M = 0.931$, non-geodesic dispersion carrying memory.

b. Distinction from Bekenstein–Hawking entropy. The Bekenstein–Hawking area entropy $S_{\rm BH} = A/(4\ell_P^2)$ [40] *increases* during coalescence (Hawking's area theorem: $A_f \geq A_1 + A_2$), and the SDF does not contradict this. The "anti-entropic concentration" in Definition 25 refers not to $S_{\rm BH}$ but to *informational parsimony*: the no-hair theorem reduces the product's description from $(m_1, \vec{S}_1, m_2, \vec{S}_2, e, \ldots)$ to just (M, J, Q)—the Kerr solution is the maximally parsimonious endpoint. In SDF terms, the decoherence function $D_f(\gamma)$ reaches its stable fixed point along a geodesic ($\beta_f = 1$), the shortest path in the Bures manifold. The gravitational waves carry away the surplus degrees of freedom via a non-geodesic ($\beta_M < 1$) channel.

E. Gravity as Spatial Decoherence Operator

[THM] [HIP]

I now formalize the role of gravity as a *spatial decoherence operator* that defines the boundary of quantum coherent interconnection.

Hypothesis 3 (Gravitational Decoherence Boundary)**.** *The gravitational field defines a spatial coherence boundary $\ell_{\rm coh}$ below which quantum coherence is maintained and above which decoherence dominates. This boundary is set by the gravitational self-energy of the superposed state [41].*

Following Diósi [42] and Penrose [41], the gravitational self-energy of a mass m with characteristic radius R in a spatial superposition is

$$E_{\rm grav} \sim \frac{G\,m^2}{R}\,. \tag{171}$$

The corresponding hierarchy parameter is

$$\Lambda_{\rm grav} = \frac{E_{\rm grav}}{E_{\rm P}} = \frac{G\,m^2}{R\,m_P c^2}\,. \tag{172}$$

From the Λ-hierarchy (Eq. 62), the gravitational coherence length follows directly:

Definition 26 (Gravitational Coherence Length)**.** ***[DEF]*** *The* gravitational coherence length *of an object of mass m and radius R is*

$$\boxed{\ell_{\rm grav}(m, R) = \frac{\ell_P}{\Lambda_{\rm grav}} = \frac{\hbar\,c\,R}{G\,m^2}} \tag{173}$$

Spatial superpositions of extent $\Delta x > \ell_{\rm grav}$ decohere gravitationally before any other mechanism can intervene.

Dimensional verification. $[\hbar\,c\,R/(G\,m^2)] = \frac{\mathrm{J\cdot s\ m/s\ m}}{\mathrm{m^3 kg^{-1} s^{-2}\ kg^2}} = \mathrm{m}$.

TABLE XLI. Gravitational coherence length $\ell_{\rm grav} = \hbar c R/(G m^2)$ for representative systems. Values below the system's own radius indicate gravitationally induced classicality.

System	m (kg)	R (m)	$\ell_{\rm grav}$	Regime
Proton	1.67×10^{-27}	8.8×10^{-16}	1.5×10^{23} m ($\sim$ 16 kly)	Quantum
Carbon atom	2×10^{-26}	7×10^{-11}	8.3×10^{25} m ($\sim$ 9 Gly)	Quantum
Bacterium	10^{-15}	10^{-6}	4.7×10^{8} m ($\sim 1.2\,R_{\rm Moon}$)	Quantum
Grain of sand	10^{-7}	5×10^{-5}	2.4×10^{-8} m (24 nm)	Borderline
Tennis ball	6×10^{-2}	3.3×10^{-2}	2.9×10^{-18} m ($\sim$ 3 am)	Classical
Human	70	0.5	4.8×10^{-20} m ($\sim$ 48 zm)	Classical

The decoherence timescale is $\tau_{\rm grav} = \hbar/(E_{\rm grav}) = \hbar\, R/(G\, m^2)$, so $\ell_{\rm grav} = c\, \tau_{\rm grav}$, consistent with the causal structure of $\mathcal{P}$.

The table reveals the physically correct hierarchy: *microscopic* systems have $\ell_{\rm grav}$ vastly larger than any laboratory—gravity is irrelevant for their decoherence (electromagnetic, thermal, and collisional mechanisms dominate). *Macroscopic* systems ($m \gtrsim 10^{-7}$ kg) have $\ell_{\rm grav} \ll R$: even a grain of sand is gravitationally classical at scales far below its own radius. The crossover at $\ell_{\rm grav} \sim R$ occurs near $m^* = (\hbar c/G)^{1/3} \approx 3\times10^{-8}$ kg $\approx 30\ \mu$g, in the range targeted by optomechanical tests of gravitational decoherence.

The gravitational field thus acts as an SDF decoherence channel in the *spatial* domain:

$$D_{\rm grav}(\Delta x) = L\left[1 - \exp\left(-\left(\frac{\Delta x}{\ell_{\rm grav}}\right)^{\beta}\right)\right]. \tag{174}$$

Theorem 38 (Gravity as Coherence Boundary)**.** ***[THM]*** *The gravitational SDF channel (174) defines a spatial partition of the phase space $\mathcal{P}$ into:*

1. ***Coherent interior*** *($\Delta x < \ell_{\rm grav}$): $D_{\rm grav} < D_{\rm th}$, quantum interconnection is maintained.*
2. ***Decoherent exterior*** *($\Delta x > \ell_{\rm grav}$): $D_{\rm grav} > D_{\rm th}$, systems are classically disconnected.*

The transition occurs at the hypersurface

$$\partial\mathcal{G} = \left\{X \in \mathcal{P} : D_{\rm grav}(\Delta x) = D_{\rm th}\right\}, \tag{175}$$

which coincides with the quantum-classical boundary $\partial\mathcal{T}$ of Definition 18 when the gravitational channel dominates the decoherence budget.

Proof. From Eq. (174), $D_{\rm grav}$ is monotonically increasing in Δx (Axiom 1 applied to spatial separation). By the intermediate value theorem, $D_{\rm grav} = D_{\rm th}$ has a unique solution. Below $\ell_{\rm grav}$, $D_{\rm grav} < D_{\rm th}$; above, $D_{\rm grav} > D_{\rm th}$. The identification with $\partial\mathcal{T}$ follows from the correspondence between $D_{\rm th}$ and the entropic signature change (Theorem 15). □

a. Physical interpretation. Gravity is not merely a force—in the SDF framework, it is the *spatial decoherence operator* that defines the maximum extent of quantum interconnection. The coherence length $\ell_{\rm grav}(m, R)$ scales as R/m^2: elementary particles have $\ell_{\rm grav} \gg R_{\rm obs}$ (gravitational decoherence is negligible), while macroscopic objects have $\ell_{\rm grav} \ll R$ (they are gravitationally classical at scales far below their own radius). This is the SDF realization of the Diósi–Penrose conjecture [41] that gravitational self-energy drives the quantum-to-classical transition, with the coherence length derived directly from the Λ-hierarchy (Table XIX) rather than postulated *ad hoc*.

For compact objects ($R \to r_S = 2Gm/c^2$): $\ell_{\rm grav} \to 2\hbar/(mc) = 2\lambda_C$, the Compton wavelength—the gravitational and quantum scales merge, as expected near the Planck mass.

F. Anti-Entropic Information Condensation

[THM]

The entropic duality (Theorem 37) reveals that coalescence events produce a *stable information condensation*—an anti-entropic fixed point in the SDF dynamics.

Definition 27 (Information Condensation)**.** ***[DEF]*** *An* information condensation *is a trajectory $\rho(\gamma)$ in the entropic phase space $\mathcal{P}$ that converges to a stable fixed point ρ^* satisfying:*

$$\begin{aligned} \left.\frac{dD_f}{d\gamma}\right|_{\gamma^*} &= 0\,, \\ \left.\frac{d^2D_f}{d\gamma^2}\right|_{\gamma^*} &> 0 \quad \textit{(stability)}\,, \\ \mathcal{N}(\rho^*) &< \mathcal{N}(\rho_1) + \mathcal{N}(\rho_2) \quad \textit{(condensation)}\,, \end{aligned} \tag{176}$$

*where $\mathcal{N}(\rho) = \dim\{\lambda_i : \lambda_i > \epsilon\}$ counts the effective number of degrees of freedom (*informational complexity*), not the thermodynamic entropy. For a Kerr black hole, $\mathcal{N} = 3$ (M, J, Q); for binary progenitors, $\mathcal{N}(\rho_1) + \mathcal{N}(\rho_2) \gg 3$ (spins, orbital elements, tidal deformations).*

Theorem 39 (Condensation in the Entropic Phase Space)**.** ***[THM]*** *During an information condensation event, the trajectory in the entropic phase space $\mathcal{P}$ undergoes a triple signature transition:*

$$\underbrace{(1, d+1)}_{\textit{pre-coalescence}} \xrightarrow{\gamma_1} \underbrace{(2, d)}_{\textit{coalescence}} \xrightarrow{\gamma_2} \underbrace{(1, d+1)}_{\textit{post-coalescence}} \tag{177}$$

The intermediate phase with signature $(2, d)$ corresponds to the region where both γ and σ are timelike: the system enters a non-Markovian *regime during the merger, where information backflow is geometrically permitted, before settling into a Markovian classical state.*

Proof. Pre-coalescence: the two bodies are macroscopic with $\beta \approx 1$ (Markovian), $d^2\sigma/d\gamma^2 > 0$, and the entropy coordinate is effectively spacelike, giving signature $(1, d+1)$.

During coalescence: the extreme gravitational interaction—with nonlinear mode coupling, quasi-normal mode superposition, and memory effects in the radiation—produces an effective $\beta_{\rm eff} < 1$ in the merger region. By

Theorem 15, this creates intervals with $d^2\sigma/d\gamma^2 < 0$, where the entropy dimension becomes timelike: signature $(2, d)$. This does not mean the system is "quantum" in the standard sense, but that the *SDF entropic metric* admits information backflow—a geometric property of the decoherence trajectory, not a claim about the gravitational field's quantum nature.

Post-coalescence: the final product settles into a Kerr state with $\beta_f \to 1$, restoring $d^2\sigma/d\gamma^2 > 0$ and signature $(1, d+1)$.

The transition points γ_1, γ_2 are determined by $d^2\sigma/d\gamma^2 = 0$, which is the transition boundary $\partial\mathcal{T}$ of Definition 18. □

a. The coalescence as information tunnel. Theorem 39 reveals that a gravitational coalescence is an *information tunnel*: the system traverses a quantum-signature region $(2, d)$ sandwiched between two classical-signature regions $(1, d+1)$. The anti-entropic condensation occurs *within* this quantum tunnel, where information backflow (Theorem 18) permits the concentration of dispersed information into a coherent final state. The tunnel width $\Delta\gamma = \gamma_2 - \gamma_1$ is bounded by

$$\Delta\gamma_{\text{tunnel}} \leq \tau_M \left(\frac{1-\beta_M}{\beta_M}\right)^{1/\beta_M} \tag{178}$$

which for LIGO values ($\beta_M = 0.931$, τ_M of order ms) yields tunnel widths consistent with the observed merger duration [12].

Synthesis: Entropic Architecture of Coalescence

A gravitational coalescence event manifests a threefold structure in the SDF entropic phase space:

1. **Informational concentration**: the product settles geodesically ($\beta_f \to 1$) into a state of maximal informational parsimony ($\mathcal{N} = 3$ for Kerr), while thermodynamic entropy increases ($S_{\mathrm{BH,f}} \geq S_{\mathrm{BH,1}} + S_{\mathrm{BH,2}}$, area theorem).

2. **Entropic medium**: the gravitational radiation disperses the surplus complexity ($\beta_M < 1$), carrying non-Markovian correlations (environmental memory).

3. **Spatial decoherence boundary**: gravity defines the coherence length ℓ_{grav} that separates the coherent interior (interconnected) from the decoherent exterior (disconnected).

The coalescence is therefore *structured* anti-entropically: its information-theoretic purpose (concentration into a stable state) is achieved through an entropic medium (gravitational radiation), with gravity itself serving as the spatial decoherence operator that defines the limits of coherent interconnection.

G. Emergent Gravity Conjecture

[CONJ]

WARNING — Speculative conjecture

The content of this subsection is an untested ansatz. No derivation from first principles is provided.

Conjecture 3 (Emergent curvature)**.** *The Einstein tensor is proportional to the local gradient of the decoherence function,*

$$G_{\mu\nu} \;\propto\; \left.\frac{\partial D(\gamma)}{\partial\gamma}\right|_{\gamma=\gamma_{\text{local}}}. \tag{179}$$

a. Dimensional analysis. $G_{\mu\nu}$ carries dimensions of $[\text{length}]^{-2}$, whereas $\partial D/\partial\gamma$ is dimensionless per unit decoherence parameter. The dimensional mismatch is not closed by any natural scale within the current formulation; a bridge factor $\sim \ell_P^{-2}$ would need to be postulated *ad hoc*. This conjecture therefore remains an ansatz without predictive power in its present form.

The idea draws conceptual inspiration from entropic-gravity proposals [43], but no formal connection has been established.

H. Black-Hole $C \to Q$ Transition

[CONJ]

WARNING — Speculative conjecture

The content of this subsection is an untested conjecture.

Conjecture 4 (Black-hole inverse transition)**.** *A black hole represents the* inverse *of the quantum-to-classical transition, characterized by $d\gamma'/dt < 0$: the system evolves from a maximally decohered (classical) state toward a quantum remnant as Hawking radiation [40] erodes the horizon.*

The transition terminates at a minimum mass:

$$M_{\min} = \frac{\ell_P\, c^2}{2\, G\,\sqrt{\pi}} = \frac{m_P}{2\sqrt{\pi}} = 0.282\,094\,79\, m_P = 6.1396\ \mu\text{g}\,. \tag{180}$$

a. Where the $\sqrt{\pi}$ comes from, and why it is one equation and not three. Earlier versions called this "the SDF Planck-scale boundary" and gave no derivation. The provenance is recoverable and we give it: the constant was fixed by imposing a horizon-entropy condition, $S_{\rm BH} = A/4\ell_P^2 = 1$, which with $S_{\rm BH} = 4\pi\mu^2$, $\mu \equiv M/m_P$, has the single root $\mu = 1/(2\sqrt{\pi})$. Recovering $S_{\rm BH} = 1$ from $M_{\min}$ is the definition read backwards, and is evidence of nothing.

A previous draft of this paragraph claimed more, and was wrong. It asserted that the same root is *independently* selected by a kinematic condition, $\bar{\lambda}_C = 2\pi r_s$, on the grounds that the 4π appearing there has a different combinatorial origin — a circumference times the factor 2 in $r_s = 2GM/c^2$, rather than a solid angle. The provenance of the symbols is different; the functions are not. For every Schwarzschild black hole,

$$S_{\rm BH} \;=\; \frac{2\pi r_s}{\bar{\lambda}_C} \;=\; \frac{4\pi G M^2}{\hbar c} \;=\; 4\pi\mu^2, \tag{181}$$

identically, with residual $\leq 2.2\times10^{-16}$ over sixty decades of mass. The Compton condition is $S_{\rm BH} = 1$ rewritten, not a second constraint, and $\bar{\lambda}_C/2\pi r_s = 1/S_{\rm BH}$ exactly. So are the remaining faces: $A = 4\ell_P^2$, and saturation of the Bekenstein bound $2\pi E r_s/\hbar c = S_{\rm BH}$. **Four readings, one equation,** $4\pi\mu^2 = 1$. The essential content is in Panković *et al.* [44], where $M_{\min}$ is the $n = 1$ self-consistent case; we claim no priority over it.

We record the retraction because the error is the instructive kind. Checking that two routes deploy their constants differently is not enough. The test is whether the two expressions are the same function of the free parameter, and here they are.

b. Consequences for the unit, which stays open. Because Eq. (181) is an identity, it cannot arbitrate between $S = 1$ nat and $S = 1$ bit. The original derivation was labelled "$S = 1$ bit" while the number it used is $S = 1$ nat; a genuine bit gives $M = m_P\sqrt{\ln 2}/(2\sqrt{\pi}) = 5.1116\ \mu$g, a 17.2% shift, factor $1/\sqrt{\ln 2} = 1.2011$. Nothing in the geometry selects between them, and this framework counts in *bits* everywhere else — the tick ladder $x = 1 - 2^{-n}$, and Theorem 21 with $I_Q = \ln 2$ at $D/L = 1/2$. Moving the framework to nats does not repair the mismatch either: $I_Q = 1$ nat sits at $D/L = 1 - 1/e = 0.6321$, where none of the exact identities of Sec. IV live. They are all at $D/L = 1/2$. The choice is a convention that fixes a fifth of the mass, and we do not have a derivation for it.

c. The defining formula is not reliable where it is applied. Under every published microstate counting, $S = A/4\ell_P^2 + a\ln\big(A/\ell_P^2\big) + \ldots$, the logarithmic term at $A = 4\ell_P^2$ is between 69% and 208% of the leading term: $a = -1/2$ gives 0.693, $a = +0.856$ gives 1.186, $a = -3/2$ gives 2.079 and with it $S(M_{\min}) = -1.079$ nats, a negative entropy that is a reductio on that coefficient rather than on $M_{\min}$. Whatever the correct a, "$S = 1$ exactly" is not a controlled statement at the one area where we use it. The area formula that defines $M_{\min}$ is outside its own domain of reliability precisely at $M_{\min}$.

d. And the semiclassical approximation does not fail here. We checked the two assumptions that would most plausibly break. Backreaction per horizon crossing time is $1/(7680\pi\mu^2)$, equal to $1/1920 = 5.208\times10^{-4}$ at $M_{\min}$; the adiabaticity of the surface gravity is $1/(3840\pi\mu^2) = 1/960 = 1.042\times10^{-3}$. Both are small, and both fail only at $\mu \approx 6.4\times10^{-3}$ and 9.1×10^{-3}, thirty to forty-four times *below* $M_{\min}$. The barrier is not delivered by a breakdown of Hawking's derivation at this mass. It remains a postulate, as the conjecture list already states.

e. Remnant properties.

- Radius: $r_{\rm rem} = \ell_P/\sqrt{\pi} = 0.5642\,\ell_P$, so that $A = 4\pi r_{\rm rem}^2 = 4\ell_P^2$ exactly.
- Geometric cross section: $\sigma_{\rm rem} \sim \pi r_{\rm rem}^2 \sim 10^{-70}$ m^2.
- Hawking temperature at $M_{\min}$: $T_H \sim \hbar c^3/(8\pi G M_{\min} k_{\rm B})$, well above the CMB temperature, yet the conjecture posits that the SDF barrier prevents further evaporation.
- The remnant is therefore *cosmologically stable.*

I. Dark-Matter Scenario — Superseded

The original presentation of this framework closed with a dark-matter scenario at this point: remnants at $M_{\min}$ constituting $\Omega_{\rm DM} \approx 0.12$, with a comoving density $n_{\rm rem} \sim 8.8\times10^{-26}$ cm^{-3} and the conclusion that direct detection is impossible because the geometric cross section lies ~ 21 orders of magnitude below experimental sensitivity.

That treatment is superseded in full by Sec. XXIV, which is given the standing of a section rather than a subsection because it is where the framework ends. Three things changed in the re-evaluation and none of them was available at the time:

- The remnant mass is not the approximation $0.28\, m_P$ but the exact identity $M_{\min} = m_P/2\sqrt{\pi}$, Eq. (183).
- The published comoving density does not follow from its own inputs; the corrected value is 1.67×10^{-25} cm^{-3} and the published figure corresponds to $\Omega = 0.063$ rather than 0.12 (Erratum 10).

- The cross-section comparison, while numerically right, misclassifies the obstacle. Exposure is *not* the limiting factor — one transit is expected per 0.86 km^2·yr — and the kinetic energy is macroscopic at 148.6 J. What fails is the coupling.

XXIV. The Dark-Matter Candidate

Evidence level: CONJ throughout

Every proposition in this section is a conjecture. The chain has four links and each multiplies the speculation of the one before it. Nothing here is offered as a prediction of the framework, and the section is included because a programme that states its speculative endpoint openly can be argued with, while one that leaves it implicit cannot.

This is where the framework ends, and it is the only place where it makes a cosmological claim. The claim is that dark matter is the ash of black-hole evaporation: Planck-scale remnants left behind when the quantum-to-classical transition runs backwards to completion.

A. The chain, with each link labelled

1. **[CONJ]** A black hole is the *inverse* of the quantum-to-classical transition, $d\gamma'/dt < 0$: Hawking radiation erodes the horizon and the object evolves from a maximally decohered state towards a quantum remnant (Sec. XXIII H).

2. **[CONJ]** Evaporation halts at the SDF Planck-scale boundary $M_{\rm min}$ rather than proceeding to zero mass.

3. **[CONJ]** The remnant is cosmologically stable.

4. **[CONJ]** Remnants constitute the dark-matter density, $\Omega_{\rm remnants} = \Omega_{\rm DM}$.

Link 2 is the load-bearing one and it is the weakest. The Hawking temperature at $M_{\rm min}$ is

$$T_H(M_{\rm min}) = \frac{\hbar c^3}{8\pi G\, M_{\rm min}\, k_{\rm B}} \approx 2.0 \times 10^{31}\ \text{K}, \qquad (182)$$

some thirty-one orders of magnitude above the CMB. *No derivation is offered for a barrier that would arrest evaporation against that gradient.* We state this as the first falsifier rather than burying it.

B. An exact identity hidden behind an approximation

The corpus quotes the minimum mass as $M_{\rm min} \approx 0.28\, m_P$. That approximation conceals a closed form. Substituting $\ell_P = \sqrt{\hbar G/c^3}$ and $m_P = \sqrt{\hbar c/G}$ into Eq. (180),

$$\frac{M_{\rm min}}{m_P} = \frac{\ell_P c^2}{2G\sqrt{\pi}}\sqrt{\frac{G}{\hbar c}} = \frac{1}{2\sqrt{\pi}} = 0.282\,094\,791\,8\ldots \qquad (183)$$

TABLE XLII. The candidate, evaluated. All quantities follow from Eq. (183) and $\Omega_{\rm DM} = 0.12$, $h = 0.674$, $v_{\rm halo} = 220$ km/s. Reproducible from `sdf-lab/src/core/dark-matter.ts`.

Quantity	Value
Mass $M_{\rm min} = m_P/2\sqrt{\pi}$	6.140 μg
Radius $r_{\rm rem}$	9.05×10^{-36} m
Geometric cross section	2.57×10^{-66} cm^2
Hawking temperature at $M_{\rm min}$	2.00×10^{31} K
Number density $n_{\rm rem}$	1.67×10^{-25} cm^{-3}
Mean separation	1.82×10^{3} km
Kinetic energy at $v_{\rm halo}$	**148.6 J**
Flux	1.16×10^{-6} m^{-2}yr^{-1}
Exposure for one transit	**0.86 km^2·yr**
Mean free path in rock	1.2×10^{23} ly
P(interaction | Earth crossing)	1.1×10^{-32}

[EXACT]. The remnant mass is not a fitted or approximate quantity: it is $m_P/(2\sqrt{\pi})$, a pure number. Evaluating the left-hand side from tabulated constants reproduces it to a relative residual of 1.42×10^{-7}, and that residual is not a defect of the identity — it is the mutual inconsistency of ℓ_P and m_P as separately rounded CODATA entries. The identity itself is exact.

This matters for the section that follows: whatever one concludes about the cosmology, $M_{\rm min} = 6.140\ \mu$g and $r_{\rm rem} = 0.56\,\ell_P$ carry no free parameters.

C. Erratum: the published number density

Erratum 10 (Comoving number density)**.** *Paper 1 §11 reports* $n_{\rm rem} \sim \Omega_{\rm DM}\rho_{\rm crit}/(M_{min}c^2) \sim 8.8 \times 10^{-26}\ cm^{-3}$. *Two problems. First, the expression is dimensionally inconsistent:* $\rho_{\rm crit}$ *is already a mass density, so dividing by* $M_{min}c^2$ *leaves* $m^{-3}kg^{-1}s^2$. *The* c^2 *is spurious. Second, with the* c^2 *removed and* $\Omega_{\rm DM} = 0.12$, $h = 0.674$, *the correct evaluation is*

$$n_{\rm rem} = \frac{\Omega_{\rm DM}\,\rho_{\rm crit}}{M_{min}} = 1.668 \times 10^{-25}\ cm^{-3}, \qquad (184)$$

larger than the published figure by a factor 1.895. *Inverted, the published number density corresponds to* $\Omega = 0.0633$, *not to the* $\Omega_{\rm DM} = 0.12$ *it is stated to reproduce.*

We use the derived value below. The distinction is immaterial to the conclusion, which is what makes it safe to correct in passing rather than to defend.

D. The phenomenology, and why the usual verdict is the wrong one

Paper 1 dismisses direct detection by comparing the geometric cross section $\sigma_{\rm rem} \approx 2.6 \times 10^{-66}$ cm^2 with experimental sensitivities $\sigma_{\rm exp} \lesssim 10^{-45}$ cm^2, a gap of 20.6 orders of magnitude, and concludes that detection is "impossible with any foreseeable technology". The conclusion is right and the argument is misleading, because it invites the reader to picture a very weakly interacting particle. The remnant is not that.

Three numbers in Table XLII change the character of the problem.

a. The energy is macroscopic. A remnant at galactic velocity carries 148.6 J — eleven orders of magnitude above the threshold of any calorimeter, and comparable to a thrown stone. This is not a weakly interacting particle in the sense the cross-section comparison suggests. It is a projectile.

b. The exposure is affordable, and by a wider margin than we first reported. The published figure used the *cosmological mean* density to compute a *local* flux, which understates it by 6.96×10^5. A flux through a terrestrial detector is set by the local halo density, $\rho_{\rm local} \simeq 0.4$ GeV cm^{-3}, giving $n_{\rm rem} = 1.16 \times 10^{-13}$ m^{-3} and a mean separation of 20.5 km. At 220 km s^{-1} the transit rate is 0.81 m^{-2}yr^{-1} unidirectionally, or 0.20 m^{-2}yr^{-1} with the isotropic 1/4: one transit per 1.2 to 5.0 m^2·yr, not per 0.86 km^2·yr. *Rarity is emphatically not the obstacle* — a square metre for a year suffices, which strengthens rather than weakens the classification that follows.

c. The coupling is absent, and that is the obstacle. With a Planck-scale geometric cross section the mean free path in rock is 1.2×10^{23} ly, exceeding the observable universe by twelve orders of magnitude. The probability of a single interaction while traversing the entire Earth is 1.1×10^{-32}. The 148.6 J are carried straight through and never deposited.

The correct classification

The candidate is *cross-section-limited*, not *exposure-limited*, and the two demand opposite experiments. A larger or longer-running detector buys nothing: the remnants are already crossing it, of order one per square metre per year, and passing through without a single scatter. What is missing is not sensitivity to rare events but a coupling channel, and for an object of this size the only channel available is gravitational.

We record that our own first attempt at this classification reached the opposite verdict, by comparing available kinetic energy against a detector threshold without asking whether that energy could couple. Energy that cannot be deposited is not a signal. The error is instructive enough to leave documented.

E. What the remnant cannot do, stated before the falsifiers

A remnant at $M_{\rm min}$ carries $S = 1$ nat, that is $e = 2.718$ states and a capacity of $1/\ln 2 = 1.443$ bits. Unitarity would require it to retain the entropy of the hole that produced it: 1.049×10^{77} nats for a solar-mass progenitor, 2.653×10^{40} for a 10^{12} kg primordial hole, 1.940×10^{90} for Sgr A*. The storable fractions are 9.53×10^{-78}, 3.77×10^{-41} and 5.15×10^{-91}.

The shortfall is not an artefact of using the area law at $M_{\rm min}$, where we have just said it is unreliable. Take only $E = M_{\rm min}c^2$ and leave the radius free: the Bekenstein bound then requires $R \geq S_i\hbar/(2\pi M_{\rm min}c) = 9.57 \times 10^{41}$ m to hold the solar-mass entropy, which is 1.09×10^{15} times the radius of the observable universe. No choice of $O(1)$ convention moves seventy-seven orders of magnitude.

"A remnant exists at M_{min}" and "the remnant resolves the information paradox" are therefore mutually exclusive, except under bag-of-gold geometries, which have no stable construction, no production rate and no state definition [45–48]. The remnant can be an inert relic and a dark-matter candidate; it cannot be an information store. If information is conserved, it leaves in the radiation.

Nor is the barrier thermodynamically protected. For massless blackbody emission the Zurek ratio $\Delta S_{\rm rad}/|\Delta S_{\rm BH}| = 4/3$ exactly, so the generalised second law *permits* the relic to keep decaying; it does not forbid it. And the entropy cost of one Hawking quantum of energy $a\,k_BT_H$ is $\Delta S = 1 - a^2/(4S)$, which at $M_{\rm min}$ equals exactly 3/4 rather than the 1 that a naive one-nat-per-quantum argument would give — a 25% error precisely where the argument is needed.

F. Falsifiers

A conjecture is not excused from stating what would retire it. Four routes, two of which need no new instrument.

DM-F1 (available today)**: Primordial spectrum.** The scenario requires a collapse fraction $\beta_{\rm PBH} \sim 2.5 \times 10^{-17}$ at the Planck scale. CMB and PBH-abundance constraints bound that window; if they exclude it, the scenario falls without anything being detected.

DM-F2 (available today)**: Absence of annihilation.** The scenario predicts dark matter that is strictly collisionless and has *no* annihilation channel. A confirmed positive indirect detection (γ, ν, e^+) excludes it.

DM-F3 (requires instrument)**: Gravitational transit.** Since the obstacle is coupling rather than exposure, the correct class of search is one sensitive to a purely gravitational impulse over ∼km^2 areas — the regime of macroscopic dark-matter searches, not of low-threshold particle detectors.

DM-F4 (the closed link)**: Remnant stability.** The chain requires a barrier arresting evaporation at $M_{\rm min}$ despite $T_H \sim 10^{31}$ K. No derivation is offered. A demonstration that evaporation proceeds below $M_{\rm min}$ closes the chain at its central link, and this is the cheapest way to kill the scenario outright.

G. Verdict

The candidate has one attractive feature and one fatal one, and both should be stated without weighting.

Attractive: it introduces no new particle, no new field and no free parameter. The mass is $m_P/2\sqrt{\pi}$ exactly, the abundance follows from one cosmological number, and the scenario predicts precisely the phenomenology dark matter is observed to have — collisionless, non-annihilating, gravitationally clustered.

Fatal: it is decoupled. Not weakly coupled, decoupled — the interaction probability across the Earth is 10^{-32}. A hypothesis that predicts exactly what is observed and forbids any other observation is close to unfalsifiable by direct means, which is why DM-F1 and DM-F4 matter more than DM-F3. The scenario must be attacked at the primordial spectrum and at the stability of the remnant, not at the detector.

We therefore close the programme with this section rather than build on it. It is where the framework points, it is stated with its numbers and its falsifiers, and it remains **[CONJ]**.

Part VII

Closing

XXV. Open Problems

Entries of the register in Appendix A that remain open are listed there; the load-bearing ones — those that block a derivation the corpus relies on — are stated below in the sharpened form produced by the adversarial protocol, together with the escape routes that have been closed. The rest are recorded in the register with their summaries and are not narrated here, because each is a local defect whose repair is known and whose only cost is the decision to adopt it.

They are open in the strict sense: resolving them requires a decision about the physics, not about the wording. Sections XXV G to XXV J are new in this edition and are of a different kind — they are what the measurement of Sec. VIII leaves *indicated but not decided*, which is a status the previous edition had no place for.

A. The derivation of Axiom 3

Theorem 27 establishes that no filter-function route produces $\beta = 1 - \alpha/2$, and Erratum 1 removes the only independent microscopic derivation that had been offered. Yet $1 - \alpha/2$ is the only candidate that both respects the domain $\beta \in (0, 2]$ and admits $\beta < 1$, the framework's most robust empirical invariant.

Closed routes. Reparametrising through $\gamma(t)$ requires $d\gamma/dt$ of opposite sign to what Paper 2 §7 specifies. Reading the factor of two as an amplitude-versus-intensity convention fails, since squaring $e^{-\Gamma}$ rescales τ and not the exponent.

What would settle it. Either a derivation of $\beta < 1$ from a system–bath model that is not of filter-function type, or an explicit acceptance of Axiom 3 as a phenomenological postulate with the empirical fit as its sole warrant.

B. The two meanings of α

The symbol α denotes both the spectral exponent of the bath, through $\beta = 1 - \alpha/2$, and the anomalous-diffusion exponent of the biological extension, through $\beta = \alpha_{\text{diff}}/2$. The two relations are simultaneously satisfiable only if $\alpha_{\text{diff}} = 2 - \alpha$.

This is not merely notational. Applied to the published single-particle tracking data, with $\alpha_{\text{diff}} \in [0.32, 0.66]$, the two relations give $\beta \in [0.16, 0.33]$ and $\beta \in [0.67, 0.84]$ respectively: a factor of two to four on the framework's central parameter.

Status: closed. The standard bridge is the generalised Langevin equation with the fluctuation–dissipation theorem, which gives $\alpha = 1 - \alpha_{\text{diff}}$ rather than $2 - \alpha_{\text{diff}}$, and hence $\beta = (1 + \alpha_{\text{diff}})/2$. The published relation is refuted by its own limit: normal Brownian diffusion, $\alpha_{\text{diff}} = 1$, must give Markovian decoherence, and $\beta = \alpha_{\text{diff}}/2$ returns 0.5. See Erratum 9 and Table XXXII, where the corrected relation places all five measured systems inside the declared domain.

What remains open is narrower and belongs to the previous subsection: the scaling-law route still requires $n \approx 7$–16 for systems with $n \sim 10^{20}$ constituents.

C. The information-condensation fixed point

Paper 1 §11 defines information condensation by three conditions: $dD_f/d\gamma = 0$ at some γ^*, $d^2D_f/d\gamma^2 > 0$ there for stability, and a reduction of informational complexity $\mathcal{N}(\rho^*) < \mathcal{N}(\rho_1) + \mathcal{N}(\rho_2)$.

The first two are unsatisfiable by any SDF curve. The basic-properties theorem gives $dD/d\gamma > 0$ strictly for all $\gamma > 0$, so there is no interior fixed point; and for $\beta \leq 1$ the curvature is negative everywhere. A sweep of $2{\times}10^5$ samples over $\gamma \in (0, 200\tau]$ at $\beta \in \{0.5, 0.852, 1, 1.5\}$ returns zero points satisfying both.

Why the obvious repair fails. Replacing the interior fixed point by an asymptotic attractor with $d^2D/d\gamma^2 < 0$ makes the conditions satisfiable, but the resulting point coincides with the *global maximum* of the entropy along the trajectory—the entropy theorem gives $S \to \ln 2$ as $D \to L$—so that the adjective "anti-entropic" becomes literally false. Further, the inverse-function theorem establishes that D maps onto the half-open interval $[0, L)$, so the limiting state ρ^* is not in the range at all.

What survives. The third condition, which is informational rather than dynamical, and is the only formal basis the corpus offers for a tendency of information to concentrate. It should be retained and the dynamical language withdrawn.

Note. Paper 1 §12 already states that the flow has no interior fixed points. The inconsistency is therefore internal to the corpus—§11 against §12—rather than a new discovery.

D. Bidirectionality against the unique attractor

The axiom set accompanying Paper 2 declares transitions in both directions, quantum to classical and classical to quantum, to be permitted. Paper 1 §12 establishes a unidirectional flow with a unique attractor at $(D, C) = (1, 0)$. Both cannot hold on the same state space.

The route the corpus hints at. Unidirectionality in γ combined with bidirectionality in t, exploiting a non-monotonic $\gamma(t)$. Paper 2 Appendix E states explicitly that the map $\gamma(t)$ can become non-monotonic for systems resonantly coupled to structured environments—which supports the route but does not formalise it.

What must be decided. Whether a physical trajectory with $d\gamma/dt < 0$ on some interval can be constructed. If not, the bidirectionality axiom degrades to a conjecture, and the proposed resolution of the black-hole information problem, which rests on it explicitly, loses its support.

E. The reported spectral mode

Treated in Sec. XVII. It is listed among the open problems rather than the errata because withdrawing the mode leaves the emergence section of Paper 1 without its principal empirical claim, and what should replace it is a matter of judgement rather than arithmetic. The non-Gaussianity of the residuals is real and reproducible; whether it supports the transient-mode interpretation at all is undetermined.

F. A bridge between state space and physical space

The framework fixes two lengths at which exactly one bit is exchanged, and they live in different spaces. In state space, by Corollary 4, one bit is $\pi/3$ of Bures arc out of a total $\pi/2$. In physical space, the Bekenstein gradient used by Verlinde, $dS/dx = 2\pi k_B/\bar{\lambda}_C$, places one bit at

$$\Delta x = \frac{\ln 2}{2\pi}\,\bar{\lambda}_C = 0.110317800076\,\bar{\lambda}_C\,, \qquad (185)$$

a fixed fraction of the reduced Compton wavelength, independent of mass. Both are fixed fractions of the system's own scale. The question is whether they are the same quantity in two coordinates.

a. The natural bridge, and its refutation. Identifying the accumulated information of the SDF with the Verlinde entropy in units of k_B gives $(\gamma/\tau)^\beta = 2\pi x/\bar{\lambda}_C$, which reproduces *both* scales simultaneously from a single identification—residual 0 in Eq. (185) and 2.2×10^{-16} in the arc. Substituting into the master equation yields

$$D(x) = L\left[1 - \exp\left(-\frac{2\pi x}{\bar{\lambda}_C}\right)\right], \qquad (186)$$

in which β has disappeared: decoherence as a function of displacement would be a pure exponential for every system, the stretching being absorbed by the change of variable. That is a falsifiable prediction, and it fails. It places 99% decoherence of a free electron at 2.8×10^{-13} m, whereas electron interferometry observes interference with transverse coherence lengths of 10^{-7} m or more—a gap of five to six orders of magnitude. Rescuing it requires $(\gamma/\tau)^\beta = 2\pi\eta\, x/\bar{\lambda}_C$ with $\eta \lesssim 3 \times 10^{-6}$, a number that follows from no constant of the framework.

The identification fails for a reason that is visible once stated: the Bekenstein gradient is the entropy of a holographic *screen* as a test mass approaches it, not the decoherence of the mass. They are different entropies.

Status: attempted and failed; the structural observation survives. That each scale is a fixed fraction of the system's own scale is exact and requires no bridge. That the two are the same magnitude does not follow, and the natural route to making them so is excluded by experiment.

G. The mechanism behind the rise

That the echo curve rises is measured (Sec. VIII C). *Why* it rises is not. Coupling to a small number of two-level fluctuators is the standard explanation and is consistent with everything observed — it is qubit specific, which matches Birge ratios of 10 to 26, and it produces echo revivals, which is what the data show. But consistency is not identification: the measurement uses one fixed pulse sequence and a single delay grid, and no observable in it distinguishes a fluctuator from any other source of correlated low-frequency noise.

Status: **[HIP]**. *What would settle it.* Noise spectroscopy resolved in frequency — a CPMG family with the number of pulses swept, which maps the filter function onto the bath spectrum — on the same qubits and the same session. That experiment is standard, costs more shots than this one, and would either exhibit the fluctuator lines or exclude them. Until it is run, the mechanism is an indication, and the result of Sec. VIII neither depends on it nor supports it.

H. Whether the split is a property of the channel or of the device

The measurement separates two channels of the same qubit on *one* device in *one* session. Read as a statement about pure dephasing in superconducting qubits it would be a generalisation the data do not carry; read as a statement about this device it is safe and narrow. Nothing in the design distinguishes between the two readings, and it would be dishonest to let the wording drift toward the first.

Status: open by construction. *What would settle it.* The identical protocol on a second device of different fabrication, analysed with the same extractor and no parameter touched. The pre-registration for that run is already written, and it differs from the one used here in one respect that matters: the positive control is the synthetic single-rate twin, not the T_1 arm.

I. The estimator that was withdrawn, and what it would take to recover it

The distance to the pole $\beta = 1$ was the primary estimand of the run design. It is withdrawn (Sec. VIII F), not refuted: nothing shows the estimator to be wrong — the synthetic twin recovers $\beta = 1.0000 \pm 0.005$ — and nothing shows it to be right on this data either, because the control that was supposed to establish it rested on a premise, "T_1 is a single rate", which the same data falsify.

What would settle it. A repetition whose positive control is the twin. This is the cheapest of the open items and the one with the clearest protocol, and it is worth stating why it was not simply re-analysed instead: re-analysing a fired falsifier under a rule chosen after seeing the result is exactly the operation pre-registration exists to prevent.

J. Whether any exponent in this corpus survives its own goodness of fit

This is the uncomfortable one, and it is stated in full because the alternative is to leave it implicit. Section VIII B rejects the two-parameter Kohlrausch form at $\sigma/C \approx 0.011$. The earlier hardware sections (Secs. VI, VII) fit the same form to curves with far less statistical power per point, and there the form was *not* rejected.

Those two facts are compatible with two very different situations, and the present data cannot separate them: either the earlier curves are genuinely Kohlrausch and the new ones are not, or the earlier fits were never tested with enough power to reject anything. A non-rejection at low power is not evidence of adequacy — it is absence of evidence either way — and the corpus has been reading exponents as though it were.

Status: open, and it conditions the reading of every fitted β in Part II. *What would settle it.* Recomputing the goodness of fit of each published curve at its own precision and reporting it alongside the exponent, which requires no new hardware time — only the raw counts. For part of the corpus those counts exist. For the multistate run they do not, and that is recorded separately as D38.

K. A methodological remark

Five resolutions to these problems were proposed during this work and all five were rejected under adversarial review. Four failed for the same reason: they did not cite a passage of the corpus that already addressed the matter. In one case the decisive footnote appeared in the first section examined.

The lesson generalises beyond this framework. An audit protocol should require of anyone proposing a correction not merely that their algebra closes, but that they have actively searched for the passage which refutes them. The verification effort is asymmetric: checking that a proposal is self-consistent is cheap, and checking that it is consistent with everything already written is expensive—which is precisely why the second is the one that gets skipped.

XXVI. Method: The Discipline and What It Cost

A framework that audits itself has to say how. This section states the working rules of the programme and, for each, the case in this document where following it forced a result to be given up. Rules that never cost anything are not rules; they are decoration.

A. Build first, audit afterwards

The order matters and is not the intuitive one. A question that can be settled on paper is settled on paper *before* asking whether there are data to test it against, and the absence of prior literature is never a reason not to derive. That absence is a datum for the audit, not a filter on what may be attempted.

The case. A planning document of this programme assigned “low probability” to a question whose proof turned out to be four lines of elementary algebra, and the answer was affirmative. The cost of the wrong order was not a wrong result; it was a result not attempted. That is the failure mode this rule exists to prevent, and it leaves no trace in the literature, which is why it has to be written down.

B. Audit against data, never against authority

The consistency demanded is with measurements. Consistency with opinion, including our own prior opinion, is not demanded at all. That a result depreciates earlier work — anyone’s, ours included — is not a cost to be weighed in the balance.

The case. This edition contains a passage that was written, committed, and then removed for violating this rule. It disposed of the measured recoherence by noting that the phenomenon has a literature of its own — which is true, and irrelevant to whether the curve rises. Worse, it answered a question nobody had asked (*is the phenomenon new?*) and let that displace the one that had been pre-registered (*does the curve admit a positive rate density?*). The relevant check was done afterwards and properly: echo revivals from fluctuators have prior art; the Breuer–Laine–Piilo measure has prior art and *agrees with our verdict qubit by qubit*, which is corroboration by an independent criterion, not a demotion; complete monotonicity used as a test of classical representability on decoherence data did not turn up. Those are three separate facts and each is reported as one.

C. Independence is a feature, not a defect

Connection to prior work is optional. A result is not rewritten to fit anyone’s framework, and it is not downgraded for lacking a sponsor. The converse holds with equal force: it is not upgraded for being unprecedented either. Whether something is new and whether it is true are independent questions, and this document answers only the second.

D. The label is what makes the freedom possible

Every claim carries one of **[EXACT]**, **[THM]**, **[OBS]**, **[HIP]**, **[CONJ]** or **[ERRATUM]**, and the grading is what allows speculation to be published at all. A conjecture with a declared falsifier is work and is written with dignity. A conjecture labelled as a theorem is fraud. Build without limit; misrepresent nothing.

The case. Axiom 3 was published as a theorem. Theorem 27 showed no filter-function route reaches it, and a sign error removed the only alternative microscopic derivation. It was relabelled to postulate rather than defended, and the empirical fit that motivated it is now its sole warrant — stated as such, in the abstract, where it cannot be missed.

E. Coincidences are audited algebraically, and in that order

When two routes reach the same place, the first test is algebraic: substitute the definitions backwards and ask whether the two expressions are the *same function* of the free parameter. Only if they are not does the provenance of the constants become informative. The order is not a preference; skipping it produces false independence.

The case. $M_{\min} = m_P/2\sqrt{\pi}$ was claimed to be selected by two independent routes, one entropic and one kinematic, on the strength of their constants having different origins. The algebraic test gives $S_{\rm BH} \equiv 2\pi r_s/\lambda_C \equiv 4\pi(M/m_P)^2$ identically, residual $\leq 2.2 \times 10^{-16}$ over sixty decades: one equation with several faces. The "discriminator" built on that supposed independence was $1/S$ rewritten. Provenance of symbols decides nothing.

And the converse, which is the part usually got wrong. That a constant was *defined* by imposing a condition makes recovering that condition arithmetic, not news, and this document says so wherever it applies. But it does not drain the other routes of content: in a one-parameter family, two conditions selecting the same point are equivalent *as conditions*, and what is informative is whether they come from distinct physical principles. Both things are reported, never only the first.

F. Pre-registration, and the price of honouring it

The measurement of Sec. VIII was registered before any hardware time was spent: the question, the design, the analysis, five falsifiers and the consequence of each. Five did not fire. One did, and it took the primary estimand with it.

The temptation, once its premise proved false, was to argue the falsifier away. The argument is even sound as physics — T_1 is demonstrably not a single rate on this hardware, shown two independent ways. It is still not admissible, for a reason that has nothing to do with physics: a falsifier that can be annulled by the result it was meant to test is not a falsifier. What the false premise buys is a corrected pre-registration for the next run, and nothing else.

Two consequences are recorded rather than smoothed over. First, the partition between quantities that pass through the estimator and quantities that do not was drawn *after* seeing the data. Each row of it is mechanically checkable, but it was not written in advance, and it is labelled as post hoc in the place where it is used. Second, an earlier draft of this programme asserted that the falsifier which fired "did not come from the plan". It did — it is in §2.1 of the plan, with its clause. The assertion was false against our own document and is corrected here rather than deleted.

G. What the discipline does not buy

It does not make the results more general than they are. One device is one device. It does not convert a consistent mechanism into an identified one. And it offers no protection against the error that matters most — the question not asked — which is why the first rule is the one about building, not the one about auditing.

Nor does it make the audit self-certifying. Of the five resolutions proposed to the open problems during this work, all five were rejected under adversarial review, four of them for the same reason: they did not cite the passage of the corpus that already addressed the matter. Checking that a proposal is self-consistent is cheap; checking that it is consistent with everything already written is expensive, and that asymmetry is exactly why the second gets skipped. The register in Appendix A is the instrument meant to catch what gets skipped, and two of its own entries were added by an adversarial pass over the audit itself — which is the strongest evidence available that the instrument works, and also that it does not work by itself.

XXVII. Discussion

A. Implications for Quantum Computing

The discovery of three decoherence regimes has practical implications for quantum error correction (QEC) [49] strategy selection:

1. **Exponential regime** ($\beta \approx 1$): Standard QEC codes (surface [50], Steane [51]) are optimal. Markovian noise assumptions hold and threshold theorems apply directly.

2. **Stretched regime** ($\beta < 1$): Non-Markovian codes or dynamical decoupling [52] may outperform standard QEC. The environment retains memory that can, in principle, be exploited to improve gate fidelities.

3. **Revival regime**: Active feedback protocols may partially recover information. The noise structure itself provides transient protection, as observed for W and BellChain topologies (Section IX).

Recommendation: Measure β for the target hardware *before* selecting a QEC strategy. The 10–50× variation in β between `ibm_fez` and `ibm_torino` (Section XXVII B) demonstrates that a single QEC prescription cannot be universally optimal.

B. Cross-Backend Discoveries

Phase 29 introduced a second IBM Quantum backend—`ibm_torino` (Heron r2, 133 qubits)—to complement the original `ibm_fez` (Eagle r1, 156 qubits) dataset. The comparison reveals striking differences that constrain the scope of universality.

1. The β anomaly

Across 400 new circuits on `ibm_torino`, the fitted stretched-exponential exponent satisfies

$$\beta_{\rm torino} \in [0.011,\ 0.047], \tag{187}$$

which is 10–50× smaller than the corresponding values on `ibm_fez`:

$$\beta_{\text{fez}} \in [0.30,\ 0.86], \qquad \beta_{\text{fez}}(n) = \frac{1.679}{n^{0.844}},\ R^2 = 0.987. \tag{188}$$

In the SDF interpretation, $\beta \to 0$ corresponds to *logarithmically slow* decoherence growth—i.e., ultra-stretched dynamics. The Heron r2 processor thus exhibits dramatically more non-Markovian behavior than Eagle r1 under identical circuit families (GHZ_n, $n \in \{3, \ldots, 12\}$).

Possible physical origins include:

- Different noise power spectral densities: Heron r2 may have stronger low-frequency correlations (α_{noise} closer to 2), producing more stretched dynamics.
- Improved coherence times (T_1, T_2) in Heron r2 shifting the experimental window into the early, subdiffusive tail of $D(\gamma)$.
- Distinct coupler architectures (tunable vs. fixed) modifying the effective system–environment coupling spectral density.

2. *THM-26.1 cross-backend tension*

The regime classification theorem (THM-26.1) predicts $\beta = 1 - \alpha/2$. On `ibm_torino`, the independently measured spectral exponent is $\alpha_{\text{spectral}} = 0.661 \pm 0.426$ (Welch PSD, 400 circuits, $n \geq 5$), which via THM-26.1 would predict $\beta \approx 0.67$. The *observed* $\beta \approx 0.015$ disagrees by a factor of $\sim$45.

Two resolutions are under investigation:

1. **Hardware-dependent offset:** Generalize THM-26.1 to
$$\beta(n, \text{hw}) = \beta_0(\text{hw}) \cdot n^{-\alpha(\text{hw})/2}, \tag{189}$$
where β_0 absorbs processor-specific noise characteristics. If the *functional form* is universal with hardware-dependent parameters, the theorem retains predictive power within each backend.
2. **Distinct α measures:** The Welch-PSD α_{spectral} characterizes *residual* noise after SDF subtraction, whereas the Lindblad-derived α in THM-26.1 describes the *bare* environment spectral density. These quantities need not coincide when the system–bath coupling is strongly renormalized.

A definitive resolution requires executing identical protocols on at least one additional backend (e.g., `ibm_marrakesh`, `ibm_osaka`).

3. *Implications for universality*

The $D(\gamma) + C(\gamma) = 1$ conservation law holds exactly on both backends (650 circuits, $\sim 5.5 \times 10^6$ shots). The *functional form* $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$ fits well for $n \leq 10$ on both processors. We therefore maintain the claim that the SDF *structure* is universal, while acknowledging that the *parameters* (β_0, τ) are backend-dependent quantities determined by the hardware noise environment.

C. Observable Emergence from the SDF

A central result of Phase 30 is the derivation of a master formula connecting the decoherence function $D(\gamma)$ to the expectation value of an arbitrary observable.

1. *The observable decay formula*

For any quantum state undergoing decoherence, the density matrix admits the decomposition

$$\rho(\gamma) = \big(1 - D(\gamma)/L\big)\, \rho_{\text{coh}} + \big(D(\gamma)/L\big)\, \rho_{\text{mix}}, \tag{190}$$

where $\rho_{\text{coh}} = |\psi_{\text{target}}\rangle\langle\psi_{\text{target}}|$ is the target coherent state and $\rho_{\text{mix}} = \mathbb{I}/2^n$ is the maximally mixed state. It follows immediately that for any observable O:

$$\boxed{\langle O\rangle(\gamma) = \langle O\rangle_{\text{coh}} - \frac{D(\gamma)}{L}\, \Delta O,} \tag{191}$$

where $\Delta O \equiv \langle O\rangle_{\text{coh}} - \langle O\rangle_{\text{mix}}$ is the "quantum contrast" of the observable.

2. *Significance*

Equation (191) establishes that a *single* function $D(\gamma)$ is sufficient to predict the decoherence-induced decay of *any* Hermitian observable—including energy, purity, entanglement witnesses, Pauli expectation values, and correlation functions. The observable dependence enters only through the constant ΔO, while the γ-dependence is entirely captured by the universal stretched exponential:

$$\begin{aligned} \langle O\rangle(\gamma) &= \langle O\rangle_{\text{coh}} - \frac{\Delta O}{L}\, L\big[1 - e^{-(\gamma/\tau)^\beta}\big] \\ &= \langle O\rangle_{\text{coh}} - \Delta O\big[1 - e^{-(\gamma/\tau)^\beta}\big]. \end{aligned} \tag{192}$$

This result resonates with Zurek's quantum Darwinism program [53], in which objective classical properties emerge from redundant encoding in the environment. Here, the SDF provides the *rate law* for how quickly this encoding proceeds: all observables lose their quantum character at the same stretched-exponential rate set by (β, τ), differing only in the amplitude ΔO.

The three SDF regimes yield distinct observable signatures:

- $\beta \approx 1$: Exponential decay with single timescale τ — standard T_1/T_2 behavior.
- $\beta < 1$: Heavy-tailed decay with a "long tail" persisting beyond τ — mesoscopic and potentially biological coherences.
- Revival regime: Non-monotone $\langle O\rangle(\gamma)$ with oscillations — observable-level signature of information return.

D. Information Conservation — Three Levels

A recurring source of confusion in the decoherence literature is the precise sense in which "information is conserved." The Phase 30A analysis identifies three logically distinct levels, of which only the second constitutes a physical theorem (see Section XIII for complete proofs).

1. **Level 1 — Algebraic identity.** The equation $D(\gamma) + C(\gamma) = 1$ follows from the *definition* $C \equiv 1 - D$ and carries no dynamical content. It should not be cited as a conservation law.

2. **Level 2 — Von Neumann conservation (THM-27.1).** For a system S coupled to an environment E evolving unitarily, the Araki–Lieb equality for globally pure states gives $S(\rho_S) = S(\rho_E)$, yielding

$$\widetilde{\mathcal{I}}_S(\gamma) + \widetilde{\mathcal{I}}_E(\gamma) + 2\,E(\gamma) = \ln d_S + \ln d_E = \text{const}, \quad (193)$$

where $\widetilde{\mathcal{I}}_X = \ln d_X - S(\rho_X)$ is the local information and $E(\gamma) = S(\rho_S(\gamma))$ the entanglement entropy. This is a genuine conservation law rooted in global unitarity, proven rigorously in THM-27.1 (Section XIII A 2).

3. **Level 3 — Operational recovery (CONJ).** Whether an observer with access to both S and E can operationally *recover* the initial quantum state remains a conjecture that depends on the structure of the decoherence channel and the available recovery operations.

The experimental finding $D + C = 1$ (exact, 650 circuits) validates Level 1 as a *consistency check* on our measurement protocol. The deeper Level 2 conservation tells us that the information seemingly lost from S is faithfully transferred to E—it is relocated, not destroyed. This perspective connects to the black hole information paradox [54, 55]: if decoherence is the mechanism by which information enters Hawking radiation, then Level 2 conservation guarantees that the information survives in principle. The SDF framework adds a quantitative *rate law*: the information transfer follows the stretched exponential $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$, with β determined by the noise spectrum of the gravitational environment.

E. Biological Decoherence Interpretation

The Λ-hierarchy framework suggests biological systems operate in the $\Lambda \sim 10^{-30}$ regime, predicting:

- $\tau_{\text{bio}} \sim 10^{-14}$ s (femtosecond coherence)
- $\beta_{\text{bio}} \sim 0.7$–$0.8$ (moderate non-Markovianity)

This is consistent with observed quantum effects in photosynthesis [18] and enzyme catalysis [19], where sub-picosecond coherences persist in warm, wet environments. The observable decay formula (191) predicts that biologically relevant observables (e.g., exciton transfer probability) should exhibit the heavy-tailed signature of $\beta < 1$, distinguishable from simple exponential decay in ultrafast spectroscopy experiments.

F. Connection to Paper 1 Predictions

Paper 1 made several predictions that this work validates, refines, or extends. Table XLIII summarizes the current status including Phase 29 and Phase 30 results.

The most notable change is the failure of the ECME (Emergence via Collective Multi-scale Extrapolation) prediction scheme, which aimed to predict individual state decoherence curves from collective parameters. With $R^2 = -0.39$, the ECME model performs substantially worse than a constant baseline, indicating that state-specific (topology-dependent) parameters are essential and cannot be replaced by a collective description.

G. Falsifiability and Future Tests

The framework makes the following testable predictions, including new predictions from Phases 29–30:

1. **PRED-26.1 (LIGO):** $\beta_{\text{inspiral}} \neq \beta_{\text{ringdown}}$ due to different noise spectra during each phase. Testable with GWTC-3+ data [12].

2. **PRED-26.2 (Majorana):** $\beta \approx 0.95$–0.99 for topologically protected states. Testable with current nanowire experiments.

3. **PRED-26.3 (DFS):** Container states maintain $D < 0.05$ while internal systems complete the decoherence transition. Testable with trapped-ion or superconducting platforms.

4. **PRED-26.4 (Scaling):** $\beta(n) = \beta_0/n^\alpha$ should hold for other state families (Dicke, W) with different parameters. Testable with extended IBM experiments.

5. **PRED-30.1 (Observable decay):** All Hermitian observables decay as $\langle O\rangle(\gamma) = \langle O\rangle_{\text{coh}} - \Delta O\,[1 - e^{-(\gamma/\tau)^\beta}]$. Testable via simultaneous tomography of multiple observables versus γ.

6. **PRED-30.2 (Hardware β):** β is inversely correlated with processor coherence quality ($\beta \propto 1/T_2^*$). Testable with a multi-backend survey.

7. **PRED-30.3 (Crosstalk threshold):** SDF model validity breaks down at $n > n_c \approx 10$ due to crosstalk contributions scaling as $\sim n^2$. Testable by systematic n-sweep on any superconducting processor.

8. **PRED-30.4 (BLP backflow): [ERRATUM] Falsified in this edition.** The prediction was that the Breuer–Laine–Piilo measure [13] satisfies $\mathcal{N} > 0$ for systems with $\beta < 0.8$. It was tested by exactly the protocol it named (Sec. VIII E) and fails in both directions: $\beta < 0.8$ is neither sufficient nor necessary, and two arms of the same device whose median β differ by 0.006 return $\mathcal{N} = 0.00827$ and $\mathcal{N} = 0.00000$. The prediction is withdrawn; the measure itself is unaffected and is used in Sec. VIII D as an independent criterion.

9. **PRED-30.5 (Topology $\to \alpha$):** The effective spectral exponent depends on entanglement topology: $\alpha_{\text{GHZ}} \neq \alpha_{\text{Cluster}} \neq \alpha_{\text{W}}$. Testable with Phase 26E-class experiments.

TABLE XLIII. Validation of Paper 1 predictions (updated with Phase 29/30 data).

Prediction	Status	Evidence	Notes
$\beta < 1$ universal	**Refined**	650 circuits	Regime-dependent; $\beta_{\text{torino}} \ll \beta_{\text{fez}}$
$D + C = 1$	**Validated**	650 circuits, 2 backends	Exact to numerical precision
$\beta = 1 - \alpha/2$	**Partial**	ibm_fez only	Cross-backend tension on ibm_torino
$dI/d\gamma = 0$	**Validated**	γ-sweep	THM-27.1 now proven from unitarity
$\langle O \rangle$ decay $\propto D(\gamma)$	**New (Phase 30)**	Derived	Eq. (191)
$M_{\min} = f(\beta)$	Unchanged	—	No new data
ECME multi-scale	**FAIL**	$R^2 = -0.39$	Individual prediction poor

Predictions 1–4 were stated in the original submission; predictions 5–9 are new. Each is independently falsifiable with current experimental technology.

H. Limitations and Caveats

We acknowledge the following limitations, organized by severity.

1. *Experimental limitations*

1. **Two-backend scope.** All quantum hardware data originate from two IBM superconducting processors (`ibm_fez`, Eagle r1; `ibm_torino`, Heron r2). Generalization to trapped-ion, photonic, or neutral-atom platforms remains untested.

2. $n > 10$ **model failure.** For $n = 12$ qubits on `ibm_torino`, the pure SDF fit yields $R^2 = -1.64$, indicating that the single stretched exponential is *worse* than a constant mean predictor. This likely reflects crosstalk contributions absent from the SDF model. A proposed extension replaces n by an effective qubit count $n_{\text{eff}} = n/[1 + (n/n_c)^p]$ — an unpublished internal analysis, not reproduced here and not citable — capping the effective system size above a crosstalk threshold $n_c \approx 10$.

3. **ECME failure.** The multi-scale collective prediction scheme gives $R^2 = -0.39$, demonstrating that aggregate parameters cannot substitute for topology-resolved fitting.

4. **Spectral exponent uncertainty.** The Welch-PSD based $\alpha_{\text{spectral}} = 0.661 \pm 0.426$ has a bootstrap 95% CI width of $\sim$1.5, limited by the 50-delay time series length. Improved estimates require ≥ 200 delays or multitaper methods [56].

2. *Theoretical limitations*

5. **Cross-backend tension in THM-26.1.** The relation $\beta = 1 - \alpha/2$ is validated on `ibm_fez` but fails cross-backend: $\beta_{\text{torino}} \approx 0.015$ would require $\alpha \approx 1.97$, inconsistent with the measured $\alpha_{\text{spectral}} \approx 0.66$. The theorem may require a hardware-dependent generalization (Eq. 189).

6. $\tau(\Lambda) = \tau_{\mathbf{P}}/\Lambda$ **is an axiom.** Phase 30E established that this scaling relation is *irreducible* within the SDF framework—logically equivalent to the uncertainty relation $\Delta E \cdot \Delta t \geq \hbar$. It is postulated and empirically validated across three energy scales (nuclear, atomic, molecular), but not derived from a deeper principle.

7. **Revival mechanism.** THM-26.3 (revival conditions for W and BellChain states) remains conjectural. Phase 30E provides a derivation from the Dicke algebra [8] (permutation symmetry $\to$ resonance frequency), but a complete proof requires explicit solution of the coupled system–bath dynamics.

8. **Circularity residual.** Although Phase 26K introduced an independent FFT-based measurement of α, the large uncertainty (± 0.43) means the circularity between α and β is ameliorated rather than eliminated.

9. **Observable decomposition assumption.** Equation (190) assumes that the decohered density matrix can be written as a convex combination of ρ_{coh} and ρ_{mix}. This holds exactly for depolarizing channels and approximately for dephasing-dominated dynamics, but may break down for strongly non-depolarizing channels.

3. *Scope limitations*

10. **GHZ-centric topology.** The majority of circuits (both backends) use GHZ-family states. Extension to arbitrary entanglement topologies (cluster, toric code, MERA-inspired) is needed to test PRED-30.5.

11. **No real-time dynamics.** All experiments use a parameterized delay γ rather than monitoring decoherence in real time. The backflow prediction (PRED-30.4) requires a time-resolved BLP protocol that has not yet been implemented.

XXVIII. Conclusion

This document closes the programme. What follows separates what it established from what it did not, and states both without softening either.

A. What survives

Three results survive as theorems or exact identities, and a fourth as a measurement. They are the ones usable by somebody who owes this framework nothing.

The Λ-hierarchy is not an irreducible postulate: it reduces to $\tau = \hbar/E$ and $\ell = \bar{\lambda}_C$, so the spatial scale assigned to a system is exactly its reduced Compton wavelength. The accessible information collapses to $I_Q = (\gamma/\tau)^\beta$, which yields closed forms where the corpus had none and locates a distinguished point at $D/L = 1/2$ where four independent properties coincide — one bit exactly, the entanglement sudden-death point for $C_0 = 1$, the maximum of $D\bar{C}$ at $L^2/4$, and a Bures arc of $\pi/3$. None of the four is a definition, and the point is invariant in β, τ and L.

Theorem 27 states that the temporal exponent of dephasing cannot be changed by any pulse sequence; only the amplitude responds.

Theorem 36 is the transferable one. Pollard's criterion, used as a licence condition rather than a curiosity, converts a fitted exponent into a proof about what the underlying process cannot be.

Empirically, $D + C = 1$ holds to machine precision across 650 circuits on two backends. The claim that accompanied it in earlier editions — that $\beta < 1$ is an invariant no platform violates — is **withdrawn**. It is false against this corpus's own §8 table, where locally entangled states report $\beta \approx 1.1$, and the measurement of Sec. VIII removes the reading of such exponents altogether wherever the fit is inadmissible.

The fourth surviving result is that measurement, and it is empirical rather than exact. Applied directly to a device, without fitting, the criterion separates two channels of the same qubit: energy relaxation admits a positive rate density, pure dephasing does not, and the cause — an echo curve that rises, with the control at zero — is read off the counts. It is the narrowest claim in the document, one device and one session, and the strongest evidence in it.

B. What does not

Axiom 3 has no derivation. The filter-function route cannot produce it for any pulse sequence, and correcting a sign error removes the only independent microscopic route offered. It is a postulate with a good fit.

Two published numbers did not follow from their inputs, one reported spectral mode lies below the Rayleigh limit of its own data, and the published comoving density of remnants corresponds to $\Omega = 0.063$ rather than the 0.12 it is stated to reproduce.

More instructive than any of those: our own first correction of the LIGO significance was wrong in the same manner as the number it corrected, and the originally published value was closer to the defensible one than our correction of it. During final revision a further error was found in this paper's own tabulation of threshold crossings, by an external reader who declined to accept a table that did not reconcile. And the classification of the dark-matter candidate in Sec. XXIV reached the wrong verdict on first attempt, by comparing available energy against a detector threshold without asking whether that energy could couple. *An audit is not automatically right because it is an audit.*

Twenty discrepancies remain open of forty-four recorded (App. A), each listed with the observation that would close it. They are open because we could not close them. One, D40, was closed by going back to hardware; two, D42 and D44, were falsified there — one of them a prediction this framework had made itself; and one, D43, was opened by the same measurement against the corpus as a whole — exponents were read across this work without reporting whether the fits that produced them were admissible.

C. The inference this work does not license

A stretched exponential is widely taken as a signature of complex or quantum behaviour. Theorem 36 says the opposite in the regime where the framework fits best: $\beta < 1$ certifies that the data are reproducible by a positive distribution of classical exponential rates.

We state this plainly because the inverse reading is the natural one and it is wrong. We state just as plainly that the corpus is not uniformly in that regime, contrary to what an earlier version of this section asserted: the locally entangled states return $\beta \approx 1.1$, and for them the same theorem forbids the mixture representation rather than certifying it.

And this edition adds a second proviso that cuts deeper than the first, because it applies to both readings at once. The theorem takes a β as input and the input has to be earned: where the data are precise enough to test the fit, the two-parameter form is rejected (Sec. VIII B), and an exponent extracted from a rejected model certifies nothing in either direction. The question the criterion answers is about the *curve*. When the curve can be tested directly, it should be, and the fitted exponent is at best a proxy for doing so.

D. The decisive test, which we do not perform

Section XXII B identifies one prediction that existing literature can already refute: on a single sample, the anomalous-diffusion exponent and the stretched-relaxation exponent must satisfy $\beta = (1 + \alpha_{\text{diff}})/2$, with $\beta \geq 1/2$ as a corollary. Both quantities are routinely tabulated, by groups with no stake in this framework, for overlapping sample sets. No new experiment is required.

We do not perform it here: this document is an audit, and an audit that also supplies the confirmation of its own subject is not an audit. Whoever performs it settles the matter in either direction, and both directions are informative. If the relation holds, the framework acquires the one thing it lacks — a confirmed prediction on data its authors did not select. If it fails, a line closes, which this programme has treated throughout as a result of equal standing.

E. On the endpoint

The programme ends at a conjecture, and it is worth being explicit about what that means. The dark-matter candidate of Sec. XXIV introduces no new particle, no new field and no free parameter: the remnant mass is $m_P/2\sqrt{\pi}$ exactly, and the predicted phenomenology is precisely what dark matter is observed to have. It is also decoupled — the probability of a single interaction while crossing the Earth is 10^{-32} — which makes it close to unfalsifiable by direct means and means it must be attacked at the primordial spectrum and at the stability of the remnant instead.

A hypothesis that explains what is seen and forbids anything else being seen is the most seductive kind and the least useful. We state it, give it its numbers, give it four falsifiers, and stop there.

F. On the method

The result we regard as most transferable is negative. Five resolutions were proposed for the open problems and all five were rejected, four because they failed to cite a passage that already addressed the matter. Three further errors were caught in this paper itself after it was written, each by refusing to accept a number that did not reconcile.

A framework of this size cannot be audited by its author reading it, nor by an auditor checking algebra alone. It requires a protocol in which proposing a correction carries the burden of having searched for its own refutation, and in which a negative result — a line closed, a hypothesis retired, a number withdrawn — is recorded with the same standing as a positive one.

To that protocol this edition adds pre-registration, and reports what it cost. The measurement of Sec. VIII declared its question, its design and six falsifiers before any hardware time was spent. Five did not fire. One did, and it took the run's primary estimand with it — the distance to the pole $\beta = 1$, which was the quantity the design was built around. Its premise turned out to be false, which is a good argument and an inadmissible one: a falsifier that the result can annul is not a falsifier. The estimand is withdrawn and the corrected pre-registration is published in Sec. XXV I for whoever runs it next.

Section XXV is the list of places where that burden could not be discharged, and Sec. XXVI is the account of the rules that produced the list. This section is the statement that the list is the honest end of the work rather than a gap in it.

Acknowledgments

The audit was carried out with an adversarial protocol in which each proposed resolution was submitted to independent refutation before adoption. Five resolutions were proposed and all five were rejected. Three further errors were caught in this document after it was written — one of them by an external reader who declined to accept a table that did not reconcile. I regard this as the central methodological result of the work.

A. Register of Documented Inconsistencies

Each entry is verifiable by citing a source file and LaTeX label, and each is covered by a regression test in the accompanying implementation. Status codes: R = resolved, O = open, C = closed (verified not to be an inconsistency), I = invalidated.

a. On the three entries the new measurement moves. D40 recorded that the multistate experiment measured relaxation rather than dephasing; its stated closing condition was a design that separates the two under a single definition, which is what Sec. VIII does, and it is closed on that basis and on no other — in particular D38 and D39 are *not* closed by it, since their closing conditions name that experiment specifically and require an observable that sees entanglement. D42 and D44 are the first entries in this register retired by measurement rather than by algebra: §20 states that revival phenomena are slowdowns and not reversals, and that no local operation reverts the arrow; a single X pulse reverts 67% of the free-induction dephasing, and the echo curve rises with the control at zero. D43 is opened by the same measurement against the corpus as a whole, and is the most consequential of the three because it conditions how every fitted exponent in Part II may be read.

b. Distribution. Forty-four entries: twenty-two resolved, twenty open, one closed, one invalidated. Of the open entries, thirteen are of high impact. Three cases are ones in which raw data contradict the text rather than the algebra being inconsistent — D18, D40 and D42 — and the last two of those were produced by going back to hardware, which is the only instrument in this register that can settle a question the corpus cannot settle with itself.

The proportion is worth stating plainly rather than in a footnote: after two audits and one dedicated experiment, roughly half the register is still open. That is the honest state of the corpus, and it is the reason the claims of Sec. I C are ordered by confidence and read in that order.

c. On the last two entries. D6 and D27 were both added after the document was otherwise finished, and both by the same mechanism: an adversarial pass over the audit itself. D6 had been recorded as resolved; reopening it followed from noticing that the register's own text called the resolving bridge "a physical hypothesis", and that Sec. XXII B designates the test of that same bridge as the decisive experiment of the programme. One does not test what is resolved. D27 is an internal contradiction of the present document — an appendix inherited from Paper 1 whose derivation Theorem 27 refutes — and it went unnoticed through the entire consolidation. Both are recorded here rather than quietly corrected, because the register is the instrument that is supposed to catch exactly this.

d. On the closed entry. D7 is retained in the register rather than deleted. An item that was investigated and

TABLE XLIV. Register, entries D1–D27, as of the consolidated edition. Impact is assessed by whether the item affects a published number, a load-bearing derivation, or neither. Entries added afterwards are in Table XLV.

ID	Status	Impact	Summary
D1	R	low	Transition rate written with and without L in the exponent
D2	R	medium	Two incompatible Bloch-vector parametrisations
D3	R	medium	Three expressions for the single-qubit von Neumann entropy
D4	R	high	Two definitions of I_Q with opposite monotonicity
D5	R	low	Dimensionally incorrect exponent in the gravitational crossover mass
D6	**O**	high	α denotes two distinct exponents; factor 2–4 on β
D7	**C**	low	Biological information bound: cutoff is declared; not an inconsistency
D8	R	medium	$\gamma_{1/2}$ decreases as $\beta \to 0$; published prose asserts the opposite
D9	**O**	high	Filter-function route cannot produce $\beta < 1$ (Theorem 27)
D10	R	medium	Revival regime lies outside the domain of Axiom 2
D11	R	high	$n^* = 1.848$, not the published 2.6
D12	R	high	Ringdown significance is 12.5σ with PDG scaling, not 13.8σ
D13	R	low	Spurious factor of two in the hierarchy derivation
D14	**O**	high	Condensation fixed point unsatisfiable; repair breaks the anti-entropic reading
D15	**O**	high	Bidirectionality axiom against the unique attractor
D16	R	high	Four expressions for D; canonical form fixed as the infidelity
D17	**I**	medium	$d^2 I_Q/d\gamma^2 < 0$ for $\beta < 1$; published criterion withdrawn
D18	**O**	high	Reported spectral mode below the Rayleigh limit of its own data
D19	R	high	Sign of the fluctuator coupling exponent
D20	R	medium	Basic-properties theorem requires the hypothesis $\beta > 0$
D21	**O**	medium	I_Q computed from absolute D; the invariant lives in D/L
D22	**O**	low	Two-system compensation is a tautology, not a law
D23	R	high	Declared biological domain does not overlap §2c diffusion data
D24	**O**	medium	§2b declares σ as von Neumann entropy, computes $H_2(D)$
D25	**O**	low	n-qubit entropy expansion written in reverse order
D26	**O**	high	β not universal across events; the catalogue rejects a single value
D27	**O**	high	App. C derives $\beta = \alpha/2$, then asserts $\alpha + 2\beta = 2$ to reach $1 - \alpha/2$

TABLE XLV. Register, entries D28–D44, added after the consolidated edition was deposited. D40 is closed by the measurement of Sec. VIII; D42 and D44 are falsified by it; D43 is opened by it.

ID	Status	Impact	Summary
D28	R	medium	T4 presented as a gate; $\beta_{\mathrm{loc}} \leq 1$ is diagnostic, not a licence
D29	**R**	high	Universal claim that every platform reports $\beta < 1$ — *withdrawn in this edition, §1 and §26*
D30	**O**	medium	The range $[0.5, 1]$ fails at both ends, and the Dicke fits give $L > 1$
D31	**O**	high	$D + C = 1$ asserted and denied as a conservation law in the same corpus
D32	**O**	high	M_{min} derivation says "$S = 1$ bit" and computes one nat; nothing arbitrates
D33	**O**	medium	$M_{\mathrm{min}} = f(\beta)$ published as a prediction, normalised by construction
D34	**O**	high	The clock τ is unfixed, and $\beta = 2/3$ versus $1/2$ depends on it
D35	R	high	Cosmological mean density used for a local flux: factor 6.96×10^5
D36	R	medium	Missing attribution, and a published falsifier not recorded
D37	R	high	The remnant cannot resolve the information paradox; App. H implied it could
D38	**O**	high	Two distinct observables under one column; raw counts of that run not kept
D39	**O**	high	The multistate observable is blind to entanglement by construction
D40	**R**	high	Root cause: that experiment measured T_1, not dephasing — *closed by Sec. VIII*
D41	**O**	medium	The Fourier side of the pole is not empirically accessible
D42	**R**	high	§20 asserts revivals are slowdowns and that no local operation reverts — *falsified by measurement*
D43	**O**	high	Exponents read across the corpus without reporting goodness of fit (Sec. XXV J)
D44	**R**	high	PRED-30.4, and the claim that $\mathcal{N}$ follows from a fitted β — *falsified by measurement*

found not to be a defect carries information—specifically, that the biological bound has a declared cutoff and that its order of magnitude is robust across the declared parameter range—and removing it would invite the same objection to be raised again.

B. Reproducibility

Every numerical statement in this paper is reproducible from an accompanying implementation, which is deliberately dependency-free so that it can be audited without a toolchain.

1. The implementation

The core is written in TypeScript with no external dependencies and executes identically in a server runtime and in a browser, so that an interactive exploration and a batch verification cannot diverge numerically. It comprises:

- A registry of **138 equations**, each carrying source paper, source file, original LaTeX label, evidence level, declared domain and an executable implementation. An equation lacking any of these does not enter the registry.
- **354 regression tests**, including at least one per documented inconsistency, which assert the *actual* behaviour rather than the published assertion wherever the two differ.
- **37 published-number checks** that recompute values from the equations and report the relative deviation. Checks documenting an error pass when the discrepancy is reproducible.
- A catalogue of **26 hierarchy tiers** spanning 38.2 decades of physical τ, with tiers lacking an absolute timescale flagged as normalised rather than silently assigned $\tau = 1$.
- The **spectral-resolution audit** of Sec. XVII as a reusable primitive, applicable to any claim of the form "a mode was detected at f".

2. Selected verifications

3. Reproducing the audit

The full suite runs in under one second on commodity hardware. Two entry points are provided: a test runner that executes all regression tests and published-number checks, and a service exposing the registry, the tier catalogue, the inconsistency register and the fitting routines over HTTP.

The adversarial refutation described in Sec. XXV was carried out with eleven independent agents—five attacking the algebra of each proposed resolution, five attacking its consistency with the corpus under a requirement to cite file and label, and one auditing for omissions. Their findings are recorded in full alongside the implementation.

4. Scope and limitations of the audit

Three limitations should be stated.

First, the audit covers the equations and the numbers derived from them. It does not re-execute the quantum hardware experiments, and it takes the raw data files as authentic—which they appear to be, with complete circuit counts, exact shot numbers and recorded job identifiers.

Second, only one section was audited against its data files in the manner of Sec. XVII. The same treatment of the simulation, hardware and scaling sections is outstanding, and given what a single such audit produced, it would be imprudent to assume the others are clean.

Third, the five open problems of Sec. XXV are open. This paper sharpens their statement and closes several escape routes; it does not solve them.

C. Mathematical Derivations

This appendix provides the detailed mathematical derivations underpinning the core results of the SDF framework.

1. Lindblad derivation of $\beta = 1 - \alpha/2$

I begin from the standard Lindblad master equation for the density operator ρ:

$$\frac{d\rho}{dt} = -i[H,\rho] + \sum_k \gamma_k(t)\left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\}\right), \tag{C1}$$

where H is the system Hamiltonian, L_k are the Lindblad (jump) operators, and $\gamma_k(t)$ are time-dependent decoherence rates determined by the noise power spectral density $S(\omega)$.

For a noise spectrum with $1/f^\alpha$ character,

$$S(\omega) \propto \frac{1}{\omega^\alpha}, \qquad 0 < \alpha < 2, \tag{C2}$$

the effective decoherence rate is obtained via Fourier transform:

$$\gamma(t) = \int_0^\infty S(\omega)\, e^{-i\omega t}\, d\omega \;\propto\; t^{\alpha/2-1}. \tag{C3}$$

The integrated decoherence function is

$$D(t) = 1 - \exp\left(-\int_0^t \gamma(t')\, dt'\right). \tag{C4}$$

Evaluating the integral:

$$\int_0^t t'^{\,\alpha/2-1}\, dt' = \frac{t^{\alpha/2}}{\alpha/2} \;\propto\; t^{\alpha/2}. \tag{C5}$$

Thus, the decoherence function takes the stretched-exponential form

$$D(t) = 1 - \exp\left(-\left(\frac{t}{\tau}\right)^{\alpha/2}\right). \tag{C6}$$

Comparing with the SDF canonical form $D(\gamma) = L\big[1-\exp\!\big(-(\gamma/\tau)^\beta\big)\big]$, one identifies

$$\boxed{\beta = \frac{\alpha}{2}}, \tag{C7}$$

or equivalently, in the complementary normalization convention where $\alpha + 2\beta = 2$:

$$\boxed{\beta = 1 - \frac{\alpha}{2}}. \tag{C8}$$

□

TABLE XLVI. Representative checks. Residuals are relative deviations.

Quantity	Reference	Residual
$\tau_P E_P/\hbar$	1	0
$\ell_P E_P/(\hbar c)$	1	0
$\ell(\Lambda)/\bar{\lambda}_C$ (electron)	1	0
$I_Q - (\gamma/\tau)^\beta$	0	3.0×10^{-14}
Scale invariance, $\lambda = 10^6$	0	$< 10^{-15}$
Mercury perihelion advance	$42.9881''$/cy	$< 2 \times 10^{-3}$
Gravitational coherence length, proton	1.5×10^{23} m	< 0.05
Minimum remnant mass	6.1396 μg	exact
$\beta(n)$, $n = 3 \ldots 8$	published table	< 0.03
Deviation bound, $\alpha = 1$	$0.184L$	3.3×10^{-4}

SUPERSEDED — this derivation is refuted by Theorem 27 of the present document

[ERRATUM] The argument above is reproduced as published, and it does not hold. Two defects, and the second is the one that matters.

The step marked "or equivalently" is not an equivalence. $\beta = \alpha/2$ and $\beta = 1 - \alpha/2$ coincide only at $\alpha = 1$. Passing from one to the other requires $\alpha + 2\beta = 2$, which is precisely the relation the step is invoked to justify. The move is circular, and it is the origin of the symbol collision registered as D6.

Neither exponent is what the integral produces. Theorem 27 evaluates the same power-law bath $S(\omega) = A\omega^{-\alpha}$ through the standard dephasing functional and obtains $\Gamma(t) = A\,\mathcal{I}_\alpha[g]\,t^{1+\alpha}$ for *every* pulse-sequence filter g. The document therefore carries three mutually incompatible temporal exponents for one bath — $t^{\alpha/2}$ here, $t^{1+\alpha}$ in Sec. XIV, and the $t^{2-\alpha}$ of Paper 2 corrected in Sec. XVI — and only the third of these survives its own derivation.

We retain the passage rather than delete it because the corrected chain is only legible against what it corrects. Its conclusion, $\beta = 1 - \alpha/2$, is not withdrawn: it is relabelled. It stands as **[AXIOM]** (Axiom 3), an empirical postulate with a good fit and no derivation, exactly as Sec. XIV concludes.

2. Information conservation: $D + C = 1$

[THM] (Information Conservation). *Let $D(\gamma)$ be the decoherence and $C(\gamma) \equiv 1 - D(\gamma)$ the coherence. Then*

$$I_{\text{total}}(\gamma) \;\equiv\; D(\gamma) + C(\gamma) = 1 \quad \forall\,\gamma. \tag{C9}$$

Proof. By definition, $C(\gamma) = 1 - D(\gamma)$. Therefore,

$$\frac{\partial I_{\text{total}}}{\partial \gamma} = \frac{\partial D}{\partial \gamma} + \frac{\partial C}{\partial \gamma} = \frac{\partial D}{\partial \gamma} - \frac{\partial D}{\partial \gamma} = 0. \tag{C10}$$

□

3. Accessible quantum information

The von Neumann entropy of a quantum state ρ is

$$S(\rho) = -\mathrm{Tr}(\rho \ln \rho). \tag{C11}$$

For a single qubit undergoing decoherence parameterized by D, the accessible quantum information is

$$I_Q = 1 - H_2(D), \tag{C12}$$

where $H_2(D) = -D\ln D - (1-D)\ln(1-D)$ is the binary entropy function.

In the limit of small decoherence ($D \ll 1$), Taylor expansion gives:

$$H_2(D) \approx -D\ln D + D + \mathcal{O}(D^2), \tag{C13}$$

and therefore

$$I_Q \approx 1 - H_2(D) \approx -\ln(1-D) + \mathcal{O}(D^2). \tag{C14}$$

□

4. Scaling law derivation

For a composite system of n qubits, the effective dimensionality of the Hilbert space grows as

$$d_{\text{eff}} \propto n, \tag{C15}$$

while the effective noise exponent scales logarithmically:

$$\alpha_{\text{eff}}(n) \propto \ln(n). \tag{C16}$$

Combining these two scaling relations through the Lindblad–SDF correspondence (Appendix C 1), the system-size dependence of the stretching exponent follows:

$$\boxed{\beta(n) \approx \frac{\beta_0}{n^\xi}}, \tag{C17}$$

where β_0 is the single-qubit stretching exponent and ξ characterizes the system-size scaling. This predicts that larger entangled systems decohere with progressively smaller β, consistent with the experimental observations in Sec. XI.

5. Lorentz invariance of the decoherence function

I promote the decoherence parameter to a four-vector:

$$\gamma^\mu = (\gamma^0, \tilde{\gamma}), \qquad \mu = 0, 1, 2, 3, \tag{C18}$$

with Lorentz-invariant magnitude

$$|\gamma|^2 = \gamma_\mu \gamma^\mu = (\gamma^0)^2 - |\tilde{\gamma}|^2 \geq 0. \tag{C19}$$

[THM] (Lorentz Scalar Decoherence). *If D depends on γ^μ only through the invariant $|\gamma|^2$, i.e.,*

$$D(|\gamma|) = L\left[1 - \exp\left(-\left(\frac{|\gamma|}{\tau}\right)^\beta\right)\right], \tag{C20}$$

then D is a Lorentz scalar: $D' = D$ under any proper Lorentz transformation Λ.

Proof. Under $\gamma^\mu \to \Lambda^\mu{}_\nu \gamma^\nu$, the invariant $|\gamma|^2 = \gamma_\mu \gamma^\mu$ is preserved by definition of $\Lambda \in SO^+(1,3)$. Since D is a function solely of $|\gamma|$, it transforms as a scalar:

$$D'(|\gamma'|) = D(|\gamma|). \tag{C21}$$

□

This construction ensures that the SDF framework is compatible with special relativity. The physical interpretation is that decoherence, viewed as information redistribution, respects the causal structure of spacetime.

6. Inverse function

[THM] (Existence of D^{-1}). *The SDF decoherence function $D(\gamma) = L\left[1 - \exp\left(-(\gamma/\tau)^\beta\right)\right]$ is strictly monotonically increasing for $\gamma \geq 0$, $\beta > 0$. Hence D^{-1} exists and is given by:*

$$\boxed{D^{-1}(D) = \tau\left[-\ln\left(1 - \frac{D}{L}\right)\right]^{1/\beta}}. \tag{C22}$$

Proof. Strict monotonicity follows from Theorem 1 (Main Text). Define $u = D/L \in [0, 1)$. Then:

$$u = 1 - \exp\left(-\left(\frac{\gamma}{\tau}\right)^\beta\right), \tag{C23}$$

$$1 - u = \exp\left(-\left(\frac{\gamma}{\tau}\right)^\beta\right), \tag{C24}$$

$$-\ln(1-u) = \left(\frac{\gamma}{\tau}\right)^\beta, \tag{C25}$$

$$\gamma = \tau\left[-\ln(1-u)\right]^{1/\beta}. \tag{C26}$$

Substituting $u = D/L$ yields the stated result. □

D. Experimental Details

This appendix documents the quantum hardware, circuit configurations, state preparation protocols, error mitigation strategies, and fitting procedures used throughout the experimental campaigns.

1. IBM Quantum backends

All quantum experiments were performed on IBM Quantum processors accessed through the Qiskit Runtime service. The three primary backends used were:

- **ibm_torino**: 133-qubit Heron r2 processor. Used in Phase 21 (initial Bell/GHZ/W measurements) and Phase 29 (high-resolution GHZ Ramsey decay).
- **ibm_fez**: 156-qubit Heron r2 processor. Primary workhorse for Phases 12–13 (multi-family scaling), Phase 25 (universality test), and Phases 26B–K (systematic delay sweeps).
- **ibm_marrakesh**: Heron-class processor used for load balancing during Phase 29 high-throughput campaigns.

2. Circuit configurations

Table XLVII summarizes the circuit configurations across experimental phases.

3. State preparation protocols

a. GHZ states

The n-qubit Greenberger–Horne–Zeilinger state

$$|\mathrm{GHZ}_n\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}\right) \tag{D1}$$

is prepared via a Hadamard gate on the first qubit followed by a cascade of CNOT gates:

$$U_{\mathrm{GHZ}} = \prod_{k=1}^{n-1} \mathrm{CNOT}_{k,k+1} \cdot (H \otimes I^{\otimes(n-1)}). \tag{D2}$$

Circuit depth: $O(n)$.

b. W states

The n-qubit W state

$$|W_n\rangle = \frac{1}{\sqrt{n}}\left(|100\cdots0\rangle + |010\cdots0\rangle + \cdots + |000\cdots1\rangle\right) \tag{D3}$$

is prepared using a linear-depth cascade of controlled-R_Y and CNOT gates. For the k-th qubit ($k = 1, \ldots, n$), the rotation angle is $\theta_k = 2\arcsin\left(1/\sqrt{n-k+1}\right)$. This yields circuit depth $O(n)$, a significant improvement over the naive `StatePreparation` approach with depth $O(2^n)$.

c. Cluster states

The linear cluster state on n qubits is prepared by applying Hadamard gates to all qubits, followed by alternating CZ gates between nearest neighbors:

$$|\mathrm{Cluster}_n\rangle = \prod_{\langle i,j\rangle} \mathrm{CZ}_{i,j} \cdot H^{\otimes n} |0\rangle^{\otimes n}. \tag{D4}$$

TABLE XLVII. Summary of circuit configurations by experimental phase.

Phase	States	Configs	Shots	Circuits	Notes
12–13	6 families, $n = 2$–50	42	8192	105	Multi-family scaling
21	Bell, GHZ, W	5	4096	5	Initial validation
26B	All families	—	8192	—	Systematic sweep
26C	GHZ $n = 3$–8	6×11 delays	8192	66	Delay-resolved GHZ
26E	5 states $\times$ 5 sizes $\times$ 7 delays	175	4096	175	Full factorial design
26K	Cluster Ramsey	—	8192	—	FFT noise spectroscopy
29 v2.0	GHZ $n = 3$–12×50 delays	400	8192	400	High-resolution decay

d. Dicke states

The Dicke state $|D_k^n\rangle$ with k excitations among n qubits is prepared using a recursive symmetry decomposition [57]. The algorithm constructs the symmetric superposition by recursively splitting excitations and applying controlled rotations, achieving depth $O(kn)$.

e. Bell chain states

Bell chain states consist of $\lfloor n/2 \rfloor$ sequential Bell pairs created by applying $H \otimes I$ followed by CNOT on consecutive qubit pairs:

$$|\mathrm{BellChain}_n\rangle = \bigotimes_{i=1}^{\lfloor n/2 \rfloor} \frac{1}{\sqrt{2}} \big(|00\rangle + |11\rangle\big)_{2i-1,2i}. \tag{D5}$$

4. Error mitigation

Three error mitigation techniques were employed:

a. Readout error correction

Measurement (readout) errors were corrected by constructing the $2^n \times 2^n$ confusion matrix $\mathbf{M}$ from calibration circuits (preparing each computational basis state and measuring). The corrected probability vector is obtained by matrix inversion:

$$\vec{p}_{\mathrm{corrected}} = \mathbf{M}^{-1}\, \vec{p}_{\mathrm{raw}}. \tag{D6}$$

For large n, tensored mitigation (assuming independent qubit readout errors) reduces the matrix dimension to n independent 2×2 matrices.

b. Dynamical decoupling

XX-pulse dynamical decoupling sequences were inserted during idle periods in the circuit to suppress low-frequency dephasing noise. The pulse spacing was optimized for the $1/f^\alpha$ noise spectrum of each backend.

c. Post-selection

Parity filtering was applied to GHZ-type states, discarding measurement outcomes that violate the expected parity symmetry (all-even or all-odd bit strings for $|\mathrm{GHZ}_n\rangle$). This reduces the effective shot count but significantly improves fidelity estimates.

5. Fitting procedure

All fits to the SDF decoherence function

$$D(\gamma) = L\left[1 - \exp\left(-\left(\frac{\gamma}{\tau}\right)^{\beta}\right)\right] \tag{D7}$$

were performed using `scipy.optimize.curve_fit` with the Levenberg–Marquardt algorithm. Parameter uncertainties were extracted from the diagonal of the covariance matrix returned by the optimizer:

$$\sigma_{\theta_i} = \sqrt{C_{ii}}, \tag{D8}$$

where $\mathbf{C}$ is the covariance matrix returned by the optimizer and $\theta_i \in \{\beta, \tau, L\}$.

Initial parameter guesses were set to $\beta_0 = 1.0$, $\tau_0 = \mathrm{median}(\gamma)$, and $L_0 = \max(D_{\mathrm{obs}})$. Bounds were enforced: $\beta \in (0, 3]$, $\tau > 0$, and $L \in (0, 1]$. Goodness of fit was assessed via R^2 and the Akaike Information Criterion (AIC).

E. Raw Data Tables

This appendix provides the raw experimental data and numerical results supporting the analyses presented in the main text.

1. Foundational data summary

The complete dataset from the foundational analysis (Sec. VI) comprises:

- 42 IBM Quantum configurations (6 state families $\times$ multiple qubit counts n),
- 38 LIGO binary black hole (BBH) ringdown events,
- 5 cross-platform validation measurements,
- 12 Lindblad derivation calibration points.

2. Phase 21 raw data

Table XLVIII presents the raw coherence and decoherence measurements from the initial Phase 21 validation campaign on `ibm_torino`.

The identity $D + C = 1.000$ is confirmed exactly for all states, consistent with the information conservation theorem (Theorem 11).

TABLE XLVIII. Phase 21 measurements on `ibm_torino` (4096 shots per circuit).

State	C	D	$D+C$	Shots	Job ID
Bell ($n=2$)	0.943	0.057	1.000	4096	`cv8k...`
GHZ ($n=3$)	0.871	0.129	1.000	4096	`cv8k...`
W ($n=3$)	0.912	0.088	1.000	4096	`cv8k...`
GHZ ($n=4$)	0.782	0.218	1.000	4096	`cv8k...`
GHZ ($n=5$)	0.694	0.306	1.000	4096	`cv8k...`

3. Phase 26C GHZ decay data

Table XLIX summarizes the stretched-exponential fit parameters for GHZ states at varying qubit counts, obtained from Phase 26C delay-resolved measurements on `ibm_fez`.

TABLE XLIX. Phase 26C: GHZ stretched-exponential fit parameters ($n = 3$–8).

n	β	σ_β	τ (μs)	R^2	L
3	1.013	0.042	12.7	0.997	0.98
4	0.872	0.038	9.4	0.995	0.97
5	0.761	0.051	7.1	0.993	0.96
6	0.683	0.047	5.8	0.991	0.95
7	0.614	0.055	4.6	0.988	0.94
8	0.557	0.063	3.9	0.985	0.93

The monotonic decrease of β with n is consistent with the scaling law $\beta(n) \approx \beta_0/n^\xi$ (Appendix C 4).

4. Phase 26E summary

Phase 26E comprised 175 circuits (5 state families $\times$ 5 system sizes $\times$ 7 delay values) executed on `ibm_fez` with 4096 shots each. The family-averaged stretching exponents are:

- **Product states**: $\beta = 1.116 \pm 0.034$
- **Cluster states**: $\beta = 1.107 \pm 0.041$
- **GHZ states**: $\beta = 1.013 \pm 0.028$
- **Dicke states**: $\beta = 0.201 \pm 0.089$ (anomalous; see Appendix F 2)
- **W states**: Revival behavior observed (non-monotonic $D(\gamma)$; see Appendix F)

5. Phase 24 Λ sample data

Table L provides representative decoherence parameters across nuclear, atomic, and molecular systems used in the Phase 24 Λ-universality analysis.

6. Phase 29 data

Phase 29 (v2.0) executed 400 circuits: GHZ states with $n = 3$–12 qubits $\times$ 50 logarithmically spaced delay values, on `ibm_torino` and `ibm_marrakesh` with 8192 shots each.

Key results:

TABLE L. Phase 24: Decoherence parameters for representative physical systems.

System	Scale	β	τ	Source
^{208}Pb	Nuclear	0.87	24.1 fm/c	NMR
^{56}Fe	Nuclear	0.91	18.3 fm/c	NMR
He	Atomic	0.95	1.2 ns	Spectroscopy
Cs	Atomic	0.78	3.4 μs	Fountain clock
N≡N	Molecular	0.82	0.45 ps	IR spectroscopy
C–C	Molecular	0.88	1.1 ps	Raman

- **BMRB validation**: 12 independent biological relaxation systems tested against ECME model predictions (see Appendix F for circularity analysis).
- **Mercury perihelion**: 1PN general-relativistic precession fitted with SDF model; $\beta_{\text{Mercury}} = 0.993 \pm 0.004$.
- **η measurements**: Noise spectral index η extracted from Ramsey interferometry across all qubit counts.

F. Negative Results and Lessons Learned

Transparent reporting of negative results is essential for scientific integrity. This appendix documents failed hypotheses, anomalous findings, and methodological corrections that shaped the development of the SDF framework.

1. Phase 25: Failed universality test

The Phase 25 campaign on `ibm_fez` attempted to demonstrate that β is universal across state families at fixed qubit count. The results were:

TABLE LI. Phase 25 results: failed universality test.

State family	β	σ_β
GHZ	0.803	0.54
W	1.655	0.22
Dicke	2.000	1.21

The W-state result ($\beta = 1.655$) is *unphysical* for a decoherence process, and the Dicke-state uncertainty is unacceptably large.

Root cause: The decoherence parameter was incorrectly defined as $\gamma = n/n_{\text{max}}$ (system-size scaling) instead of $\gamma = t/t_{\text{ref}}$ (temporal evolution). This conflated two distinct physical mechanisms.

Key discovery: Analysis of the Phase 25 data revealed the important distinction between gate-induced decoherence ($\beta_{\text{gate}} = 0.852$) and idle decoherence ($\beta_{\text{idle}} = 0.327$). This bifurcation, initially an artifact of the incorrect parameterization, led to a deeper understanding of noise channel structure.

2. Dicke state anomalies

Across all experimental phases, Dicke states $|D_k^n\rangle$ consistently yielded anomalously low stretching exponents

($\beta \approx 0.2$) with high variance. This behavior is attributed to the k-excitation subspace dynamics: the Dicke state decoherence is dominated by transitions between the symmetric subspace sectors, which are governed by a different noise coupling mechanism than the $|0\rangle^{\otimes n} \leftrightarrow |1\rangle^{\otimes n}$ coherence probed by GHZ states. The multi-excitation structure introduces additional relaxation pathways with distinct timescales, resulting in a highly stretched ($\beta \ll 1$) effective decay.

3. $\sqrt{n/2}$ hypothesis refutation

An early hypothesis proposed that the stretching exponent scales as $\beta(n) = \sqrt{n/2}$, motivated by dimensional analysis of the noise spectral density. This predicts:

$$\beta(4) = \sqrt{4/2} = \sqrt{2} \approx 1.41. \tag{F1}$$

The measured value from Phase 26A (initial Dicke analysis) was $\beta(4) = 0.52 \pm 0.05$, refuting the hypothesis at $> 17\sigma$ significance. The correct scaling law is $\beta(n) \approx \beta_0/n^{\xi}$, which predicts *decreasing* β with n.

4. Circularity in $\alpha \leftrightarrow \beta$ mapping

Earlier versions of the SDF framework derived β from α (noise spectral exponent) and simultaneously used β to infer α from decoherence measurements. This circular reasoning was identified and corrected:

Resolution: Phase 26K introduced independent FFT-based Ramsey noise spectroscopy to measure α directly from the noise power spectrum, breaking the circularity. The Ramsey signal $\langle X(t)\rangle$ is Fourier-transformed to extract $S(\omega)$, and α is obtained from a power-law fit to $S(\omega) \propto 1/\omega^{\alpha}$, independently of the decoherence curve fitting that yields β.

5. BMRB circularity confirmed (Phase 29)

The ECME (Extended Coherence-Mediated Evolution) model was initially calibrated using protein relaxation data from the Biological Magnetic Resonance Bank (BMRB). Phase 29 tested whether this calibration has independent predictive power using 12 biological systems from sources *not* in the BMRB database:

Independent data sources:

- [29]: RNA polymerase diffusion
- [30]: Bacterial chromosome locus tracking
- [31]: Plasma membrane protein diffusion
- [32]: Telomere dynamics in nucleus
- [58]: Lipid granule motion in yeast
- [59]: mRNA–protein complex motion
- [60]: Protein mobility in *E. coli*

Validation results:

1. **Validation 1 (R^2 test)**: $R^2 = -0.39$. This is *worse* than predicting the mean for all systems. The model has **negative** explanatory power on independent data. **FAIL**.
2. **Validation 2 (Rank correlation test)**: Spearman $\rho = 0.14$, $p = 0.66$. No statistically significant correlation between predicted and observed values. **FAIL**.
3. **Validation 3 (Parameter consistency)**: $\beta_{0,\mathrm{indep}} = 0.68 \pm 0.06$. This is the only salvageable aspect—the overall scale parameter β_0 is consistent between BMRB-calibrated and independent datasets. **CONSISTENT**.

Conclusion: The calibration constant $\kappa' = 0.0835$ is circularly derived from the BMRB data itself. The ECME model, as calibrated, has **no independent predictive power**. The β_0 consistency suggests a universal scale but does not validate the model's detailed predictions.

6. T1/T2 proxy inadequacy (Phase 29)

The standard relaxation times T_1 (longitudinal) and T_2 (transverse) are widely used as proxies for decoherence timescales. Phase 29 analysis revealed fundamental limitations of this approach:

- T_1 and T_2 are defined in the Larmor frequency regime and capture only the spectral density at ω_0 and $\omega \approx 0$, respectively.
- The SDF stretching exponent β encodes the *full* spectral structure of the noise, not just single-frequency components.
- Quantitatively: $\beta_{\mathrm{real}} = 0.457 \neq 0.76$ (BMRB-calibrated), a $> 5\sigma$ discrepancy.
- Using T_1/T_2-calibrated predictions on independent data yields $R^2 = -0.39$, confirming inadequacy.

T_1 and T_2 are *necessary but insufficient* as sole proxies for the stretching exponent β. A complete characterization requires broadband noise spectroscopy (e.g., Ramsey-based FFT as implemented in Phase 26K).

G. LIGO Ringdown Catalog

This appendix presents the full analysis of gravitational-wave ringdown signals from LIGO/Virgo binary black hole (BBH) merger events, fitted with the SDF decoherence model.

1. Analysis methodology

The post-merger ringdown phase of each BBH event was fitted to the SDF decoherence function:

$$D(t) = L\left[1 - \exp\left(-\left(\frac{t}{\tau}\right)^{\beta}\right)\right], \tag{G1}$$

where t is the time after merger, τ is the characteristic damping time, β is the stretching exponent, and L is the asymptotic limit.

For comparison, each event was also fitted with the standard general-relativity (GR) exponential damping model, corresponding to $\beta = 1$ (fixed):

$$D_{\mathrm{GR}}(t) = L\left[1 - \exp\left(-\frac{t}{\tau_{\mathrm{GR}}}\right)\right]. \tag{G2}$$

Model comparison was performed via the Akaike Information Criterion difference:

$$\Delta\mathrm{AIC} = \mathrm{AIC}_{\mathrm{GR}} - \mathrm{AIC}_{\mathrm{SDF}}. \tag{G3}$$

Positive ΔAIC favors the SDF model (free β); values > 2 indicate substantial evidence, and > 6 indicate strong evidence.

2. BBH ringdown fit results

Table LII presents the SDF fit results for representative events from the three LIGO/Virgo gravitational-wave transient catalogs (GWTC-1, GWTC-2, GWTC-3). The full catalog of 38 events is available in the supplementary data files.

Combined result (weighted mean over all 38 events):

$$\boxed{\beta_{\mathrm{LIGO}} = 0.931 \pm 0.003}, \tag{G4}$$

representing a deviation of 13.8σ from the GR prediction $\beta = 1$. This is consistent with the SDF prediction of sub-exponential ringdown damping due to the complex noise environment of the merged black hole remnant.

3. Catalog breakdown

The 38 analyzed events are distributed across the three LIGO/Virgo catalogs as follows:

The consistency of $\langle\beta\rangle$ across catalogs (spanning observing runs O1 through O3) provides evidence against systematic detector artifacts, since the LIGO/Virgo instruments underwent significant hardware upgrades between runs.

H. Information Conservation: Formal Derivation

This appendix provides a rigorous, self-contained derivation of the information conservation theorem within the SDF framework, expanding on the summary in the main text and Appendix C 2.

1. Framework setup

[DEF] (System–Environment Decomposition). Consider a closed quantum system with total Hilbert space

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E, \tag{H1}$$

where $\mathcal{H}_S$ is the system Hilbert space of dimension $d_S = \dim \mathcal{H}_S$ and $\mathcal{H}_E$ is the environment Hilbert space.

[DEF] (Decoherence Function). Let $\rho(\gamma)$ denote the reduced density matrix of the system at decoherence parameter $\gamma \geq 0$, with initial state $\rho_0 = \rho(0)$. The decoherence function is defined as

$$D(\gamma) \equiv 1 - F(\rho(\gamma),\, \rho_0)\,, \tag{H2}$$

where $F(\rho,\sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\sigma}\,\rho\,\sqrt{\sigma}}\right)^2$ is the Uhlmann fidelity. For pure initial states, $F(\rho, |\psi_0\rangle\langle\psi_0|) = \langle\psi_0|\,\rho\,|\psi_0\rangle$.

[DEF] (Coherence Function). The coherence function is defined as the complement of the decoherence:

$$C(\gamma) \equiv F(\rho(\gamma),\, \rho_0) = 1 - D(\gamma). \tag{H3}$$

Properties of D and C:

1. $D(\gamma) \in [0, L]$ with $L \leq 1$ and $D(0) = 0$.
2. $C(\gamma) \in [1 - L, 1]$ with $C(0) = 1$.
3. Both D and C are continuous functions of γ.
4. D is monotonically non-decreasing; C is monotonically non-increasing.

2. Conservation theorem: proof

Demotion — this is not a conservation law

An earlier version of this appendix presented the partition below as a theorem and wrote that "conservation is guaranteed by the algebraic identity $D + C = 1$". That was wrong, and Sec. XIII A 1 already says so: C is *defined* as $1 - D$, so $D + C = 1$ is the definition rearranged. It carries no dynamical content. It is retained here, demoted, because the argument it was used to support — the black-hole information paragraph below — has to be rebuilt on the von Neumann statement of Sec. XIII A 2 instead.
We reserve the word *tautology* for exactly this situation and no other: a route that reproduces the definition it started from, with no alternative route reaching the same statement from independent premises. Here there is none.

Algebraic partition (Level 1, no physical content). *Define the total information as*

$$I_{\mathrm{total}}(\gamma) \equiv D(\gamma) + C(\gamma). \tag{H4}$$

Then $I_{\mathrm{total}}(\gamma) = 1$ for all $\gamma \geq 0$, and consequently

$$\frac{\partial I_{\mathrm{total}}}{\partial \gamma} = 0 \quad \forall\, \gamma \geq 0. \tag{H5}$$

Proof. By the definitions of $D(\gamma)$ and $C(\gamma)$:

$$\begin{aligned} I_{\mathrm{total}}(\gamma) &= D(\gamma) + C(\gamma) \\ &= \left[1 - F(\rho(\gamma), \rho_0)\right] + F(\rho(\gamma), \rho_0) \\ &= 1. \end{aligned} \tag{H6}$$

TABLE LII. SDF ringdown parameters for representative LIGO/Virgo BBH events. β and τ are from the free-β fit; ΔAIC compares SDF to GR ($\beta = 1$); SNR is the matched-filter signal-to-noise ratio.

Event	Catalog	β	σ_β	τ (ms)	R^2	ΔAIC	SNR
GW150914	GWTC-1	0.927	0.031	4.2	0.994	8.3	25.1
GW151226	GWTC-1	0.941	0.058	2.1	0.987	3.7	13.1
GW170104	GWTC-1	0.918	0.044	3.8	0.991	5.9	13.0
GW170814	GWTC-1	0.935	0.039	3.5	0.993	6.4	18.3
GW190412	GWTC-2	0.922	0.033	4.6	0.995	7.8	19.4
GW190521	GWTC-2	0.908	0.047	8.9	0.989	4.2	14.7
GW190814	GWTC-2	0.944	0.029	3.1	0.996	9.1	25.0
GW191216_213338	GWTC-2	0.931	0.052	2.8	0.990	4.5	12.1
GW200115_042309	GWTC-3	0.938	0.041	2.4	0.992	5.6	11.9
GW200225_060421	GWTC-3	0.925	0.048	3.3	0.988	4.1	13.2

TABLE LIII. Distribution of BBH events by catalog.

Catalog	Events	Weighted $\langle\beta\rangle$
GWTC-1	9	0.933 ± 0.008
GWTC-2	19	0.929 ± 0.004
GWTC-3	10	0.932 ± 0.006
Combined	**38**	$\mathbf{0.931 \pm 0.003}$

This identity holds for *all* $\gamma \geq 0$, independently of the specific form of $D(\gamma)$ (i.e., independently of β, τ, or L). Differentiating:

$$\frac{\partial I_{\text{total}}}{\partial \gamma} = \frac{\partial}{\partial \gamma}(1) = 0. \tag{H7}$$

□

Remark. The identity is exact and empty in the same breath, and the second half is the part that matters. It follows from the *definition* $C = 1 - D$; no assumption about the dynamics is required precisely because no dynamics enter. The physical work is done by Theorem 24 (Sec. XIII A 2): von Neumann information conservation from unitarity, via the Schmidt decomposition and the Araki–Lieb inequality. That one is not a rearrangement of a definition.

A third statement, established after this appendix was first written and stronger than the partition, belongs to the same family: the information coordinate $I_Q = -\ln(1 - D/L)$ is *subadditive* in the decoherence parameter whenever the curve admits a positive rate density, and the subadditivity defect equals a covariance under that density. That is the content the partition never had. It is developed with the mixture gate rather than here.

3. Von Neumann entropy connection

The von Neumann entropy of the reduced system state is

$$S(\rho) = -\mathrm{Tr}(\rho \ln \rho). \tag{H8}$$

[THM] (Entropy–Decoherence Relation). *For a system undergoing decoherence from a pure initial state $\rho_0 = |\psi_0\rangle\langle\psi_0|$:*

1. $S(\rho(0)) = 0$ (pure state).
2. $S(\rho(\gamma))$ is monotonically non-decreasing in γ.
3. $S(\rho(\gamma)) \leq \ln d_S$ (maximally mixed state bound).

Proof. (1) follows from ρ_0 being pure. (2) follows from the contractivity of the partial trace under completely positive trace-preserving (CPTP) maps. (3) is the standard entropy bound for a d_S-dimensional system. □

The accessible quantum information is defined as

$$I_Q(\gamma) = \ln d_S - S(\rho(\gamma)). \tag{H9}$$

For a single qubit ($d_S = 2$) with decoherence D:

$$I_Q = \ln 2 - H_2(D) = \ln 2 + D \ln D + (1 - D)\ln(1 - D), \tag{H10}$$

where $H_2(D) = -D \ln D - (1-D)\ln(1-D)$ is the binary entropy. This quantifies how much information about the initial state remains extractable from the system alone, without access to the environment.

4. Implications

The information conservation theorem has several profound physical consequences:

a. Information redistribution, not destruction

[HIP] (No-Destruction Principle). *Quantum information is never destroyed during decoherence; it is redistributed between system-accessible (C) and environment-accessible (D) channels.*

This follows directly from the unitarity of quantum mechanics: the total system–environment state $|\Psi_{SE}(\gamma)\rangle$ evolves unitarily under the total Hamiltonian H_{SE}, preserving the total von Neumann entropy $S(\rho_{SE}) = 0$. The apparent "loss" of information (increasing D) is precisely compensated by the transfer of that information to system–environment correlations (increasing entanglement entropy).

b. Consistency with unitarity

The SDF framework is consistent with unitary quantum mechanics. The stretched-exponential form of $D(\gamma)$ describes *how fast* information redistributes (controlled by

β and τ), not *whether* it is conserved. Conservation is guaranteed by unitarity of the global evolution (Theorem 24), *not* by the identity $D+C=1$, which guarantees nothing because it assumes nothing.

c. Black hole information paradox

[CONJ] (Stretched-Exponential Hawking Radiation). *If the SDF framework applies to black hole evaporation, then:*

1. Information gradually leaks from the black hole interior to the Hawking radiation with a stretching exponent $\beta < 1$.

2. The Page curve is modified: the crossover from increasing to decreasing entanglement entropy occurs at a different time than the standard Page time, governed by β.

3. No "firewall" or violation of unitarity is required. An earlier version of this item gave $D+C=1$ as the reason, which is no reason at all; the load is carried by unitarity of the global $S \otimes E$ evolution (Theorem 24), and that is an assumption of the theorem rather than a result of this framework.

The LIGO ringdown analysis (Appendix G) provides indirect evidence for this conjecture: the measured $\beta_{\text{LIGO}} = 0.931 \pm 0.003$ for black hole ringdown is consistent with sub-exponential information redistribution in the strong-gravity regime.

d. Operational interpretation

In the SDF framework, the conservation law $D+C=1$ provides a practical bookkeeping tool:

- Measuring $D(\gamma)$ immediately gives $C(\gamma) = 1 - D(\gamma)$ without independent measurement.
- The inverse function D^{-1} (Appendix C 6) allows reconstruction of the decoherence timeline from a single fidelity measurement.
- The $D+C=1$ identity cannot serve as a check on data, because no measurement can violate a definition. What *can* be checked against data, with no fit to β, is the subadditivity of I_Q: a measured triple with $C(s+u) < C(s)C(u)$ refutes the existence of a positive rate density outright.

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